Advanced Calibration for Device Simulation User Guide

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SYNOPSYS®

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About This Guide

The Synopsys® Sentaurus™ Device tool is a quantum drift-diffusion device simulator that solves semiconductor equations in one, two, or three spatial dimensions.

Extensions for hydrodynamic transport, solving the heat conduction equation, and many other features are also available. Because of the many models and corresponding model parameters and their dependency on fabrication processes and device types, it is difficult to derive automatically a well-adjusted device simulation environment for specific simulation tasks. Therefore, preselection of models and precalibration of parameter sets are necessary. Advanced Calibration Device refers to the preselection of models and the precalibration of parameter sets.

This user guide describes the contents and use of the Advanced Calibration Device files and is designed to give users fast access to parameter sets and model selections needed for device simulation. This user guide is intended for users who are familiar with Sentaurus Device and want to achieve a higher accuracy in device simulation. See the *Sentaurus™ Device User Guide*.

For additional information, see:

- The TCAD Sentaurus release notes, available on the Synopsys SolvNetPlus site (see Accessing SolvNetPlus on page 8)
- Documentation available on the SolvNetPlus site

Conventions

The following conventions are used in Synopsys documentation.

Convention	Description
`Courier font`	Identifies text that is displayed on the screen or that the user must enter. It identifies the names of files, directories, paths, parameters, keywords, and variables.
Italicized text	Used for emphasis, the titles of books and journals, and non-English words. It also identifies components of an equation or a formula, a placeholder, or an identifier.

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Send an email message to:

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- support-tcad-eu@synopsys.com from within Europe
- support-tcad-ap@synopsys.com from within Asia Pacific (China, Taiwan, Singapore, Malaysia, India, Australia)
- support-tcad-kr@synopsys.com from Korea
- support-tcad-jp@synopsys.com from Japan

Statement on Inclusivity and Diversity

Synopsys is committed to creating an inclusive environment where every employee, customer, and partner feels welcomed. We are reviewing and removing exclusionary language from our products and supporting customer-facing collateral. Our effort also includes internal initiatives to remove biased language from our engineering and working environment, including terms that are embedded in our software and IPs. At the same time, we are working to ensure that our web content and software applications are usable to people of varying abilities. You may still find examples of non-inclusive language in our software or documentation as our IPs implement industry-standard specifications that are currently under review to remove exclusionary language.

1

Introduction to Advanced Calibration Device

This chapter introduces the use of Advanced Calibration Device in Sentaurus Device simulations.

Overview of Advanced Calibration Device

Advanced Calibration Device is a set of models and parameters that is recommended for device simulations of certain device types and technologies. This set of models and parameters is contained in text files, which standard text editors can open.

You select the standard Synopsys calibration by starting Sentaurus Device with the corresponding Advanced Calibration Device command file section and by loading the Advanced Calibration Device parameter file from that command file.

For more information about how to use the command file that defines models and the parameter file that defines the parameters of the models selected in the command file, see the *Sentaurus™ Device User Guide*.

This user guide describes the parameter file and command file sections for Advanced Calibration Device.

Location of Advanced Calibration Device Files

Each release of TCAD Sentaurus provides new Advanced Calibration Device files that include the latest set of models and parameters. The `MaterialDB` folder of the standard installation of Sentaurus Device stores the parameter files and is located at:

```
$STROOT/tcad/$STRELEASE/lib/sdevice/MaterialDB
```

The `STROOT` environment variable indicates where the Synopsys TCAD distribution has been installed.

See Table 1 on page 13 for a list of the available material-related parameter files.

Target Devices and Technologies

Advanced Calibration Device provides a set of parameters for CMOS technologies from micrometer dimensions down to the 5 nm node, including planar gate-first and gate-last technologies, nonplanar technologies, and technologies based on novel channel materials.

In addition, Advanced Calibration Device provides parameters and models for silicon smart-power and power devices (IGBT, LDMOS, NMOS, and PMOS), GaN HEMTs, InGaAs channel devices, SiC diodes, and SiC FETs.

The main process features of planar and nonplanar bulk and SOI technology nodes covered by the model and parameter sets are:

- Stress and crystallographic orientation engineering
- Poly/SiON or SiO_2 gate stacks and metal/HfO_2/SiON or SiO_2 gate stacks
- Gate-first and gate-last technologies
- Silicon, germanium, SiGe, GaN, SiC, or InGaAs channels

For these devices and technologies, Advanced Calibration Device provides model selections and parameter sets for the simulation of the following device properties:

- Band structure, electrostatic properties, and quantization
- Low-field, high-field, and ballistic transport
- Stress and mole fraction dependency of mobility

Note:

The parameter sets must be considered as initial parameter sets, that is, starting points for further calibration.

Content of Parameter Files

The information presented here relates to the models presented in Chapter 2 on page 23 and to the files stored in the `MaterialDB` folder of Sentaurus Device.

Table 1 on page 13 lists the content and features of the material-related parameter files. References and additional comments for the parameters can be found in the parameter files.

When simulating silicon or SiGe CMOS, or silicon smart-power or power devices, you must load one or more of the following files:

- `Silicon.par`
- `Germanium.par`
- `SiliconGermanium.par`
- `Siliconc100.par`
- `Siliconc110.par`
- `SiliconGermaniumc100.par`
- `SiliconGermaniumc110.par`

When simulating wide-bandgap devices, depending on the material, you must load one or more of the following files:

- `AlGaN.par`, `AlInGaN.par`, `AlInN.par`, `AlN.par`
- `Diamond.par`
- `Ga2O3.par`
- `GaN.par`
- `InGaN.par`, `InN.par`

When simulating III–V arsenide devices, you must load one or more of the following files:

- `AlAs.par`
- `AlGaAs.par`
- `GaAs.par`
- `InAs.par`
- `InGaAs.par`

When simulating SiC devices, you must load one of the following files:

- `SiliconCarbide.par`
- `4H-SiC.par/4HSiC.par/SiC_4H.par`
- `6H-SiC.par`

Table 1 Parameter files in MaterialDB folder of Sentaurus Device

Parameter file	Material	Parameters
`AlAs.par`	AlAs	• Permittivity • Heat capacitance • Thermal conductivity • Band gap and affinity • Density-of-states • Parameters in `Multivalley` section • Optical parameters Other model parameter sections are available in the parameter file but are not yet updated and reviewed.
`AlGaAs.par`	AlGaAs	• Permittivity • Heat capacitance • Thermal conductivity • Band gap and affinity • Density-of-states • Parameters in `Multivalley` section • Optical parameters Other model parameter sections are available in the parameter file but are not yet updated and reviewed.

Table 1 Parameter files in MaterialDB folder of Sentaurus Device (Continued)

Parameter file	Material	Parameters
`AlGaN.par`	AlGaN	• For most models, mole fraction dependency is interpolated linearly between the corner materials GaN and AlN • U-shape of thermal conductivity is described by a piecewise linear spline function in the parameter file • Mole fraction–dependent band gap (parabolic interpolation) • Mole fraction–dependent mass is interpolated linearly between GaN and AlN for density-of-states calculation • Mole fraction–dependent mobility with doping dependency is described by a piecewise linear spline function in the parameter file • Mole fraction–dependent high field dependency is described by a piecewise linear spline function in the parameter file • Mole fraction–dependent impact ionization is described by a piecewise linear spline function in the parameter file
`AlInGaN.par`	AlInGaN	• For most models, mole fraction dependency is interpolated linearly between the corner materials AlN, GaN, and InN
`AlInN.par`	AlInN	• For most models, mole fraction dependency is interpolated linearly between the corner materials AlN and InN

Table 1 Parameter files in MaterialDB folder of Sentaurus Device (Continued)

Parameter file	Material	Parameters
`AlN.par`	AlN	• Isotropic and anisotropic permittivity • Thermal conductivity • Lattice heat capacity • Band gap and affinity • Density-of-states and masses • Mobility with doping dependency • High field dependency • Shockley–Read–Hall • Auger • Impact ionization • Piezoelectric polarization • Optical material properties • Mechanical compliance parameters
`CobaltSilicide.par`	CoSi/$CoSi_2$	• Permittivity • Thermal conductivity • Lattice heat capacity • Resistivity • Optical material parameters
`Diamond.par`	Diamond	• Isotropic and anisotropic permittivity • Thermal conductivity • Incomplete ionization • Band gap and affinity • Density-of-states • Constant mobility • High field dependency • Impact ionization
`Ga2O3.par`	Ga_2O_3	• Permittivity • Heat capacitance • Thermal conductivity • Band structure (band gap and affinity) • Density-of-states for electrons • High-field saturation • Impact ionization

Table 1 Parameter files in MaterialDB folder of Sentaurus Device (Continued)

Parameter file	Material	Parameters
`GaAs.par`	GaAs	• Permittivity • Heat capacitance • Thermal conductivity • Band structure (band gap, affinity, and parameters for the `Multivalley` section) • Bandgap narrowing • Density-of-states • Bulk and inversion mobility for different surface and channel orientations • Quantization parameters • Optical properties Other model parameter sections are available in the parameter file but are not yet updated and reviewed.
`GaAsSb.par`	GaAsSb	• Permittivity • Heat capacitance • Thermal conductivity • Band gap and affinity • Density-of-states • Parameters in `Multivalley` section
`GaInSb.par`	GaInSb	• Permittivity • Heat capacitance • Thermal conductivity • Band gap and affinity • Density-of-states • Parameters in `Multivalley` section

Table 1 Parameter files in MaterialDB folder of Sentaurus Device (Continued)

Parameter file	Material	Parameters
`GaN.par`	GaN	• Isotropic and anisotropic permittivity • Thermal conductivity • Lattice heat capacity • Band gap and affinity • Density-of-states and masses • Bulk mobility with doping dependence • Inversion layer mobility (`IALMob`) • High field dependency • Shockley–Read–Hall • Auger • Radiative recombination • Impact ionization • Piezoelectric polarization • Optical material properties • Mechanical compliance parameters
`GaSb.par`	GaSb	• Permittivity • Heat capacitance • Thermal conductivity • Band gap and affinity • Density-of-states • Parameters in `Multivalley` section
`Germanium.par`	Germanium	• Permittivity • Heat capacitance • Thermal conductivity • Band gap and bandgap narrowing • Density-of-states • Bulk mobility • High field dependency • Optical properties Other model parameter sections are available in the parameter file but are not yet updated and reviewed.

Table 1 Parameter files in MaterialDB folder of Sentaurus Device (Continued)

Parameter file	Material	Parameters
`InAs.par`	InAs	• Permittivity • Heat capacitance • Thermal conductivity • Band structure (band gap, affinity, and parameters for the `Multivalley` section) • Bandgap narrowing • Density-of-states • Bulk and inversion mobility for different surface and channel orientations • Quantization parameters • Optical properties Other model parameter sections are available in the parameter file but are not yet updated and reviewed.
`InGaAs.par`	InGaAs	• Permittivity • Heat capacitance • Thermal conductivity • Band structure (band gap, affinity, and parameters for the `Multivalley` section) • Bandgap narrowing • Density-of-states • Bulk and inversion mobility for different surface and channel orientations • Quantization parameters • Optical properties Other model parameter sections are available in the parameter file but are not yet updated and reviewed.
`InGaN.par`	InGaN	• For most models, mole fraction dependency is interpolated linearly between the corner materials GaN and InN

Table 1 Parameter files in MaterialDB folder of Sentaurus Device (Continued)

Parameter file	Material	Parameters
`InN.par`	InN	• Isotropic and anisotropic permittivity • Heat capacitance • Thermal conductivity • Band gap and affinity • Density-of-states • Mobility with doping dependency • High field dependency • Impact ionization • Piezoelectric polarization • Mechanical compliance parameters
`InSb.par`	InSb	• Permittivity • Heat capacitance • Thermal conductivity • Band gap and affinity • Density-of-states • Parameters in `Multivalley` section

`Silicon.par`
`Siliconc100.par`
`Siliconc110.par`
	Si	• Permittivity
		• Heat capacitance
		• Thermal conductivity
		• Band gap and bandgap narrowing
		• Density-of-states
		• Quantization
		• Bulk and inversion mobility for different surface and channel orientations, and film thicknesses
		• High field dependency
		Other model parameter sections are available in the parameter file but are not yet updated and reviewed.

Table 1 Parameter files in MaterialDB folder of Sentaurus Device (Continued)

Parameter file	Material	Parameters

`SiliconCarbide.par`
`4H-SiC.par/4HSiC.par/SiC_4H.par`
`6H-SiC.par`
	SiC	• Isotropic and anisotropic permittivity
		• Isotropic and anisotropic thermal conductivity
		• Heat capacitance
		• Band gap and bandgap narrowing
		• Density-of-states
		• Bulk mobility with doping dependence
		• Inversion layer mobility (`Enormal`)
		• Inversion layer mobility (`IALMob`)
		• Anisotropic mobility
		• Isotropic and anisotropic high field dependency
		• Shockley–Read–Hall
		• Auger
		• Isotropic and anisotropic impact ionization
		• Incomplete ionization

`SiliconGermanium.par`
`SiliconGermaniumc100.par`
`SiliconGermaniumc110.par`
	SiGe (0–100%) can be used for Ge as well	• All models are mole fraction dependent
		• Permittivity
		• Heat capacitance
		• Thermal conductivity
		• Band gap and bandgap narrowing
		• Density-of-states
		• Quantization
		• Bulk and inversion mobility for different surface and channel orientations, and film thicknesses
		• High field dependency
		• Auger
		Other model parameter sections are available in the parameter file but are not yet updated and reviewed.

Special Parameter Files

The `MaterialDB/custom` directory of Sentaurus Device contains the following special parameter files that reflect certain application cases:

- `GaAs_DEEPEN.par`, `InAs_DEEPEN.par`, and `InGaAs_DEEPEN.par` contain parameters in the `Bandgap` and `Multivalley` sections extracted from hybrid functional calculations within the scope of the European Union DEEPEN project.
- For `AlN_DEEPEN.par`, `GaN_DEEPEN.par`, and `InN_DEEPEN.par`, the parameters in the `Piezoelectric_Polarization` section are included, based on the results of hybrid functional calculations. In addition, the bowing parameters are updated in the `Bandgap` section for the `AlGaN_DEEPEN.par`, `AlInN_DEEPEN.par`, and `InGaN_DEEPEN.par` files.
- `GatePolySilicon.par` is for the polysilicon gate of FETs.
- `mcmob.par` contains the default parameter set for the `MCmob` model (see Appendix A on page 109).
- `Silicon_kvm.par` and `SiliconGermanium_kvm.par` contain calibrated kinetic velocity model (KVM) parameters for better simulation of ballistic effects in short-channel devices (see the *Sentaurus™ Device User Guide*).
- `In52Al48As.par`, `In53Ga47As.par`, and `In52Al48As_In53Ga47As.par` contain material parameters specifically for $In_{0.52}Al_{0.48}As$, $In_{0.53}Ga_{0.47}As$, and $(In_{0.52}Al_{0.48}As)_x(In_{0.53}Ga_{0.47}As)_{1-x}$ alloys, respectively, which are lattice matched to InP material. To be recognized in Sentaurus Device, these material names can be added manually to the `datexcodes.txt` file:

```
Materials {
   In52Al48As {
      label   = "In52Al48As"
      group   = Semiconductor
   }
   In53Ga47As {
      label   = "In53Ga47As"
      group   = Semiconductor
   }
   In52Al48As_In53Ga47As {
      label   = "In52Al48As_In53Ga47As"
      group   = Semiconductor
   }
}
```

The mole fraction–dependent material $(In_{0.52}Al_{0.48}As)_x(In_{0.53}Ga_{0.47}As)_{1-x}$ must be defined as well in the `Molefraction.txt` file:

```
#(In52Al48As)(x)(In53Ga47As)(1-x)
In52Al48As_In53Ga47As (x=0) = In53Ga47As
In52Al48As_In53Ga47As (x=1) = In52Al48As
```

Using Advanced Calibration Device

To use Advanced Calibration Device, you can look up the parameter files and the command files that are best suited for your device and technology requirements in Chapter 2 on page 23.

The European Union projects DEEPEN and III-V-MOS have generated additional parameter sets that complement the general parameter files of Advanced Calibration Device. The parameter files from the DEEPEN project are stored in the `MaterialDB/custom` folder for AlGaN, AlInN, AlN, GaAs, GaN, InAs, InGaAs, InGaN, and InN (see Special Parameter Files on page 21).

2

Guide to Device Simulations

This chapter provides a guide for the quick selection of parameter sets and device models.

Specifying Simulation Cases

Table 2 and Table 3 present typical simulation cases, where a path is defined that you can use to select the proper device simulation parameter set and the appropriate `Physics` section for the command file. In the tables, the characters *A* to *J* indicate different paths that are described in subsequent sections.

Table 2 Specifying the simulation case: CMOS

	Node															
	> 130 nm	**32–130 nm**			**14–32 nm**				**< 14 nm**							
Selection path	A	B	C	C	B	C	C	C	E	D	E	E	I	I	E	E	E
Planar bulk and SOI	X	X	X	X	X	X	X	X									
FinFET									X		X	X	X				
Nanowire														X	X	X	X
Thin-layer SOI										X							
Si	X	X	X		X	X		X	X	X					X		
SiGe				X			X				X					X	
Ge												X					X
InGaAs													X	X			
Gate-first	X	X			X												

Table 2 Specifying the simulation case: CMOS (Continued)

	Node															
	> 130 nm	**32–130 nm**			**14–32 nm**				**< 14 nm**							
Selection path	A	B	C	C	B	C	C	C	E	D	E	E	I	I	E	E	E
Gate-first with metal gate			X	X		X	X									
Gate-last with metal gate								X	X	X	X	X	X	X	X	X	X
Without stress	X												X	X	X		
With stress		X	X	X	X	X	X	X	X	X	X	X				X	X

Table 3 Specifying the simulation case: Smart power and power

	Device					
	HVMOS	**HVMOS**	**LDMOS**	**HEMT**	**Diode**	**IGBT**
Selection path	F	J	F	G	H	F
Si	X		X			X
SiC		X			X	
GaN				X		

Accessing Parameter Files and Model Frameworks

The following sections describe the parameter files and the model frameworks. For information about handling and loading parameter files, see the *Sentaurus™ Device User Guide*.

Path A: Basic CMOS Simulation Command File Sections

Quantization uses the density gradient model of Sentaurus Device (`eQuantumPotential` and `hQuantumPotential`). However, for technologies with gate oxide thicknesses greater than 3 nm, you can neglect quantum correction without greatly affecting the simulation of the gate capacitance.

The orientation of the conducting interface is detected automatically. However, you must define the channel or the current flow direction manually using the correct mobility parameter file. The corresponding parameter files are located in the `MaterialDB` folder.

For the <110> channel direction, you must load the `Siliconc110.par` file. For the <100> channel direction, you must load the `Siliconc100.par` file.

```
Physics {Fermi}
   Physics (Material="Silicon") {
   eQuantumPotential(AutoOrientation density)
   hQuantumPotential(AutoOrientation density)
   Mobility (Enormal (IALMob(AutoOrientation)) HighFieldSaturation)
   # For PMOS, it is useful to use PhononCombination=2 in the IALMob.
   EffectiveIntrinsicDensity(OldSlotboom)
   Recombination(SRH)
   # Recombination(SRH Band2Band(Model=NonlocalPath))
                           # For off-current calculation, band-to-band
                           # tunneling models must be switched on.
}
```

Path B: Add Influence of Mechanical Stress to Physics Section Shown in Path A

Taking into account mechanical stress, you must combine the low-field mobility model with a model describing the stress influence. You can model the mobility change by mechanical stress in one of the following ways:

- Use the `eSubband` and `hSubband` models for a more physics-based calculation of the mobility enhancement by mechanical stress in silicon devices that is directly based on band structure and subbands. For the subband models, you must set the channel direction in the command file when it is *not* the <110> channel.
- Use the `MCmob` physical model interface (PMI) model (parameters are extracted from Sentaurus Device Monte Carlo) or the `SBmob` PMI model (parameters are extracted from Sentaurus Band Structure). For the `MCmob` and `SBmob` models, you must set the channel direction in the parameter file (see Appendix A on page 109).

The orientation of the conducting interface is detected automatically. However, you must define the channel or the current flow direction manually using the correct mobility parameter file. The corresponding parameter files are located in the `MaterialDB` folder.

For the <110> channel direction, you must load the `Siliconc110.par` file. For the <100> channel direction, you must load the `Siliconc100.par` file.

With subband models, you must use the following syntax:

```
Physics {Fermi}
   Physics (Material="Silicon") {
   eQuantumPotential(AutoOrientation density)
   hQuantumPotential(AutoOrientation density)
   Mobility (Enormal (IALMob(AutoOrientation)) HighFieldSaturation)
   # For the inclusion of ballistic effects, you can use the command
   # BalMob(KVM(Full) Frensley Fermi) in the Mobility statement.
   # For PMOS, it is useful to use PhononCombination=2 in the IALMob.
   EffectiveIntrinsicDensity(OldSlotboom)
   Recombination(SRH)
   # Recombination(SRH Band2Band(Model=NonlocalPath))
                           # For off-current calculation, band-to-band
                           # tunneling models must be switched on.

   eMultivalley(MLDA kpDOS -density) # hMultivalley for holes.
                                 # For holes, defining kpDOS and parfile
                                 # together results in double-counting
                                 # for holes when gamma hole valleys are
                                 # defined in the parameter file.

   Piezo (
      Model (
         Mobility (
            saturationfactor=0.2
            eSubband(Fermi EffectiveMass Scattering(MLDA))
           # eSubband(Fermi EffectiveMass Scattering(MLDA) -RelChDir110)
           # for <100> channel.
           # hSubband(Fermi EffectiveMass Scattering(MLDA)) for holes
           # For holes, the replacement of the option Fermi by the
           # option Doping can result in speedup of the simulation.
         )
         DOS(eMass hMass)
         DeformationPotential(Minimum ekp hkp)
      )
   )
}
```

With the `MCmob` model or `SBmob` model, you must use the following syntax:

```
Physics {Fermi}

Physics (Material="Silicon") {
   eQuantumPotential(AutoOrientation density)
   hQuantumPotential(AutoOrientation density)
   Mobility (Enormal (IALMob(AutoOrientation)) HighFieldSaturation)
   # For PMOS, it is useful to use PhononCombination=2 in the IALMob.
   EffectiveIntrinsicDensity(OldSlotboom)
   Recombination(SRH)
   # Recombination(SRH Band2Band(Model=NonlocalPath))
                             # For off-current calculation, band-to-band
                             # tunneling models must be switched on.
```

```
    Piezo (
       Model (
          Mobility (
             saturationfactor=0.2
             efactor(Kanda sfactor=SBmob(Type=0))    # Type=1 for holes
             #efactor(Kanda sfactor=MCmob(Type=0))   # when using MCmob.
          )
          DOS(eMass hMass)
          DeformationPotential(Minimum ekp hkp)
       )
    )
 }
```

These command file sections are designed for electron transport (NMOS). For hole transport (PMOS), replace `eSubband` with `hSubband`, `eMultivalley` with `hMultivalley`, and specify `Type=1` for `MCmob` and `SBmob`.

The mobility of unstrained long-channel devices at low-drain bias voltage does not usually depend on the channel direction for the (100) surface. Even so, the warping of selected valence bands results in an orientation-dependent mass; the measurements do not show a dependence on the channel direction for very long channels. This might be a result of the many scattering processes and randomizations of the *k*-direction in long-channel devices.

For short-channel devices, the situation is completely different. First, the saturation velocity depends on the channel direction for higher fields. Second, the ballistic transport depends on the channel direction. Third, the warping of the valence band structure becomes important because there are fewer scattering effects.

Path C: Add High-k Mobility Degradation to Physics Section Shown in Path B

The orientation of the conducting interface is detected automatically. However, you must define the channel or the current flow direction manually using the correct mobility parameter file. The corresponding parameter files are located in the `MaterialDB` folder.

You must load the following parameter files for the corresponding channel direction:

- For the silicon <100> channel direction, load the `Siliconc100.par` file.
- For the silicon <110> channel direction, load the `Siliconc110.par` file.
- For the SiGe <100> channel direction, load the `SiliconGermaniumc100.par` file.
- For the SiGe <110> channel direction, load the `SiliconGermaniumc110.par` file.

```
Physics {Fermi}

Physics (Material="Silicon") {
#or Physics (Material="SiliconGermanium") {
   eQuantumPotential(AutoOrientation density)
   hQuantumPotential(AutoOrientation density)
   Mobility (
      Enormal (
         IALMob(AutoOrientation)
         # For PMOS, it is useful to use PhononCombination=2 in IALMob.
         RPS               # Used for remote phonon scattering (RPS).
         NegInterfaceCharge (SurfaceName="s1")
                           # Used for remote Coulomb scattering (RCS)
                           # and remote dipole scattering (RDS).
         PosInterfaceCharge (SurfaceName="s1")
                           # Used for RCS and RDS.
      )
      HighFieldSaturation)   # To include ballistic effects, you can
                             # use the BalMob(KVM(Full) Frensley Fermi)
                             # command in the Mobility statement.
   EffectiveIntrinsicDensity(OldSlotboom)
                    # For SiGe with high Ge mole fraction or for pure
                    # Ge, it is recommended to switch off the Fermi
                    # correction and to use
                    # EffectiveIntrinsicDensity(OldSlotboom NoFermi)
   Recombination(SRH)
   # Recombination(SRH Band2Band)  # For off-current calculation,
                                   # band-to-band tunneling models
                                   # must be switched on.

   eMultivalley(MLDA kpDOS -density)   # hMultivalley for holes.
             # For SiGe, use eMultivalley(MLDA kpDOS parfile -density)

   Piezo (
      Model (
         Mobility (
            saturationfactor=0.2
            eSubband(Fermi EffectiveMass Scattering(MLDA))
          # eSubband(Fermi EffectiveMass Scattering(MLDA) -RelChDir110)
          # for <100> channel.
          # hSubband(Fermi EffectiveMass Scattering(MLDA)) for holes
          # For holes, the replacement of the option Fermi by the
          # option Doping can result in speedup of the simulation.
         )
         DOS(eMass hMass)
         DeformationPotential(Minimum ekp hkp)
      )
   )
}
```

The surface name specifies the surface or interface that causes the mobility degradation, for example, the HfO_2–oxide interface in the high-k gate stack of a MOS transistor. The surface must be defined in the `Math` section of the command file. For more information about high-k mobility degradation models, see Unstrained Long-Channel Low-Field Mobility on page 47.

The mobility of unstrained long-channel devices at low-drain bias voltage does not usually depend on the channel direction for the (100) surface. Even so, the warping of selected valence bands results in an orientation-dependent mass; the measurements do not show a dependence on the channel direction for very long channels. This might be a result of the many scattering processes and randomizations of the *k*-direction in long-channel devices.

For short-channel devices, the situation is completely different. First, the saturation velocity depends on the channel direction for higher fields. Second, the ballistic transport depends on the channel direction. Third, the warping of the valence band structure becomes important because there are fewer scattering effects.

Path D: Thin-Layer MOS Simulations Including Thin-Film Effects

The `Mobility` command in the `Physics` section of Path C must be replaced by:

```
Mobility (
   ThinLayer (IALMob(AutoOrientation))
    # For PMOS, it is useful to use PhononCombination=2 in IALMob.
      Enormal (
         RPS                  # Used for remote phonon scattering (RPS).
         NegInterfaceCharge (SurfaceName="s1")
                              # Used for remote Coulomb scattering (RCS)
                              # and remote dipole scattering (RDS).
         PosInterfaceCharge (SurfaceName="s1")  # Used for RCS and RDS.
      )
      HighFieldSaturation)
```

All other options and statements remain the same.

The mobility of unstrained long-channel devices at low-drain bias voltage does not usually depend on the channel direction for the (100) surface. Even so, the warping of selected valence bands results in an orientation-dependent mass; the measurements do not show a dependence on the channel direction for very long channels. This might be a result of the many scattering processes and randomizations of the *k*-direction in long-channel devices.

For short-channel devices, the situation is completely different. First, the saturation velocity depends on the channel direction for higher fields. Second, the ballistic transport depends on the channel direction. Third, the warping of the valence band structure becomes important because there are fewer scattering effects.

Path E: FinFET Simulations

You can use the same `Physics` section as shown in Path C. The `AutoOrientation` option for the density gradient model must be switched on. As a starting point for calibration, the same parameter files `Siliconc100.par`, `Siliconc110.par`, `SiliconGermaniumc100.par`, and `SiliconGermaniumc110.par` can also be used. However, because the interface quality of the fin sidewalls can be very different from the interface quality of a planar transistor, parameter adjustments are often necessary.

In general, the electron inversion mobility of FinFETs is higher at the (110) sidewalls than in planar (110) transistors [1].

Table 4 and Table 5 show the approach for modeling planar and nonplanar FinFET devices regarding orientation-related parameters.

Table 4 Parameter files for planar FETs and FinFETs with silicon channel

Device	Number of conducting planes	Conducting plane orientation	Channel orientation	Parameter file to load
Planar MOS	1	(100)	<100>	`Siliconc100.par`
Planar MOS	1	(110)	<100>	`Siliconc100.par`
Planar MOS	1	(100)	<110>	`Siliconc110.par`
Planar MOS	1	(110)	<110>	`Siliconc110.par`
FinFET	3	(100)	<100>	`Siliconc100.par`
FinFET	3	(100) (110)	<110> <110>	`Siliconc110.par`

Table 5 Parameter files for planar FETs and FinFETs with SiGe channel

Device	Number of conducting planes	Conducting plane orientation	Channel orientation	Parameter file to load
Planar MOS	1	(100)	<100>	`SiliconGermaniumc100.par`
Planar MOS	1	(110)	<100>	`SiliconGermaniumc100.par`
Planar MOS	1	(100)	<110>	`SiliconGermaniumc110.par`
Planar MOS	1	(110)	<110>	`SiliconGermaniumc110.par`

Table 5 Parameter files for planar FETs and FinFETs with SiGe channel (Continued)

Device	Number of conducting planes	Conducting plane orientation	Channel orientation	Parameter file to load
FinFET	3	(100)	<100>	`SiliconGermaniumc100.par`
FinFET	3	(100) (110)	<110> <110>	`SiliconGermaniumc110.par`

For very short gate lengths (less than 15 nm), the isotropic density gradient model can overestimate the leakage current in the transport direction and quantization effects at doping and mole fraction gradients. In those cases, the `alpha` parameter in the density gradient model can help to suppress artificial source/drain leakage and quantization effects in the transport direction. For FinFET simulations with a gate length below 15 nm, using an `alpha` in the transport direction smaller than 0.1 is a good starting point. For more information about handling and using the anisotropic density gradient model, see the *Sentaurus™ Device User Guide*.

When using `EparallelToInterface` as a driving force in the high-field saturation model, the electric field components that are normal to the channel direction and in-plane with the fin cross section can lower the mobility. This is an artifact and makes the use of this driving force very questionable.

Note:

Using the `DOS(eMass hMass)` model in the `Piezo` section and the temperature equation at the same time might result in convergence problems. Using the `NumericalIntegration` option, for example, `DOS(hMass(NumericalIntegration))`, can improve the numeric behavior.

The mobility of unstrained long-channel devices at low-drain bias voltage does not usually depend on the channel direction for the (100) surface. Even so, the warping of selected valence bands results in an orientation-dependent mass; the measurements do not show a dependence on the channel direction for very long channels. This might be a result of the many scattering processes and randomizations of the *k*-direction in long-channel devices.

For short-channel devices, the situation is completely different. First, the saturation velocity depends on the channel direction for higher fields. Second, the ballistic transport depends on the channel direction. Third, the warping of the valence band structure becomes important because there are fewer scattering effects.

Path F: Smart-Power HVMOS and LDMOS Devices

In principle, you should use Path A for silicon-based smart-power devices exhibiting a traditional polysilicon/oxide gate stack when no mechanical stress occurs in the structure. When mechanical stress occurs in the structure, you must use Path B. However, in many cases, the linear piezoresistance model might be sufficient because the mechanical stress is relatively low. For complex devices or 3D simulation structures, you might need to simplify the model complexity to reduce simulation time or to achieve convergence. You can use the following simplified `Physics` section in those cases:

```
Physics {Fermi}

Physics (Material="Silicon") {
   Mobility (Enormal (IALMob(AutoOrientation)) HighFieldSaturation)
   # For PMOS, it is useful to use PhononCombination=2 in IALMob.
   EffectiveIntrinsicDensity(OldSlotboom)
   Recombination (Auger SRH(DopingDep TemDep))
   # Recombination(Auger SRH(DopingDep TemDep) Band2Band)
      # For off-current calculation, band-to-band tunneling models
      # must be switched on.
}
```

Quantum correction is neglected here, resulting in better simulation performance. The loss of accuracy can mostly be compensated by a corresponding workfunction shift and a reduction of the gate isolation permittivity. For devices with a gate isolation oxide thickness greater than 3 nm, the influence of quantum correction decreases. The `Physics` section above is designed for I_d–V_d and I_d–V_g simulations, but it does not include the models for breakdown, substrate current, and complex electrostatic discharge simulation. In addition, you must switch on the commands needed to account for self-heating.

Parameter files and the model section have been tested for LDMOS devices. No tests or investigations have been conducted for IGBTs, thyristors, or bipolar devices.

Path G: GaN HEMT Simulations

Typically, you should use the density gradient model for quantum correction. However, no model for quantum correction is included because of missing parameters.

The values for the trap concentration and the energy level given here are typical for nitride-passivated GaN HEMTs. However, the actual values depend on the device and might differ from the values presented here. In addition to the interface traps shown in the command file `Physics` section here, bulk traps are important and must be implemented in the simulation, for example, when investigating current collapse and gate leakage effects.

For III–V nitride wide-bandgap materials, you must load the parameter files `GaN.par`, `AlGaN.par`, `AlN.par`, `InGaN.par`, and `InN.par`, which contain the material parameters.

The following `Physics` section should be used together with the parameter files:

```
Physics {
   Mobility (
      Enormal(IALMob)         # Arora and Masetti models can also be
                              # used but cannot describe the mobility
                              # in the channel inversion layer.
                              # Model parameters are in the parameter
                              # files GaN.par, AlGaN.par, and AlN.par.
      HighFieldSaturation
   )
   EffectiveIntrinsicDensity(noBandGapNarrowing)
   Piezoelectric_Polarization(strain)
   Recombination(Radiative)
   Fermi
   Thermodynamic
   Thermionic
   HeteroInterface
   eBarrierTunneling "STUN"   # For tunneling at the source contact,
                              # you must define a nonlocal mesh in
                              # the Math section.
   eBarrierTunneling "DTUN"   # For tunneling at the drain contact,
                              # you must define a nonlocal mesh in
                              # the Math section.
}

Physics (MaterialInterface="AlGaN/Nitride") {
   Traps(Donor Level Conc=1E13 EnergyMid=0.4 FromMidBandGap)
               # The values for the trap concentration and the energy
               # are calibration parameters as well as the trap
               # location and the trap type. Several combinations
               # of different trap distributions are usually
               # necessary to reproduce measured results.
   Piezoelectric_Polarization(strain activation=0)
}
```

In addition to tunneling models at the source and drain contacts, tunneling models below the gate contact, finally covering the entire gate stack, down to the GaN inversion channel, are necessary to simulate gate leakage. These tunneling models must consider both direct tunneling and trap-assisted tunneling.

Note:

The choice of the transition between the 2D electron gas and the 3D bulk situation is critical. This transition is controlled by the parameters `lcrit`, `lcrit_c`, and `S` in the parameter file.

Path H: SiC Diode Simulations

The parameter files `4H-SiC.par` and `6H-SiC.par` contain the material parameters for SiC-based wide-bandgap materials.

The following `Physics` section should be used together with the parameter files:

```
Physics {
   eMobility (
      DopingDependence(Arora)   # For 6H-SiC, Arora parameters are not
                                # available in the parameter file.
                                # Therefore, it is recommended to
                                # switch to the Masetti model.
      HighFieldSaturation
   )
   hMobility (
      DopingDependence(Arora)   # For 6H-SiC, Arora parameters are not
                                # available in the parameter file.
                                # Therefore, it is recommended to
                                # switch to the Masetti model.
      HighFieldSaturation
   )
   EffectiveIntrinsicDensity(Slotboom NoFermi)
   IncompleteIonization
   Recombination(SRH(DopingDep TempDep Tunneling) Band2Band Avalanche
      (Okuto))
                                # For forward characteristics, you can
                                # neglect the commands Band2Band and
                                # Tunneling. This can improve
                                # convergence behavior.
   Fermi
   #Thermodynamic              # Add for thermodynamic simulations.
   #AnalyticTEP                # Add for thermodynamic simulations.
   Temperature=300.0
   eBarrierTunneling "NLM" (Twoband Transmission)
               # Nonlocal mesh definition in Math section necessary.
   Traps (
      Donor Level EnergyMid=1.0 FromConductionband Conc=1e14
      eXSection=1.0e-12 hXSection=1.0e-12 PooleFrenkel
      TrapVolume=1.0e-9
      HuangRhys=0.1 PhononEnergy=0.05 eBarrierTunneling (Twoband)
   )              # Bulk traps are often necessary for the physically
                  # correct calculation of the thermal behavior of the
                  # reverse leakage current.
}

Physics (Electrode="top") {
   #MSPeltierHeat              # Add for thermodynamic simulations.
   Schottky BarrierLowering
}
```

Depending on how the mobility model was calibrated, that is, whether or not the total chemical concentration or the active concentration was used, you can add `IncompleteIonization` to the `Mobility` statement. For holes, the suggested parameters for the Arora model in the `4H-SiC.par` file have been calibrated to the total acceptor concentration. In addition, scattering on the neutral impurities is often comparable to scattering on the ionized impurities for 4H-SiC p-type layers.

Note:

Using the `IncompleteIonization` model might cause convergence problems.

To distinguish between cubic and hexagonal 4H-SiC lattice sites, you can activate the following option in the `IncompleteIonization` statement of the `Physics` section of the command file:

```
IncompleteIonization(Split (Doping="NitrogenActive" Weights=(0.5 0.5)))
```

The corresponding parameters must be defined in the parameter file. Precalibrated values are in the `4H-SiC.par` and `6H-SiC.par` files for phosphorus and nitrogen. For phosphorus, the parameters corresponding to hexagonal sites are active, and modification of the `datexcodes.txt` file is necessary to treat both sites.

The values for the trap concentration, the energy levels, and other parameters in the `Traps` command are typical. However, the actual values depend on the device and might differ from the values presented here. Especially for the leakage current in reverse mode, tunneling by defect traps can play an important role and should be investigated.

The use of the `NoFermi` option depends on the extraction of the parameters for the bandgap narrowing model. This option is applied if you use parameters for bandgap narrowing that have been extracted assuming Fermi statistics.

When including heat generation and conduction, a lumped model (thermal network) using the thermal resistance of layers and material regions that are not included in the simulation can improve the simulation results.

You can use an alternative model, the `JainRoulston` model, which is physically more suited for bandgap narrowing.

For 2D and 3D structures, anisotropic material properties for mobility, permittivity, thermal conductivity, and impact ionization must be taken into account in the simulation. This can be performed by the following statement in the `Physics` section:

```
Aniso (Mobility Poisson Temperature Avalanche)
```

The corresponding parameters must be defined in the parameter file. Precalibrated values are in the `4H-SiC.par` (same as `SiliconCarbide.par`) and `6H-SiC.par` files for mobility, permittivity, thermal conductivity, impact ionization, incomplete ionization, high-field saturation, and other models.

Path I: InGaAs FinFET Simulations

For InGaAs FinFET devices with a gate width less than 10 nm, the following quantization models of Sentaurus Device are available to model quantized distributions of carriers:

- Use the modified local-density approximation (MLDA) model with the `ThinLayer` option specified in the command file to activate size quantization with the multivalley model. For

a 2D structure, the layer thickness can be extracted automatically, and the depletion at corner regions corresponds better to the results of the Schrödinger equation using the `MaxFit` option. Setting `MaxFitWeight=0.35` is advisable to start with, although an adjustment might be required depending on fin geometries.

- Use the density gradient model with the multivalley model. The quantization parameter (γ-parameter) of the density gradient model depends on the layer thickness and the fin geometry. Therefore, it is advisable to calibrate the solutions from the Schrödinger equation in test structures with a similar channel geometry.

In addition, the inversion and accumulation layer mobility (`IALMob`) model is available to model inversion-type n-channel InGaAs FinFETs. The parameters have been calibrated to available experimental data under the condition of `PhononCombination=1`.

The parameter files `GaAs.par`, `InGaAs.par`, and `InAs.par` contain the material parameters for III–V arsenide materials.

For the MLDA model, you must use the following `Physics` section with the parameter files:

```
Physics {Fermi}

Physics (Material="InGaAs") {
   Mobility (
      eHighFieldSaturation # In case of convergence issues, you can use
                           # eHighFieldSaturation(EparallelToInterface).
      Enormal(IALMob(PhononCombination=1))
   )
   EffectiveIntrinsicDensity(BandGapNarrowing(JainRoulston) NoFermi)
   Recombination(SRH Auger)
   eMultiValley(MLDA Nonparabolicity ThinLayer)
   LayerThickness(MaxFitWeight=0.35)
}
```

For the density gradient model, you must use the following `Physics` section with the parameter files:

```
Physics {Fermi}

Physics (Material="InGaAs") {
   Mobility (
      eHighFieldSaturation # In case of convergence issues, you can use
                           # eHighFieldSaturation(EparallelToInterface).
      Enormal(IALMob(PhononCombination=1))
   )
   EffectiveIntrinsicDensity(BandGapNarrowing(JainRoulston) NoFermi)
   Recombination(SRH Auger)
   eMultiValley(Nonparabolicity)
   eQuantumPotential(Density)      # γ-parameter requires calibration.
}
```

Path J: SiC HVMOS Simulations

Typically, you should use the density gradient model for quantum correction. However, no model for quantum correction is included because of missing model parameters.

For SiC wide-bandgap materials, you must load the parameter file `4H-SiC.par` or `SiliconCarbide.par`, which contains the material parameters.

Note:

The choice of the transition between the 2D electron gas and the 3D bulk situation is critical. This transition is controlled by the parameters `lcrit`, `lcrit_c`, and `S` in the parameter file.

The following `Physics` sections should be used together with the parameter file:

```
Physics {
   Fermi
   DefaultParametersFromFile
}

Physics (Material = "SiliconCarbide") {
   IncompleteIonization
   Mobility (
      IncompleteIonization
      Enormal(IALMob(PhononCombination = 2))
      HighFieldSaturation
   )
   IncompleteIonization(Split (Doping = "NitrogenConcentration"
                               Weights = (0.5 0.5)))
   EffectiveIntrinsicDensity(BandGapNarrowing(Slotboom) NoFermi)
   Recombination (SRH(DopingDependence TempDependence) Auger)
}
```

References

[1] C. D. Young *et al*., “(110) and (100) Sidewall-oriented FinFETs: A performance and reliability investigation,” *Solid-State Electronics*, vol. 78, pp. 2–10, December 2012.

3

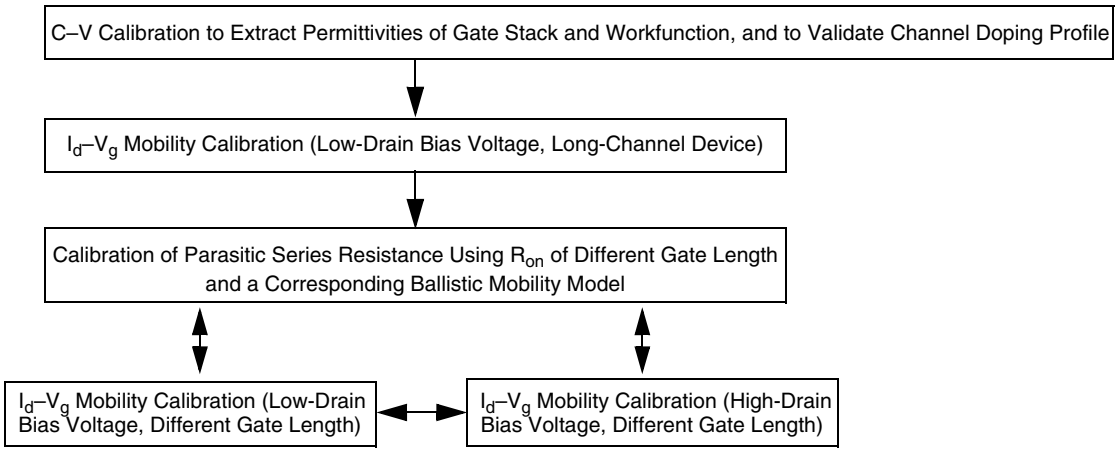
Calibration Methodology for Device Simulations

This chapter describes the basic calibration methodology used to improve the accuracy of CMOS device simulations.

Calibration of CMOS Devices

Figure 1 shows a general flow of the calibration methodology used for CMOS devices. Different measurements and layout configurations allow you to separate device simulation model parameters. However, in many cases, iterations are necessary.

Figure 1 Calibration methodology used for CMOS devices (I_d–V_g calibration can be complemented by involving mobility measurements)



The next sections describe this calibration methodology in detail. It is assumed that all process calibration issues are solved beforehand and the doping profile is correct.

C–V Calibration

Capacitance–voltage (C–V) calibration is the first step in the device calibration and is strongly connected to the doping profile and to process simulation or calibration. Because it

is assumed that the doping profile is correct, the bottom of the C–V curve (marked in Figure 2) fits the experiment already.

C–V calibration is performed on large-area MOS transistors or MOS capacitor test structures at low frequency. Therefore, in some cases, you can have slightly different workfunctions compared to smaller transistor devices.

Figure 2 C–V calibration methodology (NMOS)

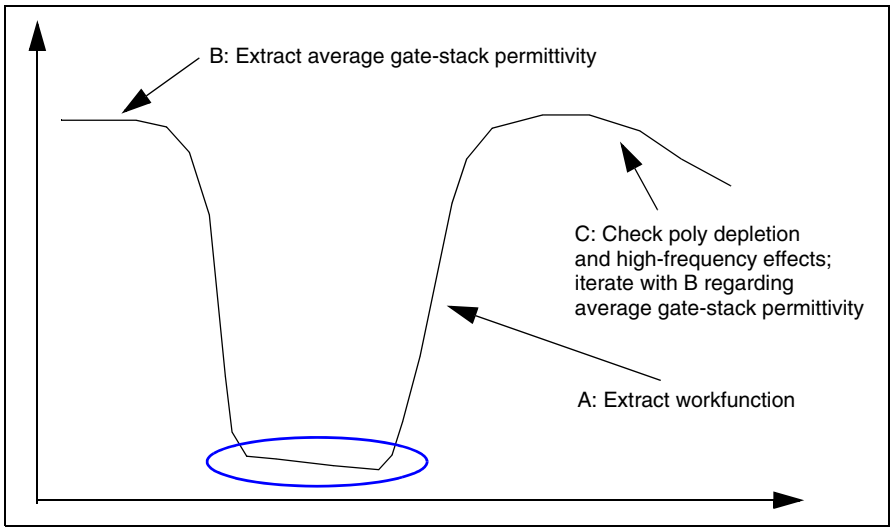


Table 6 Main steps and extracted parameters of C–V calibration

Step	Target	Parameter to extract	Validity range of parameter	Comments
A	Region of weak or moderate inversion	Workfunction	Metal gate workfunction: NMOS (4.0–4.7 eV) PMOS (4.5–5.2 eV) Polygate barrier: –0.2 V – 0.2 V	Workfunction or barrier voltages extracted from C–V and used for long-channel I–V simulation can require corrections, particularly when special C–V test structures are used.
B	Accumulation part of C–V	Average permittivity of gate isolation	Permittivity of high-k material: 14–25 Permittivity of interfacial oxide: 3.9–6	For high-k gate stacks with several material layers, first the permittivity of the interfacial oxide layer is adjusted. Second, the permittivity of the high-k material is calibrated.
C	Inversion part of C–V	Poly doping in case of polygate	Depends on process conditions	–

I_d–V_g Mobility Calibration

The calibration of long-channel devices is necessary to extract low-field mobility parameters. Typically, for this, you use the I_d–V_g characteristics of devices with a gate length greater than 500 nm, at a low-drain bias voltage smaller than 100 mV. An iterative process extracts the parameters of the mobility model in different voltage regions (see Figure 3, Figure 4, and Table 7).

Figure 3 Long-channel I_d–V_g mobility calibration methodology in linear scale

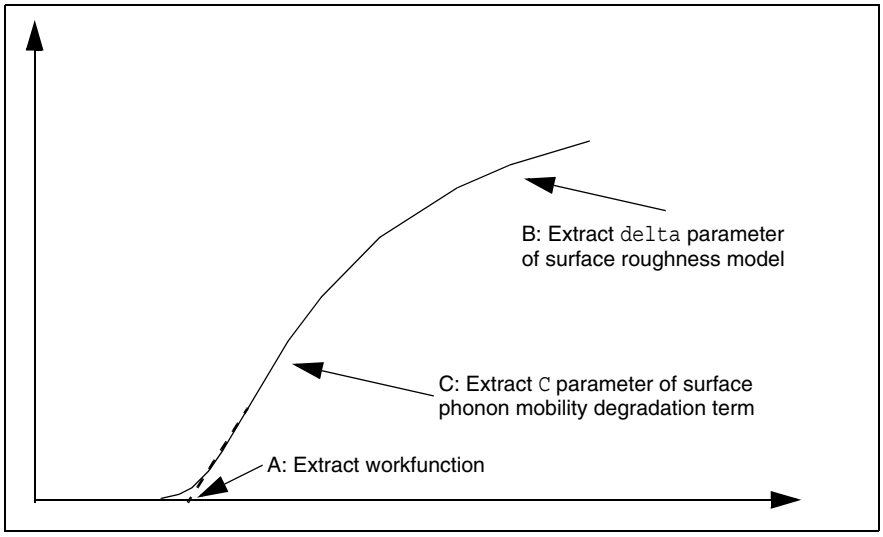


Figure 4 Long-channel I_d–V_g mobility calibration methodology in logarithmic scale

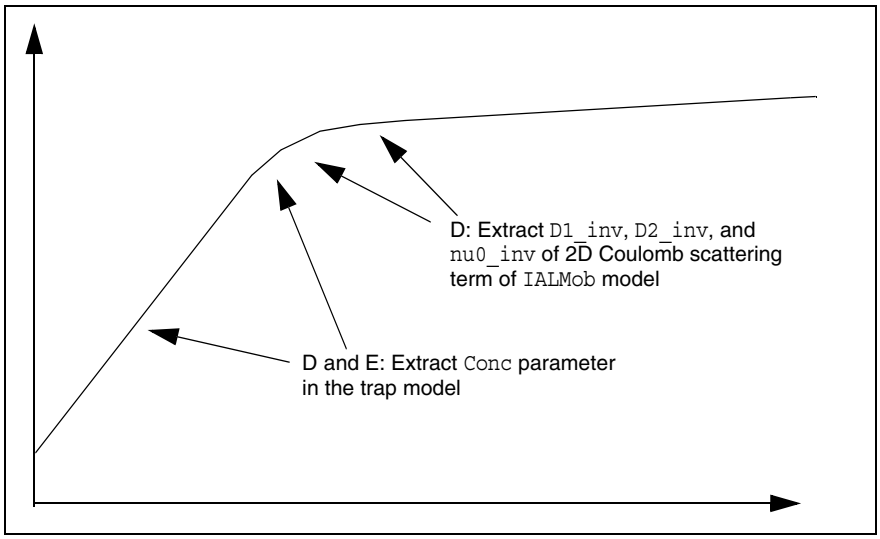


Table 7 Main steps and extracted parameters of long-channel I_d–V_g mobility calibration

Step	Target	Parameter to extract	Validity range of parameter	Comments
A	Threshold voltage	Workfunction	Depends on gate material. Stress in the metal gate, grain sizes, and orientations influence the workfunction and can result in a dependency on gate length and geometric parameters.	Take initial value for workfunction from C–V calibration. Note there can be an influence of remote Coulomb or dipole scattering (RCS or RDS), or 2D impurity Coulomb scattering on the threshold voltage, which couples workfunction and mobility extraction.
B	Strong inversion – high gate bias voltage	`delta` parameter of surface roughness model	Depends on process conditions and other parameters. No validity range is known.	Surface roughness can change when changing process conditions and surface orientations. Fine-tuning is always necessary to fit I_d–V_g characteristics.
C	Linear region of I_d–V_g curve	`C` parameter of surface phonon mobility degradation term	Depends on process conditions and other parameters. No validity range is known.	Extraction requires mostly an iteration between steps B and C and, sometimes, even between steps B, C, and D. Interface properties and material composition at the interface can change when changing process conditions. Fine-tuning is always necessary to fit I_d–V_g characteristics.

Table 7 Main steps and extracted parameters of long-channel I_d–V_g mobility calibration

Step	Target	Parameter to extract	Validity range of parameter	Comments
D	Transition between logarithmic and linear slope – threshold voltage region	1. Parameters of 2D Coulomb mobility degradation term 2. Parameters of `InterfaceCharge` model	1. No validity range is known 2. Parameters of `InterfaceCharge` mode	Calibration focuses on the Coulomb roll-off visible in the μ(E_{eff}) curve, but it also can be done to the bending of the I_d–V_g curve between subthreshold and linear region. These mobility degradation terms also affect the threshold voltage and the drain-induced barrier lowering (DIBL) of the device.
E	Subthreshold region	Parameters of `InterfaceCharge` model	No validity range is known	Calibration focuses on the Coulomb roll-off visible in the μ(E_{eff}) curve, but it also can be done to the subthreshold region. This mobility degradation term also affects the threshold voltage and the DIBL of the device.

Calibration of Parasitic Series Resistance

With the inclusion of a ballistic mobility model that gives a finite resistance for an infinitely small channel length, the calibration of the channel mobility is coupled to the calibration of the source/drain channel series resistance. Therefore, the extraction of this series resistance is most important for the calibration of the parameters of the ballistic mobility model. Iterations might be needed to update the extracted series resistance when changing parameters of the ballistic mobility model in the calibration of the channel mobility.

Figure 5 Extraction methodology of source/drain series resistance



Equation 1 shows the parameters `R0` and `Rs` forming the on-resistance:

$$R_{\text{on}} = R0 + Lg \cdot Rs \tag{1}$$

Table 8 Main step and extracted parameter of source/drain series resistance

Step	Target	Parameter to extract	Validity range of parameter	Comments
F	Measured dependency of on-resistance at low-drain bias voltage on gate length	On-resistance at zero gate length *R*0 (in Equation 1)	*R*0 is determined by contact and source/drain series resistance. The value depends on the technology node and device type, and can vary greatly. No validity range is known.	*R*0 extraction is especially important for step G (extraction of “ballistic” resistance).

I_d–V_g Mobility Calibration at Low-Drain Bias Voltage

Reference [1] discusses a possible overestimation of the current response close to equilibrium when having strong electrical built-in fields, for example, caused by p-n junctions. This effect might be connected to deviations from the Einstein relation in certain situations, but it is not considered in the calibration methodology because of convergence issues. This means, to some extent, that the calibration of the parameters of the ballistic mobility model includes this physically different effect.

Figure 6 Short-channel I_d–V_g mobility calibration methodology in linear scale, with low-drain bias voltage

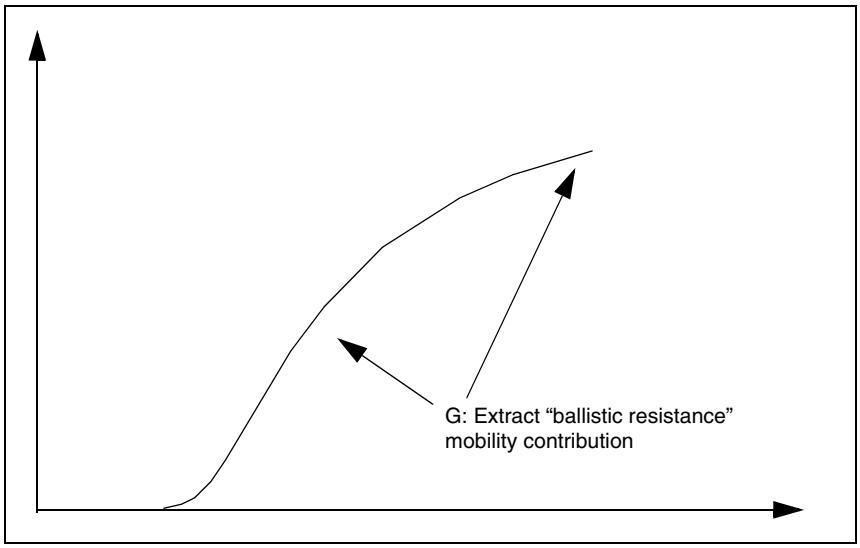


Table 9 Main step and extracted parameter of I_d–V_g mobility calibration at low-drain bias voltage

Step	Target	Parameter to extract	Validity range of parameter	Comments
G	Moderate and strong inversion	Adjust “ballistic resistance” mobility contribution using parameters of the ballistic mobility models in Sentaurus Device	No validity range	This involves iteration with step F.

I_d–V_g Mobility Calibration at High-Drain Bias Voltage

This step extracts the saturation velocity. The extraction of `vsat0` from I_{dsat} roll-off data or I_d–V_g curves at large drain bias voltage, using experiments or the Boltzmann transport equation solver, is straightforward. However, the calibration of both I_{dsat} and I_{dlin} at the same time using `vsat0`, source/drain resistance, and ballistic mobility parameters is more challenging.

Nevertheless, it is mostly possible to achieve a good agreement during the device calibration because of the inclusion of the ballistic mobility model [2].

Figure 7 Short-channel I_d–V_g mobility calibration methodology in linear scale, with high-drain bias voltage

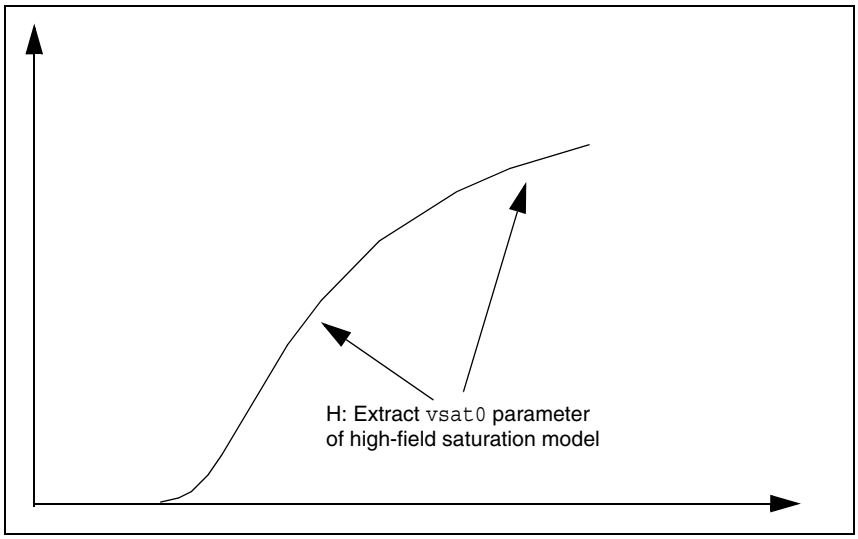


Table 10 Main step and extracted parameter of I_d–V_g mobility calibration at high-drain bias voltage

Step	Target	Parameter to extract	Validity range of parameter	Comments
H	Whole I_d–V_g curve and I_{dsat}	Adjust `vsat0`	For silicon: < 5e7 cm/s	This might involve iteration with steps F and G.

References

[1] A. T. Pham, C. Jungemann, and B. Meinerzhagen, “Modeling and validation of piezoresistive coefficients in Si hole inversion layers,” *Solid-State Electronics*, vol. 53, no. 12, pp. 1325–1333, 2009.

[2] A. Erlebach, K. H. Lee, and F. M. Bufler, “Empirical Ballistic Mobility Model for Drift-Diffusion Simulation,” in *Proceedings of the 46th European Solid-State Device Research Conference (ESSDERC)*, Lausanne, Switzerland, pp. 420–423, September 2016.

4

Descriptions of Models

This chapter presents a review of selected models of Sentaurus Device.

Overview of Models

Given the many models implemented in Sentaurus Device for the simulation of device characteristics, it is often difficult to quickly design appropriate command files for device simulation. Therefore, this chapter presents a review of selected models, which are evaluated with respect to their capabilities in simulating certain device aspects.

In the quantum drift-diffusion (QDD) approximation, electron and hole mobilities determine charge carrier transport. Advanced Calibration Device focuses on low-field mobility and high-field mobility, with and without mechanical stress, for planar and nonplanar silicon and SiGe FETs. In addition, there is a discussion of models and parameters for InGaAs, AlGaN, and SiC devices.

CMOS Devices

This section discusses models relevant to CMOS devices.

Unstrained Band Structure and Electrostatics

In Sentaurus Device, the band gap, the electron affinity, the bandgap narrowing, the density-of-states, and the dimensions of the device define the band structure. The temperature dependency of these quantities and the doping dependency of the bandgap narrowing are represented by empirical functions.

Strained Band Structure and Electrostatics

The band structure changes with the mechanical stress in the device. The dependency of the conduction and valence bands and, therefore, of the band gap and electron affinity, on

the mechanical stress or strain is described by the deformation potential model where the band structure information comes from the k·p calculation. You can also extract the stress dependency of the density-of-states from the k·p calculation.

Quantization and Band Structure

The bulk band structure changes in thin potential wells, such as inversion layers and thin films. These changes due to the quantization interact, compensate, or strengthen the stress influence on the band structure. In Sentaurus Device, the quantization is typically calculated by the density gradient model or the modified local-density approximation (MLDA) model.

Unstrained Long-Channel Low-Field Mobility

In the limit of low-driving fields, several mechanisms can degrade mobility. Typically, MISFET simulation takes into account mobility degradation by ionized dopants, surface roughness scattering, surface phonon scattering, and bulk phonon scattering. For high-k gate stacks, you must consider additional remote scattering mechanisms such as remote Coulomb scattering (RCS), remote phonon scattering (RPS), and remote dipole scattering (RDS). A crucial point is the transition between bulk mobility and inversion or accumulation layer mobility. Because of the transition from 3D to 2D charge carrier gas, the scattering behavior of electrons and holes changes and screening becomes different. Furthermore, the mobility model must include dependency on the channel material (for example, the germanium mole fraction), the channel direction, and the surface orientation. For thin silicon films in the range of a few nanometers, additional contributions come into effect that cause a dependency of the mobility on the film thickness.

In summary, a drift-diffusion or QDD low-field mobility model for the simulation of SiON/SiO_2 or high-k gate stack FETs without mechanical stress should have the following features:

- Mobility degradation by ionized impurities – impurity Coulomb scattering (ICS)
- Mobility degradation by the roughness of the surface or interface – surface roughness scattering (SRS)
- Mobility degradation by interaction with bulk phonons – bulk phonon scattering (BPS)
- Mobility degradation by interaction with surface phonons – surface phonon scattering (SPS)
- Mobility degradation by charges in the gate isolator (high-k gate stack) (RCS)
- Mobility degradation by different permittivity in the gate isolator (high-k gate stack) (RPS)
- Mobility degradation by dipole configurations in the gate isolator (high-k gate stack) (RDS)

- Mole fraction dependency of the low-field mobility model parameters (for example, alloy scattering)
- Transition between bulk scattering and scattering in 2D charge carrier gas for ICS
- Parameters for different channel and surface or interface orientations; channel: <100> and <110>; surface: (100) and (110)
- Mobility degradation for very thin silicon films
- Mobility in thin films, including influence of quantization
- Influence of mobility anisotropy as in SiC and GaN

Sentaurus Device offers the following low-field mobility degradation models for MISFET simulation:

- Bulk phonon-limited mobility model (BPS):

```
Physics { ConstantMobility }
```

- Mobility degradation by ionized impurities (in some cases, it contains bulk phonon-limited mobility, in the low-doping limit) (ICS):
  - Masetti model provides mole fraction–dependent model parameters:

```
Physics { Mobility ( DopingDependence (Masetti) ) }
```

  - Arora model provides mole fraction–dependent model parameters:

```
Physics { Mobility ( DopingDependence (Arora) ) }
```

  - Philips unified mobility model includes screening and carrier–carrier scattering, and provides mole fraction–dependent model parameters:

```
Physics { Mobility (PhuMob) }
```

  - University of Bologna bulk mobility model is specially calibrated for an extended temperature range and provides mole fraction–dependent model parameters:

```
Physics { Mobility ( DopingDependence (UniBo) ) }
```

  - Carrier–carrier scattering models (Conwell–Weisskopf and Brooks–Herring):

```
Physics { Mobility (CarrierCarrierScattering) }
```

- Mobility degradation at interfaces:
  - Enhanced Lombardi model includes automatic detection of surface orientation and provides mole fraction–dependent model parameters:

```
Physics { Mobility ( Enormal (Lombardi) ) }
```

- Inversion and accumulation layer mobility (`IALMob`) model includes automatic detection of surface orientation, provides mole fraction–dependent model parameters, and contains the Philips unified mobility model, the enhanced Lombardi model, and additional terms for Coulomb scattering in 2D electron gas:

  ```
  Physics { Mobility ( Enormal (IALMob) ) }
  ```

- University of Bologna surface mobility model is specially calibrated for an extended temperature range, includes screening of bulk ICS, no special model for ICS in 2D charge carrier gas, and simplified Lombardi-style model for acoustic phonon and surface roughness scattering:

  ```
  Physics { Mobility ( Enormal (UniBo) ) }
  ```

- 2D Coulomb scattering model for ionized impurities, which must not be used with the `PhuMob` model or any other `DopingDependence` model because mobility degradation is counted twice, is designed for Coulomb scattering (ICS) in 2D charge carrier gas:

  ```
  Physics { Mobility ( Enormal (Lombardi Coulomb2D) ) }
  ```

- 2D Coulomb scattering models for positive and negative charges at remote interfaces allow you to distinguish between RCS and RDS:

  ```
  Physics { Mobility ( Enormal (Lombardi NegInterfaceCharge
              PosInterfaceCharge) ) }
  ```

- Thin-layer mobility model describes mobility degradation due to finite silicon film thickness, includes dependency of phonon scattering on quantization as well as additional empirical mobility degradation terms, and includes automatic detection of surface orientation. The model can be combined with `Enormal`-like models:

  ```
  Physics { Mobility (ThinLayer) }
  ```

Because of the use of Mathiessen's rule, several combinations of these mobility models are possible. For MISFETs, you usually need to combine bulk phonon and Coulomb scattering with scattering at interfaces and surfaces. Historically, the following combinations have been used:

1. `DopingDependence(Masetti)` + `Lombardi`
2. `PhuMob` + `Lombardi`
3. `IALMob`
4. `IALMob` + `RPS` + `NegInterfaceCharge` + `PosInterfaceCharge`
5. `RPS` + `NegInterfaceCharge` + `PosInterfaceCharge` + `ThinLayer(IALMob)`

Combination 2 has been the standard for a long time. However, several problems require a change to another low-field mobility model. First, the transition between 3D bulk Coulomb scattering and Coulomb scattering in the inversion layer is not described correctly because

the `PhuMob` model, which is a bulk model, is used for the mobility degradation by scattering at ionized impurities from accumulation to strong inversion. This causes problems in describing the onset of moderate or strong inversion around the threshold voltage in the I_d–V_g curve. Second, in thin silicon layers, it is mandatory to include the dependency on the film thickness.

Note:

For these reasons, you must change the low-field mobility framework and switch to combination 3, 4, or 5. For MISFET simulation, the recommended models are combination 3 (devices without high-k gate stacks), combination 4 (devices with high-k gate stacks), and combination 5 (thin films).

Strained Low-Field Mobility

This section discusses strained low-field mobility.

Influence of Mechanical Stress

Modeling the influence of stress on the channel mobility must deal with several typical cases. These cases must be simulated accurately to minimize additional device calibration. The typical cases are as follows:

- Planar silicon channel PMOS with compressive stress up to 3 GPa in the channel direction and up to 2 GPa tensile stress normal to the silicon surface (SiGe pockets and dual stress liner (DSL))
- Planar silicon channel NMOS with tensile stress up to 2 GPa in the channel direction and up to 2 GPa normal to the silicon surface (SiC pockets, DSL, and stress memorization technique)
- Planar PMOS channel with biaxial strained SiGe channel and stress between 1 MPa and 4 GPa in the channel direction and up to 2 GPa tensile stress in the normal direction (SiGe pockets and DSL)
- Nonplanar silicon channel NMOS (FinFET) with mixed compressive or tensile stress components of up to 2 GPa depending on the configuration (for example, with SiC pockets wrapped around the fin and metal gate as a liner replacement)
- Nonplanar silicon channel PMOS (FinFET) with mixed compressive or tensile stress components of up to 2 GPa depending on the configuration (for example, with SiGe pockets wrapped around the fin and metal gate as a liner replacement)

- Nonplanar SiGe or Ge channel NMOS (FinFET) with mixed compressive or tensile stress components of up to 4 GPa depending on the configuration
- Nonplanar SiGe or Ge channel PMOS (FinFET) with mixed compressive or tensile stress components of up to 4 GPa depending on the configuration

You must simulate stress configurations for several wafer orientations (such as (001) and (1$\underline{1}$0)), channel directions (such as <100> and <110>), and channel materials (such as silicon and SiGe). Typically, all three stress components (s_{xx}, s_{yy}, and s_{zz}) are important. A model validity check, which is limited to uniaxial or biaxial stress configurations, is not sufficient. In addition, the response of mobility degradation components (such as phonon scattering, Coulomb scattering, and remote scattering) to mechanical stress differs and requires the separate treatment of each component.

Typically, in Sentaurus Device, the total mobility is scaled with a stress enhancement factor calculated with the models discussed here. The following models in Sentaurus Device describe the influence of stress on mobility:

- Piezoresistance mobility models:
  - First-Order (Linear) Piezoresistance Mobility Model
  - Second-Order Piezoresistance Mobility Model
  - Nonlinear Piezoresistance Models for Electrons and Holes (MCmob and SBmob)
- Occupation-based and band structure–based models:
  - Intel Stress-Induced Hole Mobility Model (hSixBand)
  - eSubband Model for Electrons
  - hSubband Model for Holes

You can choose several additional options. For the piezoresistance mobility models, the `Kanda` option includes the dependency of the stress-induced mobility change on doping concentration. Differences in the influence of mechanical stress on minority and majority carriers can be considered by an additional fitting parameter. You can also calibrate the influence of the normal electric field in a metal–insulator–semiconductor (MIS) structure and of the germanium mole fraction on the first-order piezoresistive coefficients. Because the first-order and second-order piezoresistance mobility models are bulk models, the default piezoresistive coefficients are not accurate for inversion layers and must be adjusted. The first-order and second-order piezoresistance mobility models can be used as factor models or tensor models, thereby introducing anisotropic properties.

You can fine-tune the influence of the saturation velocity on stress-dependent mobility by setting the `SaturationFactor` parameter that controls how strongly the saturation velocity depends on stress as follows:

- Setting `SaturationFactor=0` means the mobility change is applied to the low-field mobility only.
- Setting `SaturationFactor=1` means the mobility change applies to the total mobility. This is the default in Sentaurus Device.

Because it is known from Monte Carlo simulation and from experiments that short-channel transistors show a much lower mobility enhancement due to mechanical stress, it is better to apply the mobility change to the low-field mobility only. Further adjustment of this parameter should be performed with good references only.

The occupation-based and band structure–based models characterize the dependency of occupation and band structure on mechanical stress for the calculation of the mobility. These models have their own dependency on the doping concentration, and a `Kanda`-like option is not necessary. In addition, the electric field dependency should be a result of the model itself.

First-Order (Linear) Piezoresistance Mobility Model

This model uses the piezoresistance tensor with constant coefficients to calculate the stress-induced mobility change.

Options

The model can be used as a factor (isotropic) model or a tensor (anisotropic) model. The piezoresistive coefficients can depend on the normal electric field in an inversion layer. Minority and majority charge carrier transport can be distinguished. You can include dependency on the doping concentration. Calibration of dependency of the influence of mechanical stress on the saturation velocity is possible.

Advantages

The model is well adjusted to measurements for the bulk case. It is fully transformable, so it can consider all stress configurations and current directions in a bulk situation. It is easy to use and to calibrate.

Disadvantages

It is accurate for low stress only (< 200 MPa). Tensor symmetry is broken for nonbulk conditions as in an inversion layer, which is not reflected by the model. The default piezoresistive coefficients are not accurate for inversion layers.

Second-Order Piezoresistance Mobility Model

This model uses the 6th-grade piezoresistance tensor with constant coefficients to calculate the stress-induced mobility change.

Options

The model can be used as a factor (isotropic) model or a tensor (anisotropic) model. Minority and majority charge carrier transport can be distinguished. You can include dependency on the doping concentration. Calibration of dependency of the influence of mechanical stress on the saturation velocity is possible.

Advantages

Parameters are available for the bulk case. The model is fully transformable, so it can consider all stress configurations and current directions in a bulk situation.

Disadvantages

It is accurate for low and moderate stress values only (< 500 MPa). Tensor symmetry is broken for nonbulk conditions as in an inversion layer, which is not reflected by the model. The default piezoresistive coefficients are not accurate for inversion layers.

Nonlinear Piezoresistance Models for Electrons and Holes (MCmob and SBmob)

These models use nonlinear piezoresistive coefficients and higher-order cross-term corrections that are extracted from Monte Carlo simulation, Sentaurus Band Structure simulation, or other sources (experiments and simulations available from the literature).

Options

Minority and majority charge carrier transport can be distinguished. You can include dependency on the doping concentration. Calibration of dependency of the influence of mechanical stress on the saturation velocity is possible. Germanium mole fraction dependency of the model coefficients can be switched on. Parameters for different orientations can be used and an auto-orientation option is available. You can load complete parameter sets using the Sentaurus Device parameter file.

Advantages

Parameters are available for strong inversion and for different channel materials (silicon and SiGe). The model can be extended to other conditions when there are reference tools such as Monte Carlo simulations or experiments. The model is available for different channel and wafer orientations. It is fast and can be calibrated easily.

Disadvantages

Interpolation especially for untypical stress conditions must always be validated. The physics is in the reference tool, for example, the Monte Carlo simulator. Improving the model without improving the reference tool or without obtaining better experiments is not possible, except with respect to the interpolation. The settings for the local coordinate system must be specified in the Sentaurus Device parameter file.

Intel Stress-Induced Hole Mobility Model (hSixBand)

This model uses the change of band structure and occupation to calculate stress-dependent mobility.

Options

You can switch on dependency on the doping or carrier concentration.

Advantages

The model is derived from physical assumptions.

Disadvantages

Calibration is not straightforward. Only silicon as a channel material is available. The influence of surface orientation cannot be described. The model has been developed and calibrated for uniaxial stress in the <110> direction. The combination of the component normal to the MIS plane with the in-plane stress components is not validated.

eSubband Model for Electrons

This model uses the change of band structure and occupation to calculate the mobility change.

Options

You can switch on dependency on the doping or carrier concentration. Stress-related change in the scattering can be included. Change of effective mass with stress can be added to the model. For the inversion layer, the `MLDA` option can be used.

Advantages

The `MLDA` option allows you to move from bulk to inversion layer conditions, and to take surface or channel orientation dependency into account. The model is derived from physical assumptions. The model is validated and calibrated for silicon and SiGe for different channel and surface orientations. Mole fraction dependency for SiGe is available.

Disadvantages

Calibration is not straightforward.

hSubband Model for Holes

This model uses the change of band structure and occupation to calculate stress-dependent mobility.

Options

You can switch on dependency on the doping or carrier concentration. Stress-related change in the scattering can be included. Change of effective mass with stress can be added to the model. For the inversion layer, the `MLDA` option can be used.

Advantages

The `MLDA` option allows you to move from bulk to inversion layer conditions, and to take surface or channel orientation dependency into account. The model is derived from physical assumptions. The model is validated and calibrated for silicon and SiGe for different channel and surface orientations. Mole fraction dependency for SiGe is available.

Disadvantages

Calibration is not straightforward.

Unstrained High-Field Mobility

The low-field mobility models are valid for small driving forces only. To take into account high-field effects, the additional high-field saturation model is introduced in the drift-diffusion model framework.

Note:

Calibrated mole fraction–dependent parameters are available and extensive experience exists. Numeric problems are known or are already solved.

The high-field saturation model should limit the velocity-field relation to the saturation velocity of the material when there is sufficient scattering, and it must provide a way to calibrate the influence of ballistic and quasiballistic transport on the charge carrier transport. In addition, it should allow you to calibrate the germanium mole fraction dependency of the model parameters. Furthermore, because of deficiencies in the drift-diffusion model to describe the current response close to equilibrium, a model is needed that allows you to compensate or, at least, to calibrate for this deficiency.

In summary, a drift-diffusion or QDD high-field mobility model for the simulation of SiON/SiO_2 or high-k gate stack FETs without mechanical stress should have the following features:

- Limiting the drift velocity to the saturation velocity for cases with sufficient scattering
- Possibility to calibrate the model with respect to ballistic and quasiballistic transport
- Germanium mole fraction–dependent model parameters
- Possibility to calibrate the model with respect to current response close to equilibrium
- Orientation-dependent model parameters

Sentaurus Device offers the following high-field mobility models for MISFET simulations:

- Extended Canali model is the main model for drift-diffusion simulation and can also be used with the hydrodynamic model. It has calibrated parameters for silicon and SiGe for situations where ballistic or quasiballistic transport is not present. Calibration to short-channel devices where quasiballistic transport occurs can be done using model parameters:

  ```
  Physics { Mobility (HighfieldSaturation) }
  ```

- Transferred electron model is specially designed for III–V arsenides and III–V nitrides (GaAs, GaN, and so on):

  ```
  Physics { Mobility (HighfieldSaturation
     (TransferredElectronEffect)) }
  ```

- Different driving forces can be used to improve convergence or to include nonlocal properties using the hydrodynamic model:

  ```
  Physics { Mobility (HighfieldSaturation (CarrierTempDriveBasic)) }

  Physics { Mobility (HighfieldSaturation (CarrierTempDriveME)) }
  ```

The common model for drift-diffusion or QDD simulations is the extended Canali model.

Strained High-Field Mobility

In the QDD model, the dependency of high-field mobility on mechanical stress is determined by the strain dependency of the saturation velocity. The saturation velocity covers both the strain dependency of the physical saturation velocity and the strain dependency of the quasiballistic transport. The model is again the Canali model where the parameter `SaturationFactor` controls the stress dependency.

Ballistic Mobility

For short-channel transistors with a few scattering events, ballistic transport is important. The following ballistic mobility models in Sentaurus Device map ballistic transport to a mobility and, finally, to the channel resistance:

- Simple channel length–dependent ballistic mobility model

  This model depends on the gate length. Therefore, you must specify the gate length. It adds an additional term to the low-field mobility by using Matthiessen's rule, which is essentially constant and contains only one fitting parameter that represents the source-side injection velocity. Injection from the drain side is neglected.

- Injection velocity–dependent ballistic mobility model

  This model depends on the gate length. Therefore, you must specify the gate length. It adds an additional term to the low-field mobility by using Matthiessen's rule, which is essentially constant and contains only one fitting parameter that represents the source-side injection velocity. Injection from the drain side is considered empirically and is especially important at small drain bias voltage.

- Channel length–dependent model

  This model also depends on the gate length, but it is used in the framework of the kinetic velocity model as an option.

- Kinetic velocity model

  The kinetic velocity model (KVM) is a local model that does not need the explicit specification of a gate length. It includes the thermionic emission of charge carriers from the source side and their acceleration by the electric field. **This is the best and most appropriate ballistic mobility model in Sentaurus Device.**

- Multivalley ballistic mobility model

  This model in the framework of the KVM includes the influence of stress on ballistic mobility.

Gallium Nitride HEMTs

GaN HEMTs have unique qualities that require special attention. First, there is the large band gap that results in very low charge-carrier densities and numeric problems. Second, in the on-state, the current conduction is performed using a 2D electron gas channel in the GaN layer. This has implications for the modeling of mobility. Third, most of these structures use material combinations in addition to pure GaN and AlN, such as AlGaN. The mole fraction of these alloys determines the thermal conductivity, the mobility, and the band gap. For that reason, you need mole fraction–dependent parameters that make the transition

from GaN to AlN. In general, linear or parabolic interpolation between GaN and AlN is not sufficient because of the alloy influence. Furthermore, there is the influence of strain causing polarization in the device. In addition, traps strongly influence the device operation, and the saturation velocity behavior differs from silicon.

Silicon Carbide Devices

As another wide-bandgap material, SiC shows some differences to silicon as well. There are numeric problems connected with the small densities caused by the large band gap. The anisotropy of the mobility and of the avalanche coefficients is difficult to handle numerically. Traps, contact behavior, incomplete ionization, and tunneling play more critical roles in SiC devices than in silicon devices. You must include the incomplete ionization (`IncompleteIonization`) model in the simulation because of the wide band gap of SiC. Anisotropic behavior can be considered for thermal conductivity, mobility, and impact ionization.

A new version of the inversion and accumulation layer mobility model specific for 4H-SiC is available. To activate this model for SiC device simulations, specify the `SiC` keyword in the model options in the `Physics` section of the Sentaurus Device command file:

```
Mobility(Enormal(IALMob(SiC)))
```

To include anisotropic mobility, the `Aniso(Mobility)` command should also be specified in the `Physics` section.

Silicon Smart-Power Devices

Model requirements are similar to those for silicon deep-submicron devices (see Path A: Basic CMOS Simulation Command File Sections on page 24) fabricated by gate-first processes, but with a greater variety of process and doping conditions and layout specifications. In addition, there is a need for:

- High accuracy over a wide temperature range up to very high temperatures
- Good modeling capabilities for avalanche and breakdown
- Good modeling capabilities for electrostatic discharge (ESD)

For avalanche generation, the `UniBo2` model is calibrated for a temperature range from 300 K to 773 K and is best suited for ESD and high-temperature simulations. It also is recommended for room-temperature breakdown simulations.

5

Quality of Fitting and Extraction

This chapter discusses fitting quality and extraction issues.

Low-Field Mobility of Planar CMOS Devices With Silicon Channel

This section discusses the requirements and the quality of the extraction of the parameters for the low-field mobility of planar CMOS devices.

Oxide–Silicon Interface

This section briefly explains how the low-field mobility parameter sets have been calibrated.

Data from Takagi *et al*. [1][2], Nakamura *et al*. [3], and Nayfeh *et al*. [4] has been used to calibrate mobility in the transistor channel. All data is from pure polysilicon/SiO_2 gate stacks with oxide thicknesses of 25 nm for Takagi, 2 nm for Nakamura, and 5 nm for Nayfeh.

This data does not have high-k gate stacks and, therefore, allows you to extract the model parameters of surface phonon, surface roughness, and Coulomb scattering at ionized impurities. On the other hand, process-induced variations in surface roughness and the material composition of the first atomic layers at the isolator–silicon interface can influence mobility and result in different mobilities for different gate-formation process steps. These differences cannot be reflected by a single parameter set. Therefore, the parameter set presented here is a starting point for calibration.

Usually, the extraction of low-field mobility as a function of the effective electric field E_{eff} begins with measuring the drain current in the linear regime and, from that, the mobility is calculated for each gate voltage. At the same time, the inversion charge per area N_{inv} is calculated from C–V characteristics. Because E_{eff} is not measured, it must be calculated from N_{inv} using $E_{eff} = E_{eff}(N_{depl}, N_{inv}, \eta)$. Mostly, the depletion approximation with N_{depl} as the depletion charge is used for this, where the doping is extracted from other experiments and is assumed to be constant. The result is the dependency $\mu = \mu(E_{eff})$. To obtain the universal behavior, that is, the doping-independent behavior of the mobility for strong inversion, the fitting parameter η is introduced in the equation used to determine the effective field. You must use different values for η to obtain the universal behavior for

different charge carriers and surface orientations (electrons: 1/2 for (100) and 1/3 for (110), holes: 1/3).

These standard parameters provide mostly good universal behavior for a (100) surface orientation and low-to-moderate doping. For a (110) surface orientation, high doping, or nonplanar and double-gate devices, the universal behavior is mostly not fully achieved. In general, it remains a question of whether the use of this extraction method, and especially the use of the same η in the extraction from measurement and simulation, is correct.

If $\eta = 1$, then E_{eff} corresponds exactly to the electric field at the interface in the depletion approximation. However, $\eta < 1$ lowers the field strength, thereby taking into account that the current flow not only is at the surface but also is distributed a few nanometers into the depth of the silicon. Because it is unclear whether the field and current density distribution and the local dependency of mobility on the electric field are described sufficiently in the simulation, it might be difficult to achieve universal behavior in simulations with the same η parameter as in measurement.

Often, this universality is achieved only by introducing an artificial doping dependency for surface phonon scattering (SPS) and surface roughness scattering (SRS) in the Lombardi terms for the mobility models. However, it is unclear whether this doping dependency exists at all, or whether it is only a result of this forced parameter fitting. It can result in an unusual situation where the mobility degradation in intrinsic or very low–doped regions is actually stronger than in high-doped regions and drops to completely unrealistic values.

Because of this, another extraction methodology is used for the mobility parameters. The mobility model is calibrated to the measured $\mu = \mu(N_{inv})$ curves. For this, no calibration of η is necessary, and universal behavior is not requested. After this, a check is performed with $\mu = \mu(E_{eff})$ but additional calibration is performed only if there are strong disagreements. Figure 8 to Figure 16 show the agreement between measurement and simulation.

When simulating CMOS devices with precalibrated parameter sets, you must always take into account that these parameter sets are *initial* parameter sets that can undergo additional fine-tuning and major changes. The reasons are mainly connected to the dependency of mobility degradation on the processing.

For example, the interface or surface roughness depends on the process conditions and the materials involved. Applying the parameters already extracted to modern FinFET devices with a <110> channel direction often underestimates the inversion mobility, because the surface roughness–related mobility degradation is smaller on the (110) fin sidewalls than in the planar (110) case [5].

Similarly, for surface phonon–related interface mobility, the process conditions and the materials involved introduce some uncertainties that do not allow you to derive a parameter set that is valid for several device types or technology generations. Recalibration is usually required.

The strength of remote Coulomb scattering (RCS) or remote dipole scattering (RDS) depends on the number of charges in the high-k gate stack and their position. The model

must be calibrated to measurements when switching it on. RCS and RDS have the same origin: the mobility degradation by charges in the high-k gate stack. The difference is in the presence of both negative and positive charges in the case of RDS. For remote phonon scattering (RPS), the parameters are extracted from Monte Carlo simulation. Depending on the properties of the high-k gate stack, fine-tuning might be necessary. However, because it is difficult to extract the correct parameters without having mobility measurements for different interfacial layer thicknesses and different temperatures, you should keep the parameters for RPS and focus on the calibration of RCS and RDS.

The Philips unified mobility (`PhuMob`) model is calibrated to bulk mobility measurements. Usually, fine-tuning is necessary only when performing a custom calibration with high-accuracy requirements. In addition, mobility fine-tuning cannot be separated from the calibration of the doping concentration or dopant activation. Therefore, you should keep the default parameters.

In the framework of the inversion and accumulation layer mobility (`IALMob`) model, when going to inversion layer conditions, 3D Coulomb scattering turns to 2D Coulomb scattering, the `PhuMob` model is switched off, and the 2D Coulomb scattering model takes over. The 2D Coulomb scattering model is a very simple description of reality with many fitting parameters. This model mainly influences the onset of the I–V curves. The default parameters are not very good, and additional calibration is necessary.

The stress dependency of mobility resulting from `hSubband` (holes), `eSubband` (electrons), `MCmob`, and `SBmob` simulations has been compared to the literature and in-house Monte Carlo (Sentaurus Device Monte Carlo) and Kubo–Greenwood (Sentaurus Band Structure) simulations [6][7][8].

Figure 8 *Electron mobility versus areal inversion density for (100) surface in linear scale, with (top to bottom) channel doping of 3.9e15, 2e16, 7.2e16, 3e17, 7.7e17, and 2.4e18*

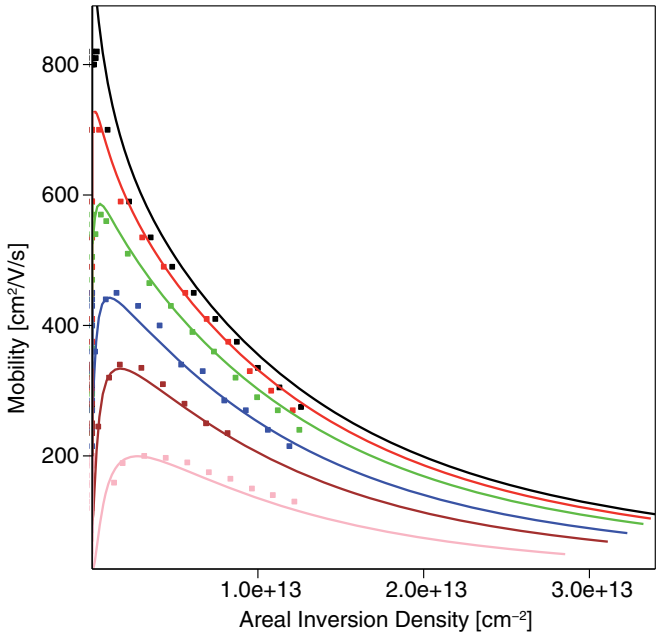


Figure 9 *Electron mobility versus areal inversion density for (100) surface in logarithmic scale with (top to bottom) channel doping of 3.9e15, 2e16, 7.2e16, 3e17, 7.7e17, and 2.4e18*

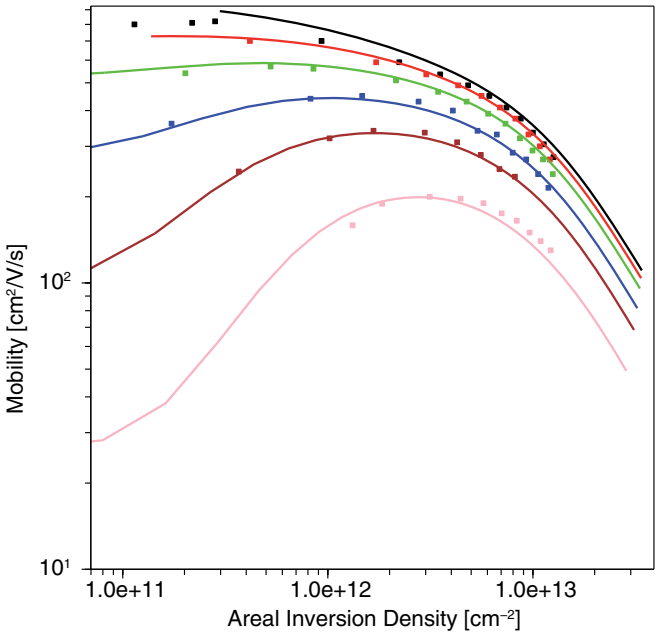


Figure 10 *Hole mobility versus areal inversion density for (100) surface in linear scale with (top to bottom) channel doping of 7.8e15, 1.6e16, 5.1e16, 2.7e17, and 6.6e17*

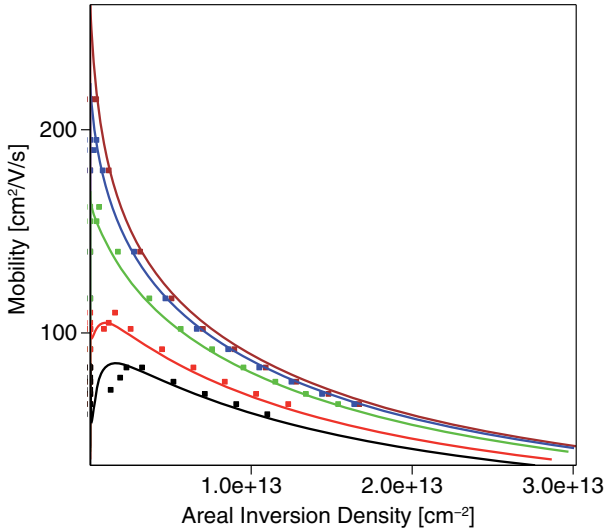


Figure 11 *Hole mobility versus areal inversion density for (100) surface in logarithmic scale with (top to bottom) channel doping of 7.8e15, 1.6e16, 5.1e16, 2.7e17, and 6.6e17*

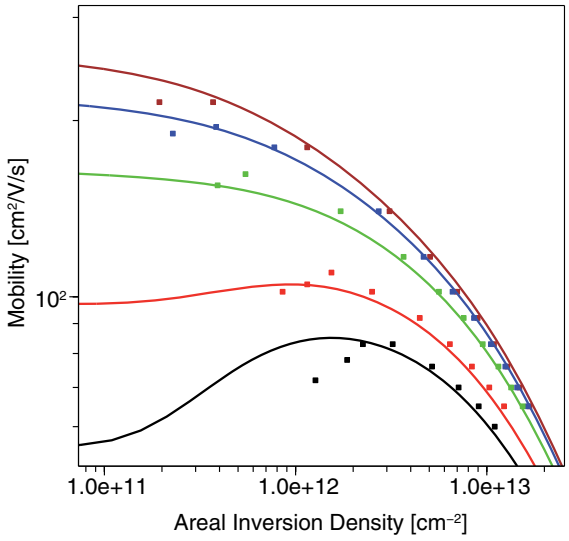


Figure 12 *Electron mobility versus effective field for (100) surface in linear scale with (top to bottom) channel doping of 3.9e15, 2e16, 7.2e16, 3e17, 7.7e17, and 2.4e18*

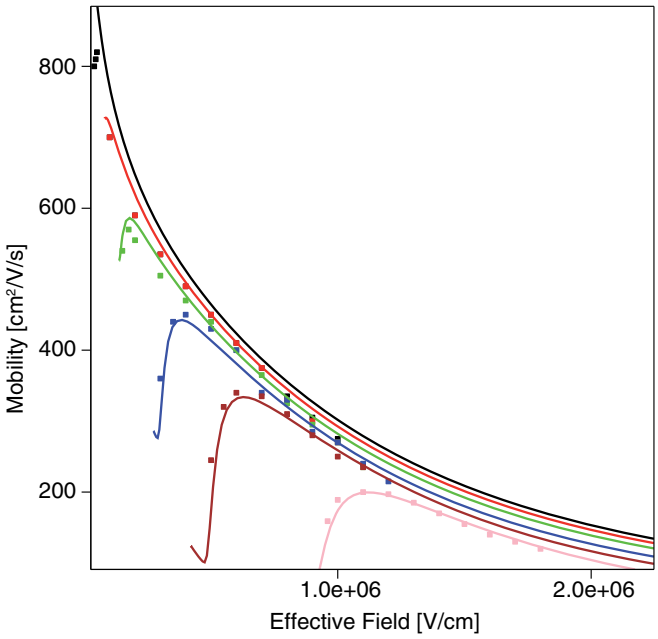


Figure 13 *Electron mobility versus effective field for (100) surface in logarithmic scale with (top to bottom) channel doping of 3.9e15, 2e16, 7.2e16, 3e17, 7.7e17, and 2.4e18*

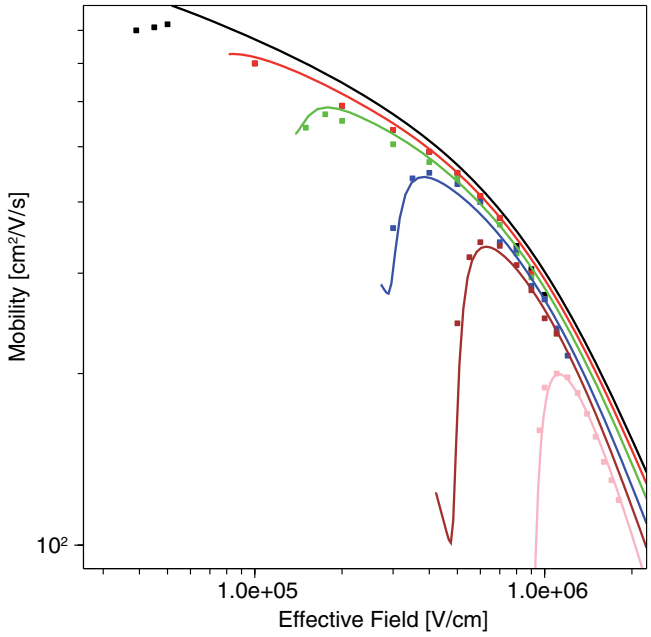


Figure 14 *Electron mobility versus areal inversion density for (100) surface as a reference and for (110)/<100> and (110)/<110> in linear scale, with channel doping between 5e17 and 9e17*

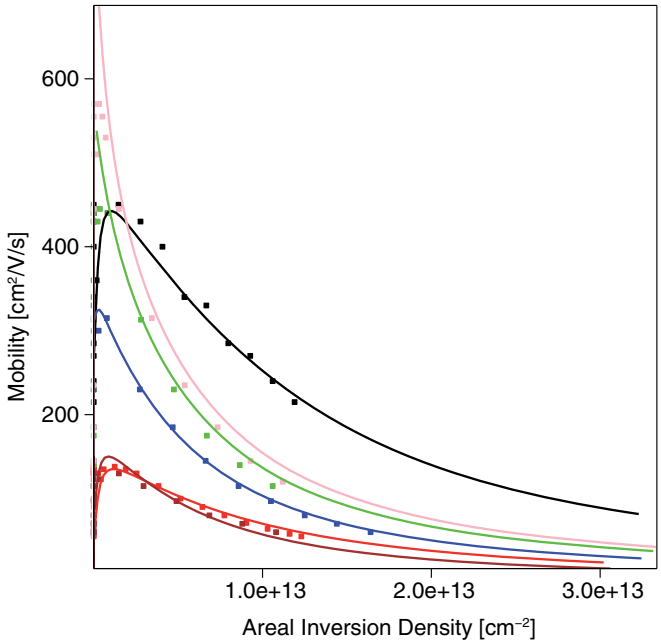


Figure 15 *Hole mobility versus areal inversion density for (100) surface as a reference and for (110)/<100> and (110)/<110> in linear scale, with channel doping between 4.5e17 and 6.6e17*

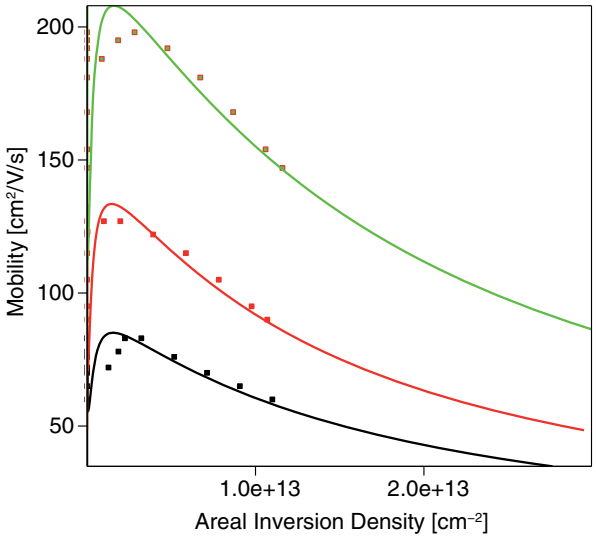
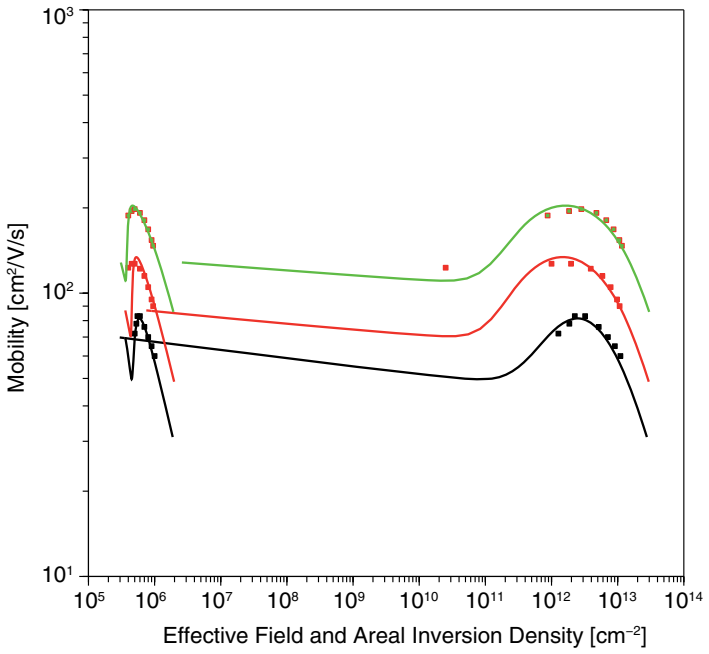


Figure 16 Hole mobility versus effective field and areal inversion density for (100) surface as a reference and for (110)/<100> and (110)/<110> in double logarithmic scale, with channel doping between 4.5e17 and 6.6e17



Influence of Metal Gate on Inversion Layer Mobility

The parameter set used to produce the results shown in Figure 8 to Figure 16 is derived from measurements on devices with a polysilicon/oxide gate stack. In devices with a metal/oxide gate stack, the channel mobility can differ because of the influence of the metal gate. The same applies to the comparison between polysilicon/HfO_2 and metal/HfO_2 gate stacks.

Usually, the transition from the polysilicon gate to the metal gate results in improved mobility. One reason for this could be the better screening of surface phonons and charges in the gate stack by the charge carriers in the metal gate [9].

Table 11 Influence of different gate materials on inversion layer mobility for electrons [9]

Gate stack	Maximum mobility [$cm^2/V/s$]	Mobility at E_{eff} = 1 MV/cm [$cm^2/V/s$]
HfO_2/Polysilicon	175	145
HfO_2/TiN	245	200

The increasing mobility caused by the metal gate can be reflected by an increase of the `IALMob` parameters `B`, `C`, and `delta` by a factor of 1.4.

Influence of High-k Gate Stack on Inversion Layer Mobility

The parameter set used to produce the results shown in Figure 8 to Figure 16 is derived from measurements on devices with a polysilicon/oxide gate stack. In devices with a high-k isolation (for example, with HfO_2), the channel mobility can degrade because of the influence of charges in the gate isolator (RCS or RDS) and RPS.

Often, the charges are arranged in dipole-like configurations. The RCS-related mobility degradation depends on the number and the arrangement of charges in the gate isolator, especially on their distance to the inversion layer. RCS depends greatly on the process conditions and the materials used in the gate stack, and so the variation between technologies and process splits and even over the wafer and between wafers can be large. Therefore, it is not possible to provide well-calibrated default parameters, and custom calibration to the process and device under consideration is usually required. However, the model should describe the major tendencies such as dependency on the thickness of the interfacial oxide or on the concentration of charges in the gate stack.

Table 12 and Table 13 list model parameters for the `RPS` model and for the `InterfaceCharge` models that fulfill these requirements.

Table 12 Recommended parameters and values for RPS model

Parameter	Value for electrons	Value for holes	Comments
`d_crit`	1.0e-7 cm	1.0e-7 cm	Distance between high-k interface and silicon interface
`l_crit`	7.7e-8 cm	7.7e-8 cm	[10]
`murps0`	124 $cm^2/(Vs)$	124 $cm^2/(Vs)$	Extracted from Sentaurus Device Monte Carlo simulation

Table 13 Recommended parameters and values for InterfaceCharge models (RCS and RDS, NegInterfaceCharge and PosInterfaceCharge models use the same parameters)

Parameter	Value for electrons	Value for holes	Comments
`c_exp`	0.5	0.5	
`E0`	1.0e-6	1.0e-6	

Table 13 Recommended parameters and values for InterfaceCharge models (RCS and RDS, NegInterfaceCharge and PosInterfaceCharge models use the same parameters) (Continued)

Parameter	Value for electrons	Value for holes	Comments
`l_crit`	7.7e-8	7.7e-8	[10]
`mu1`	250	250	

Extraction of Contact Resistance Parameters

Stavitski *et al*. [11] measured the contact resistance between PtSi and NiSi, and silicon. Based on this data, the parameters for the Schottky resistance (`SchottkyResist`) model in Sentaurus Device have been extracted (see Table 14).

Note:

No additional tests or validations have been performed with this parameter set. In addition, the parameters are not included in the parameter files in the `MaterialDB` folder.

Table 14 Parameters for the SchottkyResist model for PtSi and NiSi

Parameter	Value for electrons	Value for holes
PtSi		
`barrier`	0.8 eV	0.4 eV
`mt`	0.08	0.08
`Rinf`	3.0e-10 Ωcm^2	3.0e-10 Ωcm^2
NiSi		
`barrier`	0.6 eV	0.6 eV
`mt`	0.08	0.08
`Rinf`	1.0e-9 Ωcm^2	5.0e-10 Ωcm^2

Extraction of Parameters for Mobility Models in Accumulation Regime

In the accumulation regime, mobility is influenced by the same mobility degradation mechanisms as in the inversion regime. However, model parameters and the importance of single mobility degradation mechanisms can vary. Especially at higher background doping concentrations, the screening of Coulomb scattering differs between accumulation and inversion.

Because of the importance of the on-state and the measurement data available, the focus is on high accumulation areal densities. Reference [12] provides some specific and well explained measurements of mobilities in accumulation layers in planar bulk devices for different background doping concentrations at room temperature. These measurements have been used as a reference for the simulations with Sentaurus Device using the inversion and accumulation layer mobility (`IALMob`) model.

Despite small differences between accumulation and inversion layer mobility in the on-state claimed in [12], a calibrated parameter set is presented that improves the agreement for electrons. The parameter set also shows how to adjust the accumulation layer model parameters without changing much in the inversion regime. Table 15 shows the parameters used for calibration.

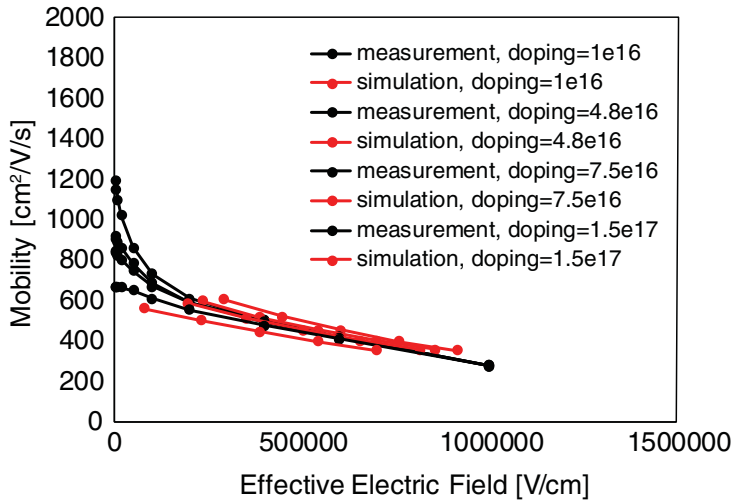
Table 15 Parameters of IALMob model used for calibration (electrons)

Parameter	Recommended value	Parameter	Recommended value
`lambda`	0.02	`alpha_sr_D`	2.0e-14
`alpha_ph2d_A`	0.0	`lambda_sr`	0.02
`alpha_ph2d_D`	2.0e-14	`lambda_sr_A`	0.0
`lambda_ph2d_A`	0.0	`lambda_sr_D`	1.0
`lambda_ph2d_D`	1.0	`D1_acc`	1200
`alpha_sr_A`	0.0	`D2_acc`	1200

Essentially with these parameter settings, the doping dependence for the inversion case remains switched off and, when there is no accumulation doping, there is no change in the results. However, with `alpha_ph2d_D` or `alpha_sr_D`, and `lambda` or `lambda_sr` larger than zero, the doping dependence for the accumulation regime is switched on. In this way, you can separate the accumulation and inversion regimes.

The parameter set in Table 15 gives reasonable agreement to the measurements in [12] (see Figure 17).

Figure 17 Mobility versus effective electric field for electrons and different donor doping concentrations; red lines: simulation and black lines: measurements from [12]



Basic Properties of Silicon Carbide Devices

Parameter values and corresponding references can be found in the `4H-SiC.par` and `6H-SiC.par` files and in [13].

All silicon carbide (SiC) polytypes are indirect semiconductors, and the maximum of the upper valence bands is located at the center (Γ-point) of the Brillouin zone. Conduction band minima of 3C-SiC, 6H-SiC, and 4H-SiC are located at point X, point U on the (L-M) line, and point M, respectively.

Photoluminescence or absorption measurements give an exciton band gap E_{gx}. There are no reliable measured values for binding energies E_x of free excitons, which are required to obtain the indirect band gap $E_g = E_{gx} + E_x$. Since measured values of E_x range from 10 mV to 80 mV, a mean direct band gap can be assumed by shifting E_{gx} by 40 mV. In the parameter file, a value of `Eg0`=3.285 eV is used.

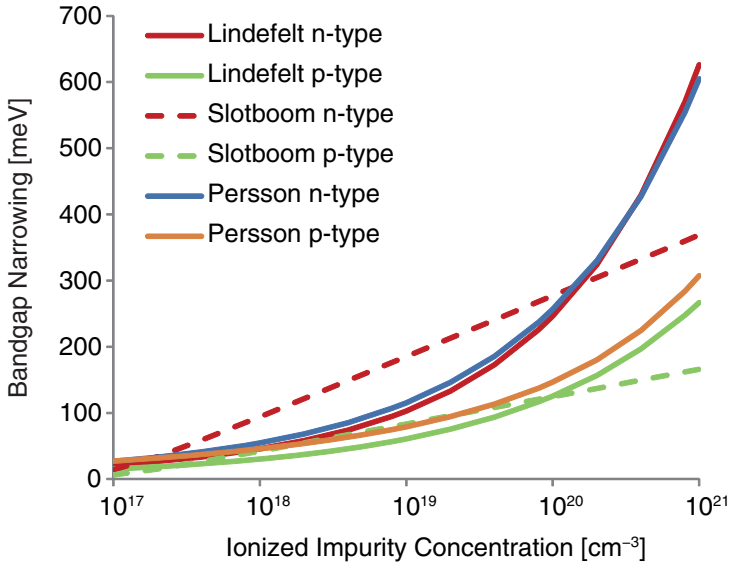
Temperature dependency of the band gap is measured for 6H-SiC in the temperature ranges from 6 K to 200 K and from 300 K to 700 K. The dependency of 3C-SiC is measured in the temperature range from 295 K to 700 K. The dependency of 4H-SiC is assumed to be similar because no experimental data is available.

The effective mass of electrons and holes has been measured by optically detected by either cyclotron resonance or Raman scattering and has been compared to theoretical calculation. The theoretical calculation reproduced well the experimental results.

There is no published measured data of bandgap narrowing. The theoretical calculation was first performed by Lindefelt [14] and followed by Persson *et al*. [15]. The former model

constitutes an extension of a semiempirical method originally developed by Jain and Roulston [16]. The latter is based on random phase approximation. Both calculations give the analytic form of band-edge displacement depending on the doping density. The analytic form corresponds to the `JainRoulston` model implemented in Sentaurus Device.

Figure 18 *Results of bandgap narrowing of 4H-SiC from [14] and [15] with the Slotboom model implemented in Sentaurus Device with calibrated coefficients*



Electron affinity is calculated assuming the linear dependency of hexagonality together with the measured band offset of the conduction band for 3C-SiC or 6H-SiC, and the calculated result of the band offset of the valence band.

Parameter sets for the Hatakeyama model [17] and the Okuto–Crowell impact ionization model are implemented in the parameter files (see Content of Parameter Files on page 11). The parameter set for the Okuto–Crowell impact ionization model along the c-axis corresponds to the one reported by Niwa *et al*. [18].

Figure 19 and Figure 20 show the comparison of the impact ionization coefficients of different references.

Figure 19 Hole impact ionization coefficient along the c-axis of 4H-SiC [17][18][19]

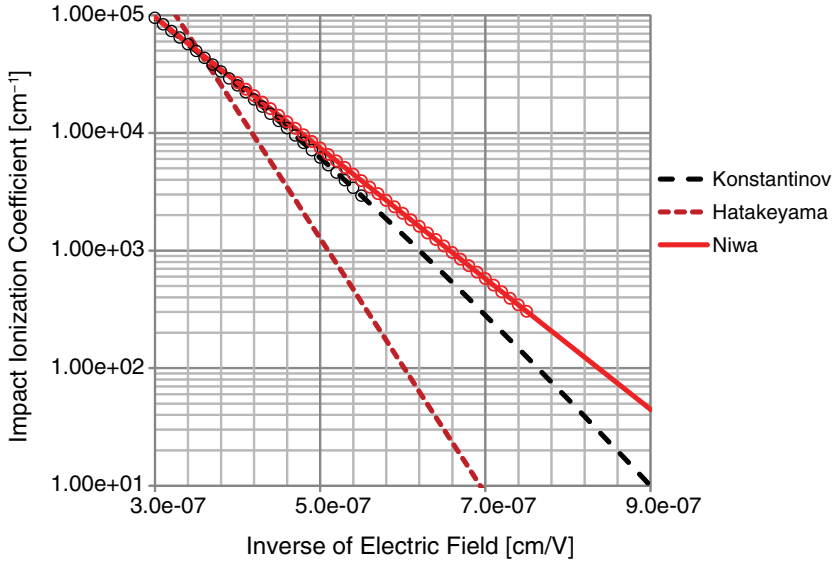
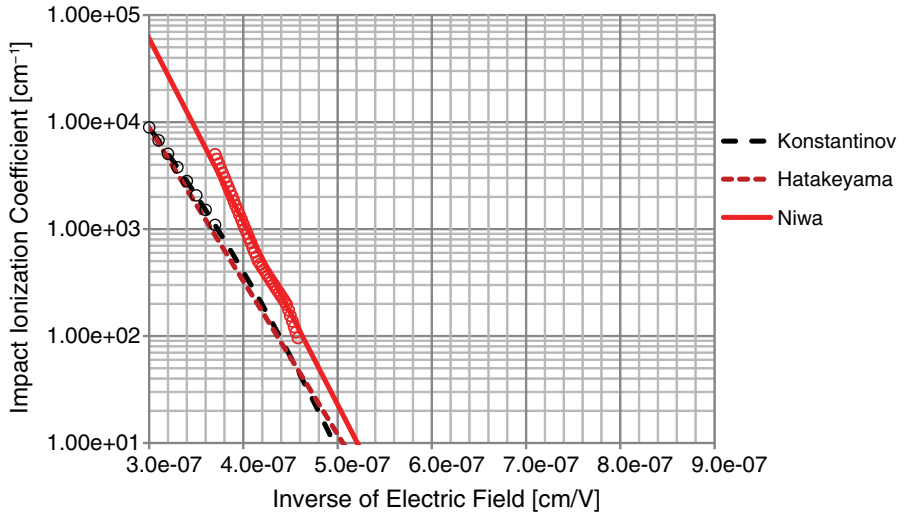


Figure 20 Electron impact ionization coefficient along the c-axis of 4H-SiC [17][18][19]



Temperature dependency of thermal conductivity of 4H-SiC and 6H-SiC perpendicular to the c-axis has been reported. At low temperatures, T^3 dependency and T^2 dependency are observed for 4H-SiC and 6H-SiC, respectively. T^2 dependency is attributed to the scattering of phonons by electrons in the impurity band. Thermal conductivity in 6H-SiC single crystals decreases with increasing electron concentration, and the slope at low temperature can be represented by the T^2 law, which confirms the dominance of electron scattering.
Temperature dependency of thermal conductivity using data from [20], [21], [22], and [23] as

cited in [24] is implemented in the parameter files (see Content of Parameter Files on page 11). Experimental data is that in the direction perpendicular to the c-axis.

Doping-dependent mobility parameter sets for the Masetti, Arora, and anisotropic inversion and accumulation layer mobility models are implemented in the parameter files (see Content of Parameter Files on page 11).

Figure 21, Figure 22, Figure 23, and Figure 24 show a comparison with different literature data.

Figure 21 Doping-dependent electron mobility at 500 K for n-doped 4H-SiC [25][26][27]

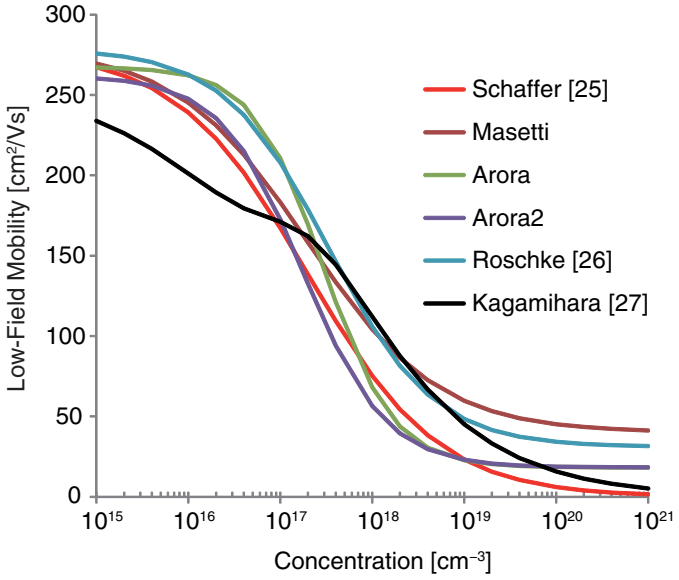


Figure 22 Doping-dependent electron mobility at 300 K for n-doped 4H-SiC [25][26][27]

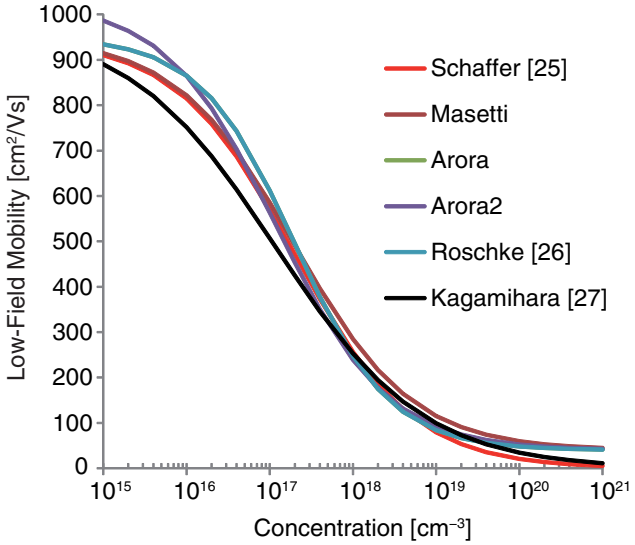


Figure 23 Doping-dependent electron mobility at 300 K for n-doped 4H-SiC [25][81][82][83][84]

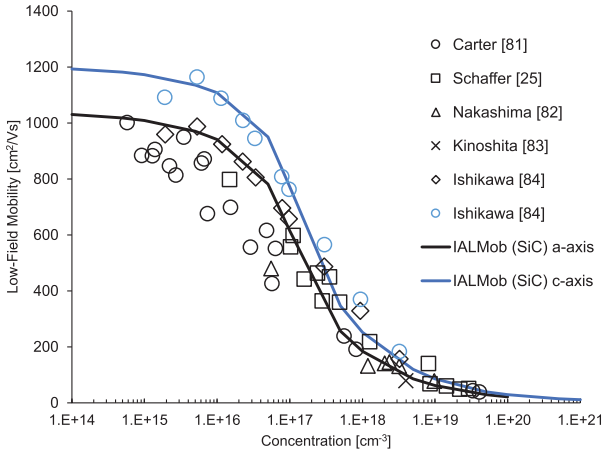
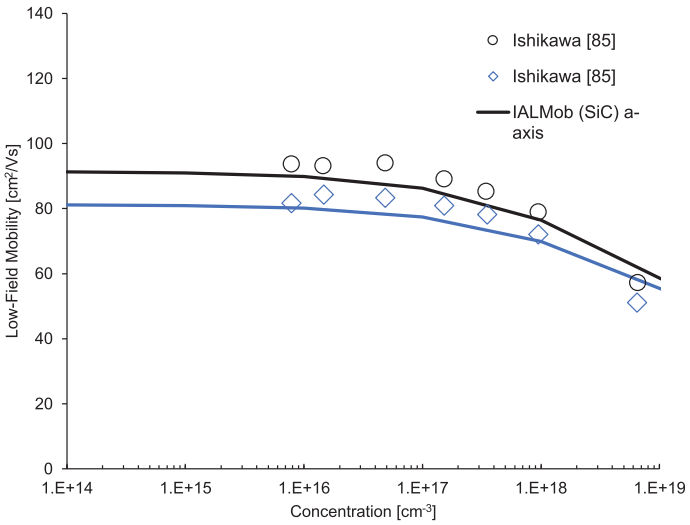
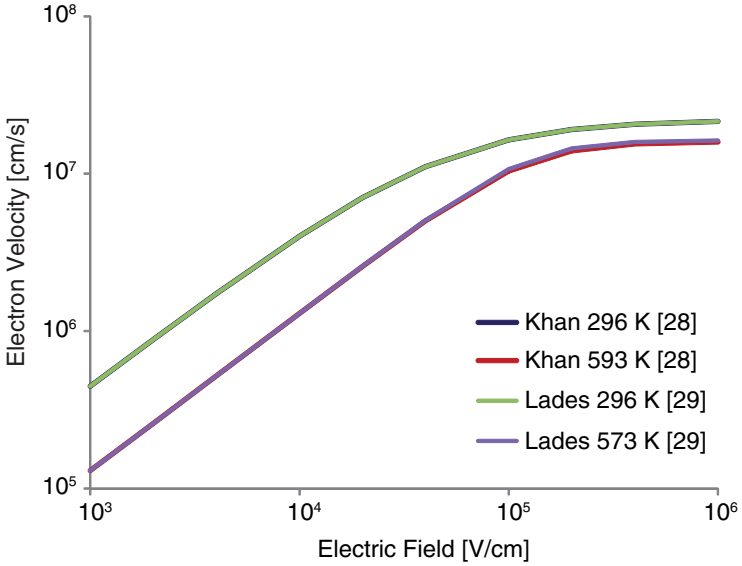


Figure 24 Doping-dependent hole mobility at 300 K for p-doped 4H-SiC [85]



For high-field saturation, all measured data refers to the current flow perpendicular to the c-axis. Parameters for the Canali model are extracted and implemented in the parameter file.

Figure 25 Drift velocity of electrons in 4H-SiC, parallel to basal plane (0001) [28][29]



A nitrogen atom residing on the C lattice site becomes the shallow donor in SiC. A phosphorus atom also is used as a shallow donor (like nitrogen) and has approximately the same ionization energy E_c – 93 meV at the cubic site and E_c – 53 meV at the hexagonal site.

However, unlike nitrogen, P mainly substitutes at a silicon site, and less than 10% of substitutional P is located at C sites.

Phosphorus has a slightly larger ionization energy at C sites than at Si sites. The ionization energy of N and P donors decreases with increasing donor concentration. Experimental data for cubic and hexagonal N donors in 6H-SiC is available. For N donors in 4H-SiC, experimental data of donor concentration dependency and fitting with empirically formula has been reported.

Aluminum and boron residing on Si lattice sites become acceptors in SiC. For Al acceptors in 4H-SiC, experimental data of acceptor doping concentration dependency and fitting with empirical formulas have been reported [30][31][32].

Figure 26 to Figure 30 show the experimental data and the fit implemented in the `SiC.par` parameter file.

Figure 26 Ionization energy for aluminum-doped 4H-SiC with extracted fit function [30][31][32]

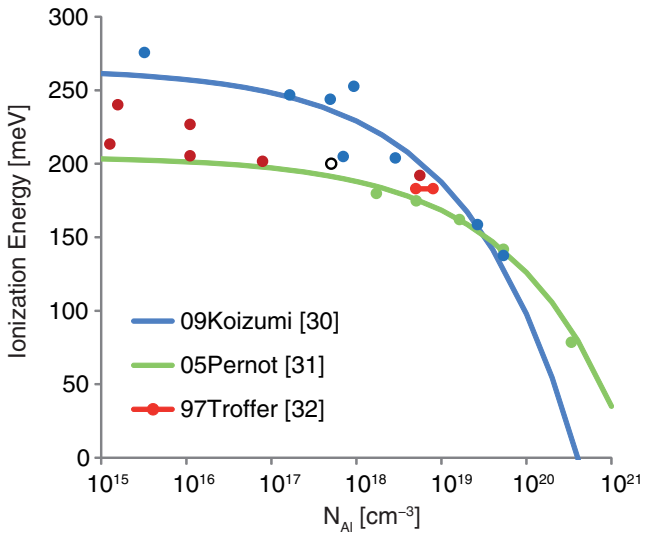


Figure 27 *Ionization energy for nitrogen-doped 4H-SiC (hexagonal site) with extracted fit function [27]*

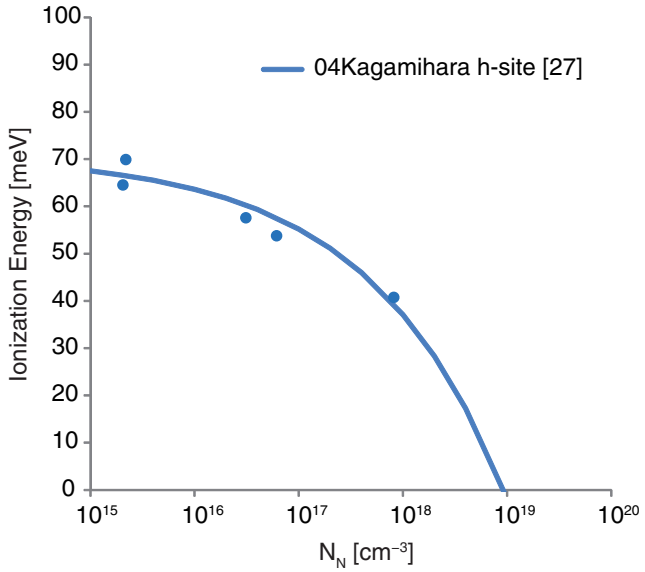


Figure 28 *Ionization energy for nitrogen-doped 4H-SiC (cubic site) with extracted fit function [27]*

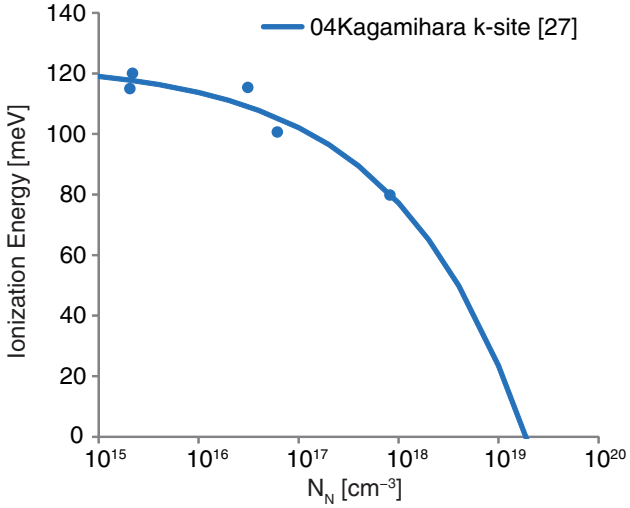


Figure 29 Ionization energy for phosphorus-doped 4H-SiC with extracted fit function

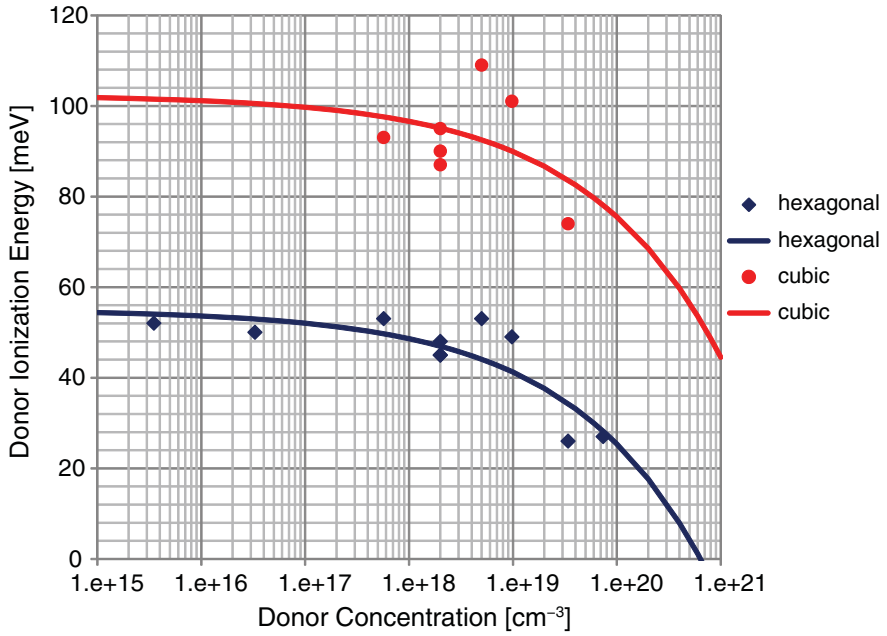
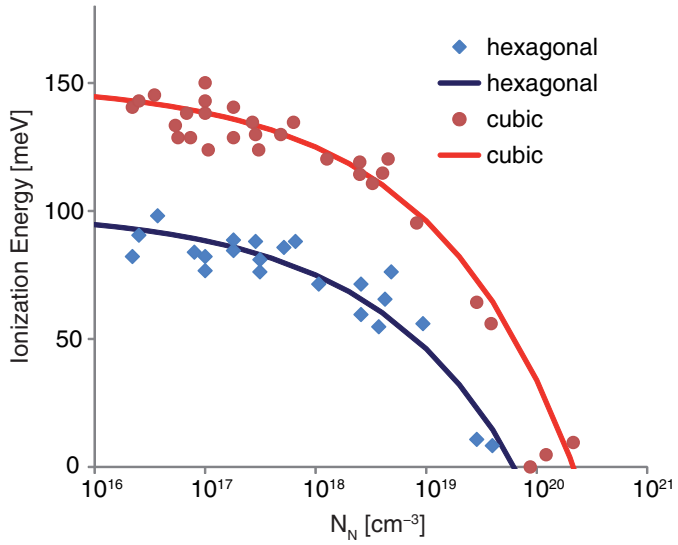


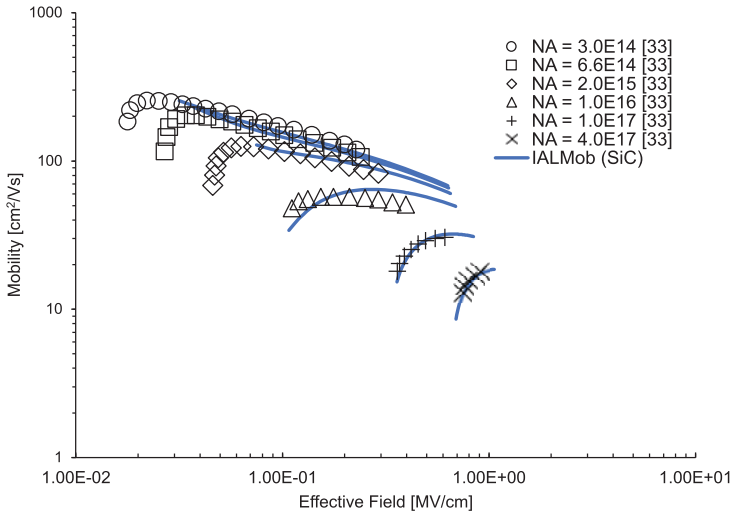
Figure 30 Ionization energy for nitrogen-doped 6H-SiC with extracted fit function



Starting from this release, an anisotropic inversion and accumulation layer mobility model `Mobility(Enormal(IALMob(SiC)))`, which includes a donor/acceptor scattering cross-section ratio function specific for 4H-SiC, is available. This model has been calibrated on measured data from [33][81][82][83][84][85] and the calibrated parameter sets are defined in the `IALMob_SiC{}` and the `IALMob_SiC_aniso{}` sections of the parameter file,

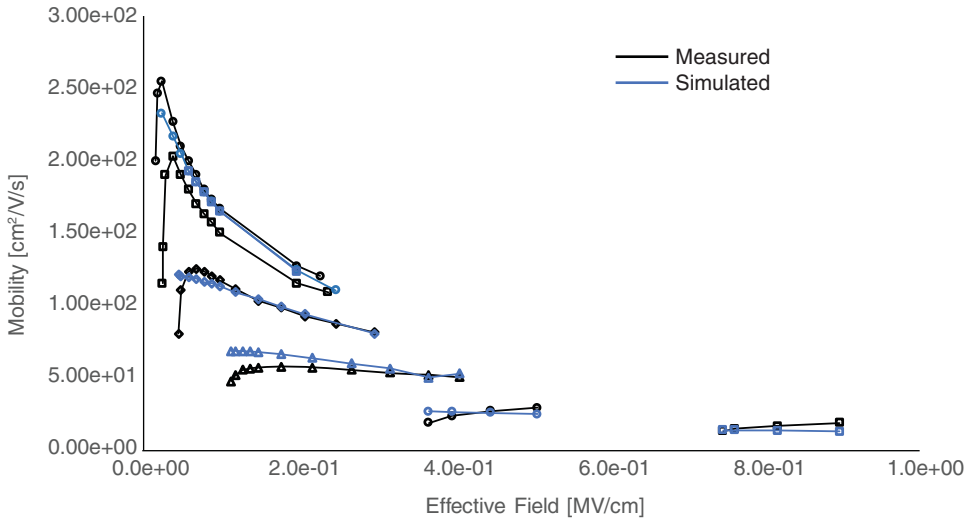
respectively. This model correctly captures the trends of doping-dependent bulk mobility as shown in Figure 23, and Figure 24.

Figure 31 Simulated and measured [33] inversion layer mobility



The inversion layer mobility parameters have been calibrated for 4H-SiC interfaces to oxide for different doping and trap concentrations as well as gate oxide formation processes. Experimental data is taken from [33], [34], and [35]. Surface phonon, surface roughness, and 2D Coulomb scattering parameters have been adjusted. The parameter files `4H-SiC.par`, `4HSiC.par`, `SiC_4H.par`, and `SiliconCarbide.par` include the calibrated parameters. Figure 31 shows the comparison between experimental data and simulation results obtained from IALMob(SiC) while simulation results only with IALMob are shown in Figure 32.

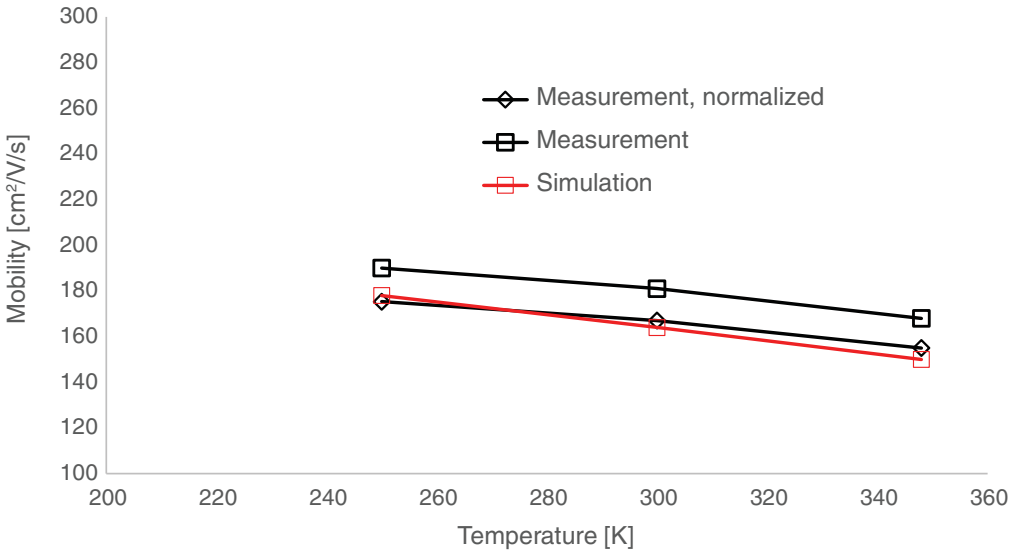
Figure 32 Simulated and measured [33] inversion layer mobility



In the calibration of the inversion layer mobility, the incomplete ionization model [36] has been included to achieve better results, especially at lower temperatures, where a large part of the dopants is not activated. To describe the temperature dependence of the mobility, the `k` parameter in the phonon scattering term, in the `IALMob` model, has been adjusted.

Figure 33 shows the comparison between experimental data [37] and simulation results. The measured mobility values over temperature have been normalized to the measured values at 300 K in the set of parameter files used in Figure 32 to compensate for fluctuations between different sets of measurements due to different trap distributions.

Figure 33 Simulated and measured [37] inversion layer mobility at different temperatures and low doping



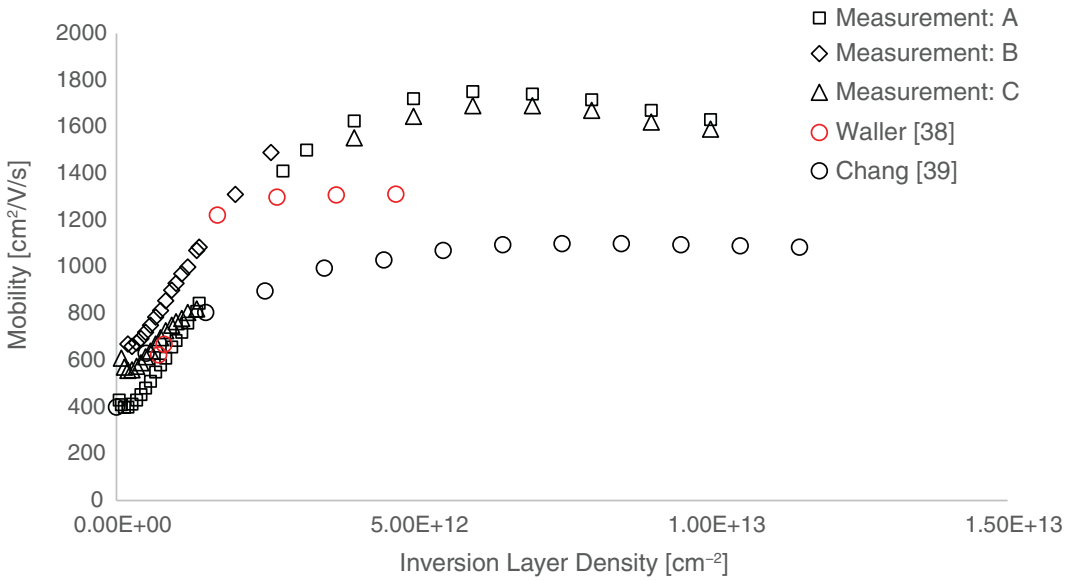
Basic Properties of Gallium Nitride Devices

The inversion layer mobility parameters have been calibrated for GaN to AlGaN or AlN interfaces.

Experimental data is taken from [38] and [39]. Surface phonon, surface roughness, and 2D Coulomb scattering parameters have been adjusted. The parameter file `GaN.par` includes the calibrated parameters. Figure 34 shows the comparison between experimental data and simulation results.

In the `IALMob` parameter set, the `s` parameter that controls the quantization is set to a higher value, assuming 2D Coulomb scattering dominates in the channel. Therefore, the transition to bulk Coulomb scattering is controlled by `lcrit_c` only.

Figure 34 Simulated and measured [38][39] inversion layer mobility



Basic Properties of Indium Gallium Arsenide Devices

Parameter values and references can be found in the `InGaAs.par` file. The following sections briefly show the results of parameter calibration and review various aspects of the calibration.

Permittivity

The permittivity values for GaAs and InAs materials have been reviewed following the data reported by Adachi [40]. Since there is no experimental data for semiconductor alloys, a linear interpolation scheme has been adopted for indium gallium arsenide ($In_{1-x}Ga_xAs$) alloys.

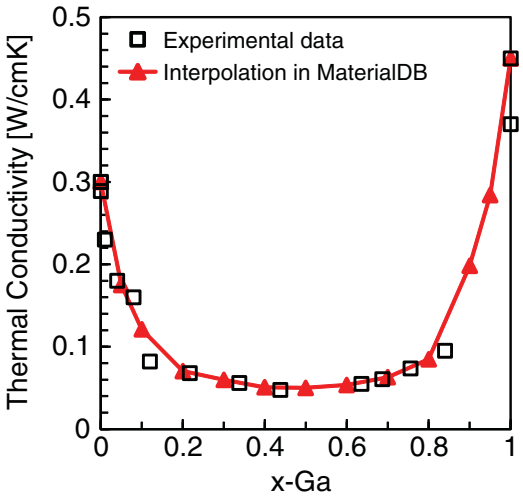
Lattice Heat Capacity

Similarly, a linear interpolation scheme for heat capacity is used for $In_{1-x}Ga_xAs$ alloys, in which the heat capacity values for GaAs and InAs at 300 K are updated with the ones reported by Adachi [41].

Thermal Conductivity

The mole fraction dependency of thermal conductivity for $In_{1-x}Ga_xAs$ alloys has been measured by Abrahams *et al*. [42] and Arasly *et al*. [43] at 300 K. Due to a random distribution of Ga and In atoms in the sub-lattice sites, the thermal conductivity values for $In_{1-x}Ga_xAs$ alloys tend to be smaller than for the corresponding binary materials. Figure 35 shows the thermal conductivity model and comparison to experimental data.

Figure 35 Thermal conductivity versus Ga mole fraction for $In_{1-x}Ga_xAs$ alloys at 300 K; experimental data is taken from [42] and [43], and binary data is taken from [44]

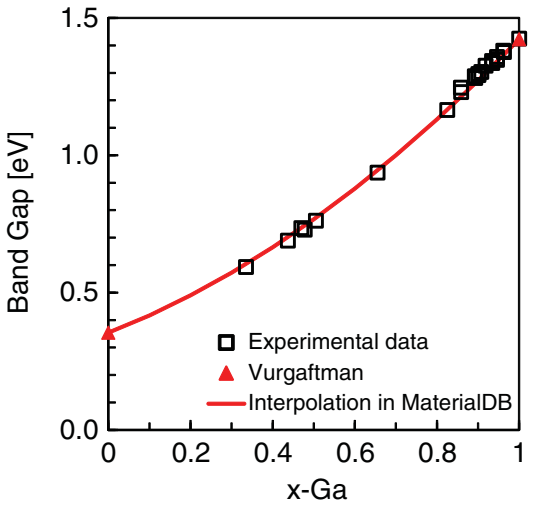


Band Gap and Electron Affinity

InGaAs ternary materials are direct bandgap materials over the entire composition range, and both the maxima of the upper valence bands and the minima of the conduction band are located at the Γ-point of the Brillouin zone. Photoluminescence and photoreflectance measurements are often used to determine the bandgap energies of $In_{1-x}Ga_xAs$ [45][46] [47]. In the case of the lattice-matched alloy to InP, $In_{0.53}Ga_{0.47}As$, which has been investigated extensively, the bandgap energy is generally recognized to be 0.816 eV at 0 K. Therefore, a temperature-independent bowing parameter of 0.477 eV is suggested [48]. As for the Varshni parameters, the values for $In_{1-x}Ga_xAs$ alloys are determined through a linear interpolation based on the ones from GaAs and InAs. Figure 36 shows the composition dependency of bandgap energy at 300 K, plotted with the new set of the bowing parameter and Varshni parameters. The revised bandgap model fits well to the experimental data.

The electron affinity values for the GaAs and InAs binary materials are taken from Adachi [40] and are further derived to obtain the values at 0 K using the revised Varshni parameters. Since little experimental data is available, linearly interpolated electron affinity values are assumed for $In_{1-x}Ga_xAs$ alloys in the parameter file.

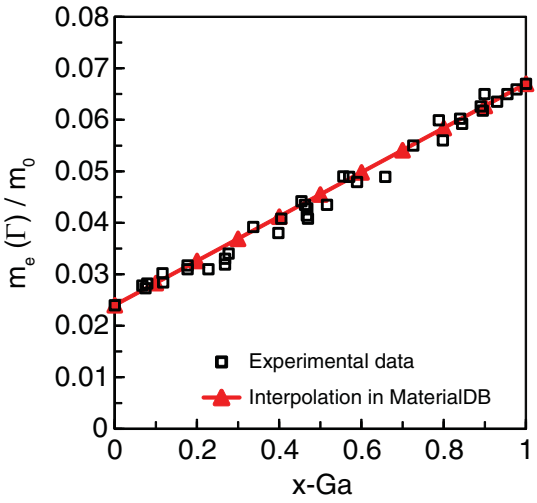
Figure 36 *Band gap versus Ga mole fraction for $In_{1-x}Ga_xAs$ alloys at 300 K; experimental data is taken from [45], [46], and [47]; binary data is taken from [48]*



Density-of-States

The electron effective masses of the Γ-valley of the $In_{1-x}Ga_xAs$ alloys have been measured extensively [41], and the results are shown in Figure 37.

Figure 37 *Electron effective mass of the Γ-valley versus Ga mole fraction for $In_{1-x}Ga_xAs$ alloys at 300 K; experimental data is reprinted from Adachi [41]*



A simple linear interpolation scheme using the corresponding GaAs and InAs binary data yields a good fit to the experimental data. In the `InGaAs.par` file, the composition

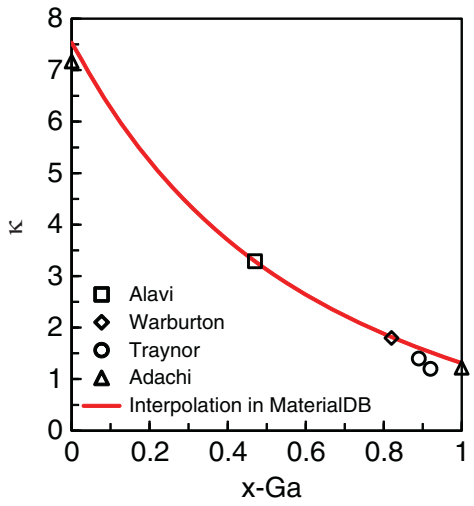
dependency of density-of-states for electrons is a set of tabulated values that are computed from the electron effective masses at 300 K using:

$$N_C(300\,\mathrm{K}) = 2.5094\times10^{19}\left(\frac{m_e}{m_0}\right)^{\frac{3}{2}} \quad (2)$$

Regarding the hole effective masses for $In_{1-x}Ga_xAs$ alloys, Vurgaftman *et al*. proposed a bowing parameter for each Luttinger valence band parameter: γ_1, γ_2, and γ_3 [48]. In measurement, $\kappa(x)$, defined as $\kappa = -1/3\gamma_1 + 2/3\gamma_2 + \gamma_3 - 2/3$, has been reported for x = 0.47 and x = 0.82–0.92 [49][50][51].

Figure 38 shows that the interpolation scheme from Vurgaftman *et al*. agrees well with the experimental data to some extent and is used to calculate the composition dependency of the overall hole effective masses, m_p.

Figure 38 κ-values versus Ga mole fraction for $In_{1-x}Ga_xAs$ alloys; experimental data is taken from [40], [49], [50], and [51]; red line represents the interpolation function suggested by [48]

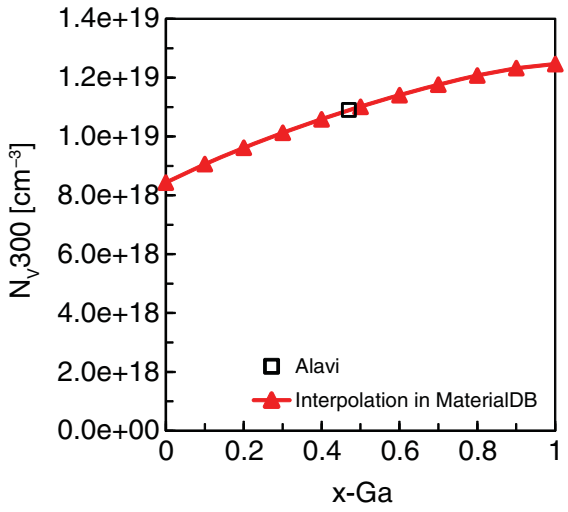


Similarly, a set of tabulated values of density-of-states for holes at 300 K is computed using:

$$N_V(300\,\mathrm{K}) = 2.5094\times10^{19}\left(\frac{m_p}{m_0}\right)^{\frac{3}{2}} \quad (3)$$

Advanced Calibration Device includes these tabulated values (see Figure 39).

Figure 39 *Valence band density-of-states at 300 K versus Ga mole fraction for $In_{1-x}Ga_xAs$ alloys; experimental data is computed based on Luttinger parameters taken from [49]; red line is computed following the interpolation scheme suggested by [48]*



Bandgap Narrowing

Doping-dependent bandgap narrowing for GaAs, InAs, and $In_{1-x}Ga_xAs$ materials is accessed using the Jain–Roulston model [52]. The coefficients for p-type doping (A_p, B_p, C_p, and D_p) in GaAs material are derived from [52], in which the quantities are refined with available up-to-date basic material parameters of GaAs.

Figure 40 shows the extent of bandgap narrowing for p-GaAs obtained with the Jain–Roulston model, and it agrees well with the experimental data measured using photoluminescence in the carrier density range from 2.7e17 to 3.9e20 cm^{-3} [53]–[57]. For p-InAs, the coefficients are retrieved in the same way, but the model validation cannot be qualified due to a lack of experimental data (see Figure 41).

Regarding n-type GaAs and InAs, the formulas in the Jain–Roulston model cannot describe the doping dependencies of bandgap narrowing. Therefore, the corresponding coefficients (A_n, B_n, C_n, and D_n) are determined preferably through curve fitting based on experimental data. As shown in Figure 40, the bandgap narrowing for n-GaAs with fitted Jain–Roulston coefficients agrees well with experimental data [53][58]. It should be noted that, for n-InAs, the bandgap narrowing value reaches nearly 0.2 eV at the carrier density of 3e18 cm^{-3} [59], and a higher density is probable, which would alter the band structure drastically. Therefore, a minimum band gap (`EgMin`) value of 0.15 eV is set in the `Bandgap` model to avoid the occurrence of an unusually low (or negative) band gap.

Since there is little experimental data reported in the literature to resolve the mole fraction dependency of bandgap narrowing for $In_{1-x}Ga_xAs$, a simple linear interpolation scheme is adopted for each coefficient in the Jain–Roulston model.

Figure 40 *Doping-dependent bandgap narrowing for p-GaAs and n-GaAs at 300 K; experiment data is taken from [53]–[58]; black line is computed following Jain–Roulston model with up-to-date GaAs basic material parameters, and red line is fitted curve based on experimental data from [53] and [58]*

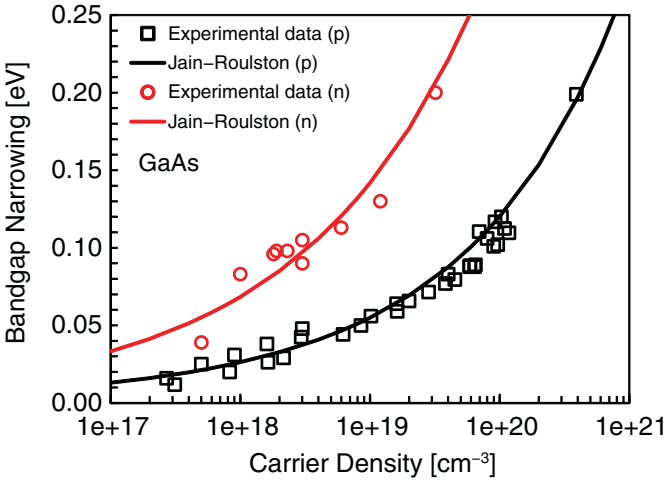
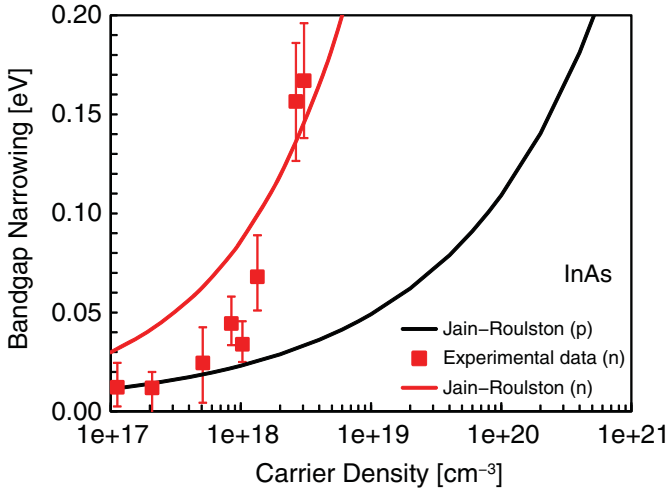


Figure 41 *Doping-dependent bandgap narrowing for p-InAs and n-InAs at 300 K; black line is computed following Jain–Roulston model with up-to-date InAs basic material parameters, and red line is fitted curve based on experimental data from [59]*



Quantization Effects

The quantization model is considered in MOS devices to obtain the correct densities and field distributions in channels. The density gradient model in Sentaurus Device would be the first choice for planar and nonplanar $In_{1-x}Ga_xAs$ device simulations, and then a calibration

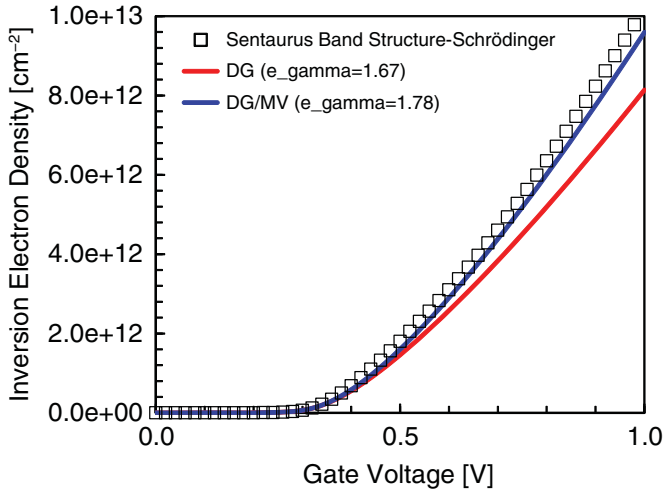
to the solutions from the Schrödinger equations is required to extract the corresponding quantization parameters.

The 1D Schrödinger solver in Sentaurus Band Structure provides a reference in terms of carrier density distribution in the inversion layer, and the fitting factor (γ) in the density gradient model then is fitted by minimizing the root-mean-square error of the charge integral over a range of applied gate voltages between these two approaches. The quantization parameter depends on the carrier polarity, the lattice orientation, the material composition, the layer thickness, the channel geometry, and the strain. The `QuantumPotentialParameters` parameter set provides the fitted γ values for simulations of unstrained $In_{1-x}Ga_xAs$ alloys, for both the bulk configuration and the double-gate configuration with channel thicknesses of 10 nm and 20 nm.

For $In_{1-x}Ga_xAs$ materials, you cannot neglect the electron occupancy in the L-valley at a higher gate voltage. Therefore, including the multivalley band structure model in the simulation allows you to determine the electron density-of-states more accurately.

The parameters for the `MultiValley` model in the `InGaAs.par` file with mole fraction dependency have been reviewed following the literature [41][48] and the band-structure calculation using the empirical pseudopotential method.

Figure 42 *Inversion electron densities for a 10 nm thick double-gate (110) MOS structure with $In_{0.53}Ga_{0.47}As$ channel: red line is simulated with the density gradient model and blue line is simulated with the multivalley and density gradient models; in both cases, e_gamma parameters have been optimized*



As shown in Figure 42, the electron charge integral in a 10 nm thick $In_{0.53}Ga_{0.47}As$ double-gate structure simulated with the multivalley and density gradient models fits better to the result obtained with the Schrödinger solver of Sentaurus Band Structure, compared to the one with the density gradient model only. A similar conclusion can be drawn when observing the carrier distributions at a 1 V gate voltage along the 10 nm channel, as shown in Figure 43.

Figure 43 *Carrier distributions at gate voltage of 1 V for a 10 nm thick double-gate (110) MOS structure with $In_{0.53}Ga_{0.47}As$ channel: red line is simulated with the density gradient model and blue line is simulated with the multivalley and density gradient models; in both cases, e_gamma parameters have been optimized*

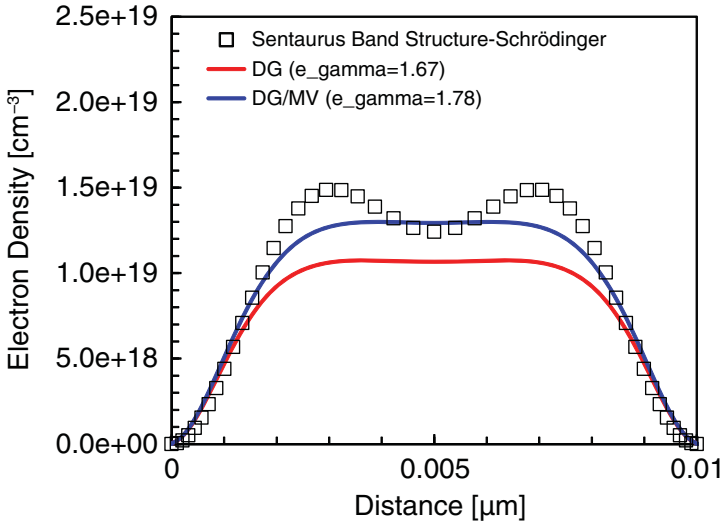


Figure 44 shows the tabulated values of fitted `e_gamma` and `h_gamma` parameters using the density gradient and multivalley models, which are provided in the `InGaAs.par` file for bulk (100) and (110) orientations. It is evident that there is no orientation dependency for the electron quantization parameter, and the compositional variation is also less distinct compared to the hole quantization parameter. The quantization parameter is also a function of layer thickness, as shown in Figure 45.

It is important to implement the corresponding quantization parameters designated for a specific layer thickness during simulations, since the discrepancy on the charge distribution is very susceptible to thin layers. For example, the `e_gamma` value optimized for a 10 nm channel width could still be used for quantum correction on a 20 nm thick channel, but with a slightly larger error. However, the same value might be only appropriate to use for thin layers down to 8 nm. If the layer thickness is less than 8 nm, you should perform a quantization calibration again to extract the correct quantization parameter.

Figure 44 *Fitted gamma values versus Ga mole fraction for bulk MOS structures with $In_{1-x}Ga_xAs$ alloys using density gradient and multivalley models*

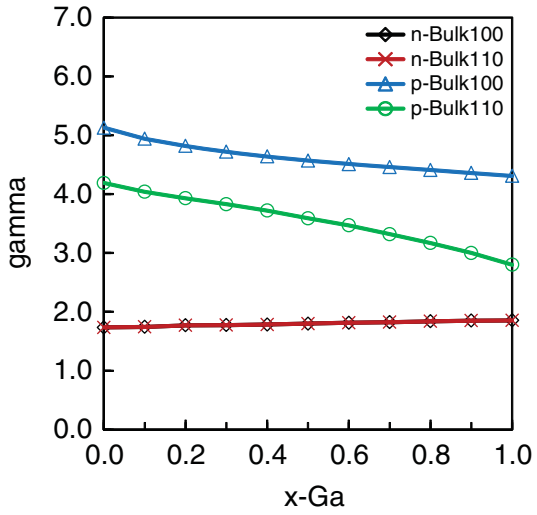
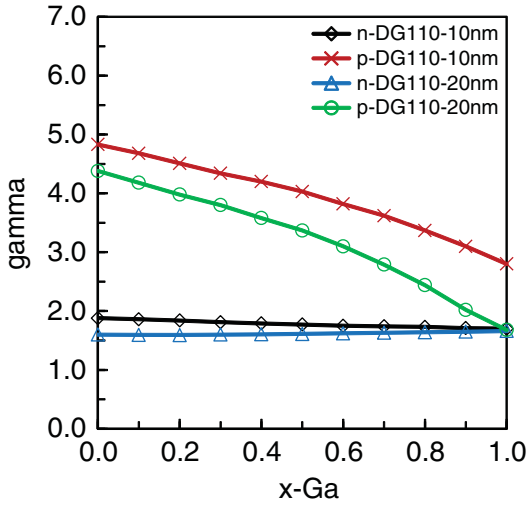


Figure 45 *Fitted gamma values versus Ga mole fraction for 10 nm and 20 nm double-gate MOS structures with $In_{1-x}Ga_xAs$ alloys using density gradient and multivalley models*



Low-Field Bulk Mobility and Its Doping Dependency

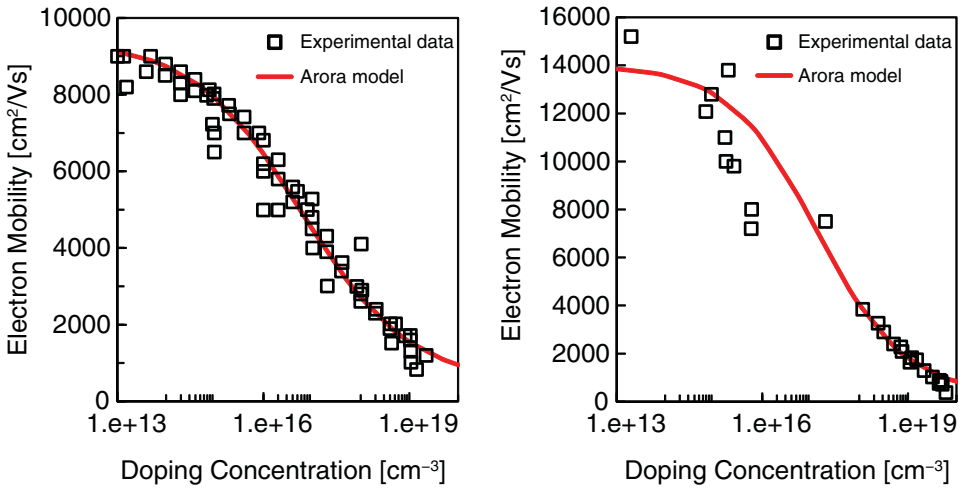
Low-field bulk mobility is a critical parameter to determine the current–voltage characteristics of InGaAs metal–insulator–semiconductor (MIS) devices, and it is dependent on mole fraction, temperature, doping, inversion charge, and channel direction for unstrained InGaAs materials.

Accurate theoretical models accounting for diverse scattering mechanisms can be complex and might apply only to limited conditions. Therefore, empirical fitting based on available experimental data can be an alternative.

The coefficients in the `ConstantMobility` and `DopingDependence` parameter sets of the `InGaAs.par` file have been reviewed and implemented following the empirical mobility model suggested in [60].

Reference [60] provides calibrated parameters in the empirical Caughey–Thomas mobility model for temperature and doping dependency, for unstrained GaAs, InAs, and $In_{0.53}Ga_{0.47}As$ materials. These parameters are transformed to the parameters of the Arora model for use in Sentaurus Device. Figure 46 demonstrates the doping dependency of electron mobility for unstrained GaAs and $In_{0.53}Ga_{0.47}As$ using the Arora model.

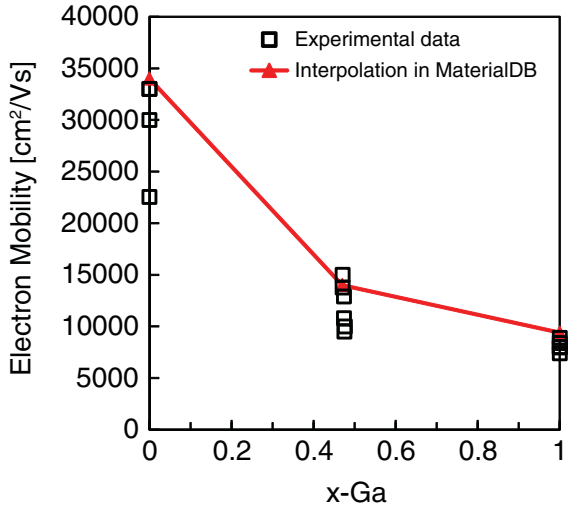
Figure 46 Doping-dependent low-field electron mobility for (left) GaAs and (right) $In_{0.53}Ga_{0.47}As$ at 300 K; red lines are computed using the Arora model with fitted parameters from [60]



The doping dependency of low-field mobility for $In_{1-x}Ga_xAs$ alloys, apart from $In_{0.53}Ga_{0.47}As$ material, is usually less investigated. Therefore, a two-sectional linear interpolation scheme is used, based on InAs, $In_{0.53}Ga_{0.47}As$, and GaAs for each parameter in the Arora model.

Figure 47 shows the compositional dependency of the low-field electron mobility for $In_{1-x}Ga_xAs$ alloys suggested in the `InGaAs.par` file with some experimental data.

Figure 47 Low-field electron mobility at 300 K versus Ga mole fraction for $In_{1-x}Ga_xAs$ alloys; experimental data is reprinted from [61]; red line is the interpolation curve suggested in the InGaAs.par file



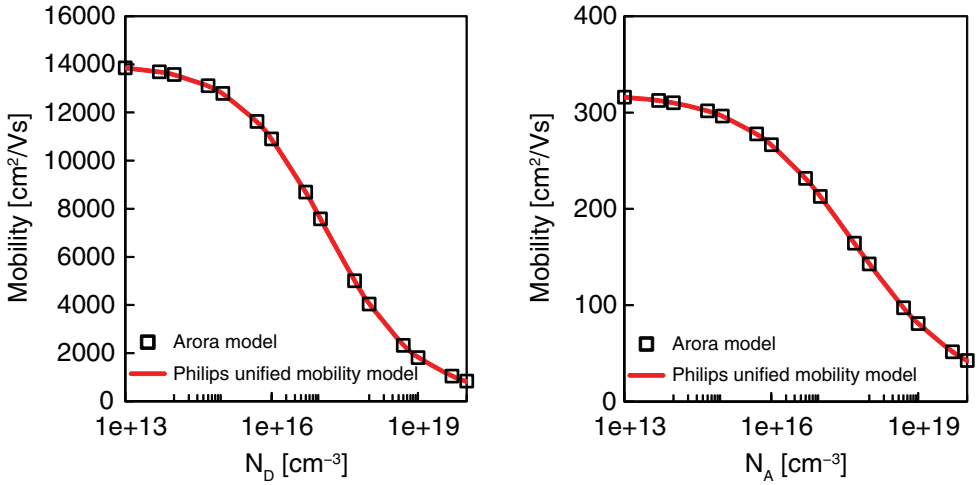
Philips Unified Mobility Model

This model describes both majority and minority carrier mobility in bulk, taking into account phonon scattering, impurity scattering, carrier–carrier scattering, and screening of scattering processes. In the `MaterialDB` folder, the parameters for majority carrier mobility in InAs, $In_{0.53}Ga_{0.47}As$, and GaAs materials have been calibrated without considering the minority impurity scattering where other coefficients, such as the clustering functions and the screening parameters, are preserved as the default ones from silicon material.

The maximum carrier mobility (μ_{max}), the minimum carrier mobility (μ_{min}), and the exponent (θ) for $In_{1-x}Ga_xAs$ alloys are transformed from the corresponding `ConstantMobility` and Arora models. Then, the reference density (N_{ref}) and the exponent (α) coefficients can be fitted to the doping-dependent mobility given by the Arora model within a doping range between 1.0e13 cm^{-3} and 1.0e20 cm^{-3} for GaAs and $In_{0.53}Ga_{0.47}As$. Since the intrinsic density of InAs is relatively larger compared to GaAs and $In_{0.53}Ga_{0.47}As$, the fitting range for InAs is limited to the range from 1e15 cm^{-3} to 1e20 cm^{-3}.

Figure 48 presents the electron and hole mobility in $In_{0.53}Ga_{0.47}As$ material at various doping levels for the Philips unified mobility model with fitted coefficients.

Figure 48 *Doping-dependent low-field mobility for (left) electron and (right) hole majority carriers in $In_{0.53}Ga_{0.47}As$ alloy at 300 K; red lines are computed with the Philips unified mobility model with fitted parameters suggested in the InGaAs.par file*



Inversion Layer Mobility

This section discusses inversion layer mobility.

Inversion and Accumulation Layer Mobility Model

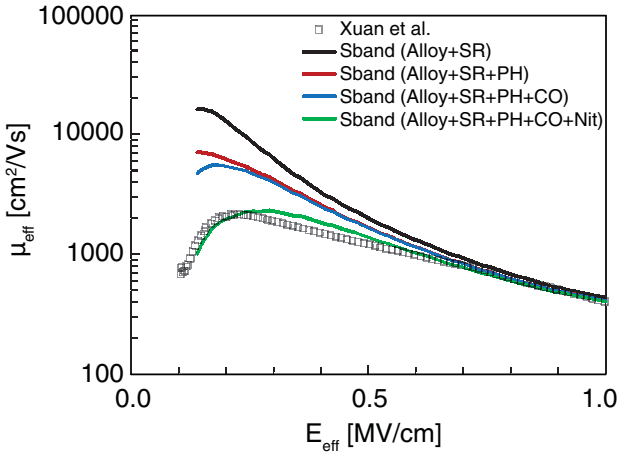
In a MOSFET device, the surface roughness and the surface phonon scattering can degrade carrier mobility at the proximity of a semiconductor–insulator interface due to a high electric field, normal to the channel interface imposed across the channel region. Since $In_{1-x}Ga_xAs$ alloys have the potential to realize n-MOSFET devices, parameter calibration with respect to available experimental data is required to establish a more accurate mobility degradation model.

This section describes how the parameters in the `MaterialDB` folder have been calibrated for the inversion and accumulation layer mobility (`IALMob`) model.

The calibration methodology involves two main steps and concerns mainly the electron mobility in the inversion layer. First, with some initial calibrations with respect to the experimental data to account for the process variations, the 1D mobility calculator of Sentaurus Band Structure provides a reference in terms of mobility in the inversion layer for the entire mole fraction range. The experimental data used in this step was reported by Xuan *et al*. [62] on an inversion-type n-channel $In_{0.53}Ga_{0.47}As$ MOSFET with 8 nm thick atomic layer deposition (ALD) Al_2O_3 as the gate dielectric. Second, the coefficients of each scattering mechanism of the `IALMob` model are extracted by comparing the mobility values between these two simulation approaches.

Figure 49 shows the dispersion curves of the mobility field in the $In_{0.53}Ga_{0.47}As$ inversion layer simulated using the 1D mobility calculator of Sentaurus Band Structure with various combinations of the alloy, surface roughness (SR), phonon (PH), and Coulomb (CO) scattering models, in which parameters Δ and Λ of the surface-roughness power-spectrum model with exponential density function have been optimized for the effective electric field in the range 0.6–1.0 MV/cm. Afterwards, the mobility offset with respect to experimental data, after considering all the available scattering models such as alloy, surface roughness, phonon, and Coulomb scattering, as well as the fixed charge trap model, can be reduced further in the range 0.1–0.4 MV/cm by adjusting the interface charge density.

Figure 49 Effective electron mobility versus effective electric field for $In_{0.53}Ga_{0.47}As$ (p-doping=1e17 cm^{-3} in channel) MOSFET with ALD Al_2O_3; simulation results are obtained using 1D mobility calculator of Sentaurus Band Structure with various combinations of scattering models, in which the surface roughness parameters (Δ=2.2e-7 cm; Λ=4.6e-8 cm) and the interface charge density (N_{it}=4.3e12 cm^{-2}) have been calibrated; experimental data is taken from [62]



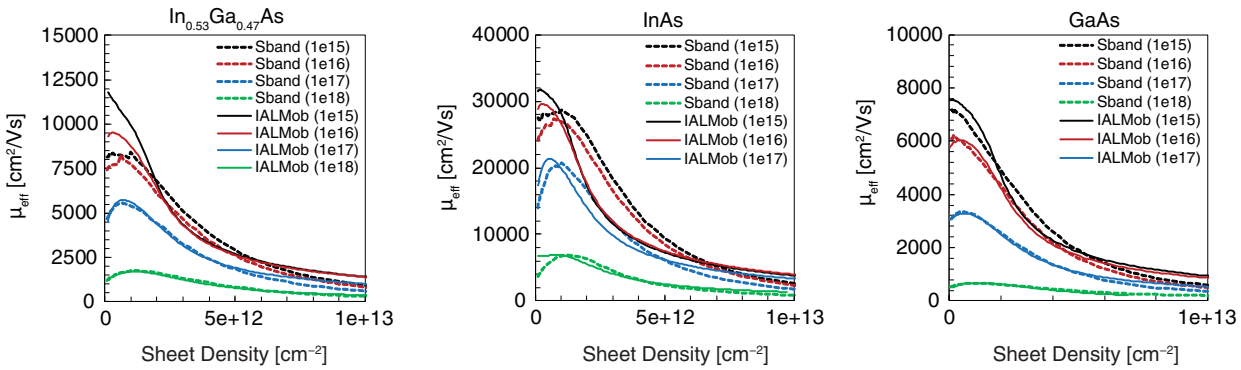
For the `IALMob` model, the calibration has been performed with the parameters of 3D phonon scattering and 3D Coulomb scattering taken from the constant mobility model and the Philips unified mobility model, respectively, and the 2D and 3D parts of phonon scattering are combined with `PhononCombination=1`. Since it is necessary to distinguish the contribution of various scattering mechanisms in the `IALMob` model, first 2D Coulomb scattering is disregarded by designating a large value in factors such as $D_{1,inv}$ and $D_{2,inv}$. Then, the resulting effective mobility values are compared to Sentaurus Band Structure results, for which only surface roughness scattering and phonon scattering are activated, for p-doping between 1e15 and 1e18 cm^{-3}.

At this step, the reviewed parameters principally include coefficients accounting for 2D phonon scattering (`B`, `C`, and λ) and those representing surface roughness scattering (δ, `A*`, and λ_{sr}), where the λ and λ_{sr} coefficients describe the doping dependencies of phonon and surface roughness scattering, respectively. Other parameters are kept as the ones for silicon material.

After the coefficients for surface roughness scattering and 2D phonon scattering are determined, the parameters for 2D Coulomb scattering ($D_{1,inv}$, $D_{2,inv}$, and $\nu_{1,inv}$) can be fitted with respect to Sentaurus Band Structure results when Coulomb scattering is implemented.

Figure 50 compares the effective electron mobility versus sheet density at various p-doping levels in the channel region between the `IALMob` model with calibrated parameters and Sentaurus Band Structure results for $In_{0.53}Ga_{0.47}As$, InAs, and GaAs. Regarding the interpolation scheme for the `IALMob` model in $In_{1-x}Ga_xAs$ alloys, a two-sectional linear curve is used, based on $In_{0.53}Ga_{0.47}As$, InAs, and GaAs for each parameter previously mentioned.

Figure 50 Effective electron mobility versus sheet density for (left) $In_{0.53}Ga_{0.47}As$, (middle) InAs, and (right) GaAs MOSFET with ALD Al_2O_3 at various p-doping levels in the channel region; dotted lines are obtained using 1D mobility calculator of Sentaurus Band Structure in combination with surface roughness, phonon, and Coulomb scattering models, and the alloy scattering model is further considered for $In_{0.53}Ga_{0.47}As$; solid lines are obtained using the IALMob model with calibrated parameters suggested in the InGaAs.par file



Interface Charge Mobility Model

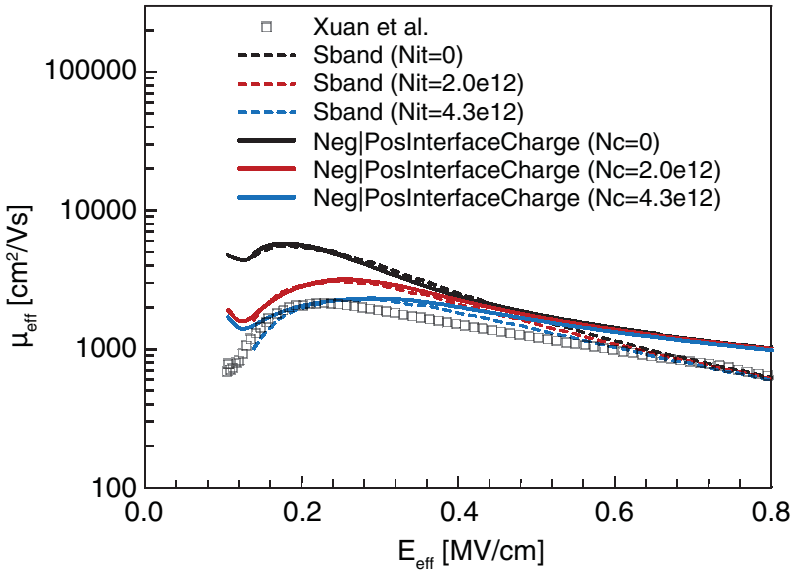
Following the results of the mobility calibration for the `IALMob` model, the mobility degradation model due to interface charges has been fine-tuned, including the `NegInterfaceCharge` and `PosInterfaceCharge` models in Sentaurus Device. As shown in Figure 49, the Sentaurus Band Structure mobility calculator can fit well to the experimental data from Xuan *et al*. [62] when interface charge density (N_{it}) is fitted to be 4.3e12 cm^{-2}.

To calibrate parameters such as μ_l, C_{trans}, l_{crit}, and E_0 for the `NegInterfaceCharge` and `PosInterfaceCharge` models, the mobility values for two more N_{it} levels, 0 and 2.0e12, are also computed using Sentaurus Band Structure.

Figure 51 presents the curves of mobility versus field for $In_{0.53}Ga_{0.47}As$ material when implementing the `IALMob` model and the negative and positive interface charge mobility models with calibrated parameters with respect to Sentaurus Band Structure results. Similar

to other mobility models, the interpolation scheme for the `NegInterfaceCharge` and `PosInterfaceCharge` models in $In_{1-x}Ga_xAs$ alloys uses a two-sectional linear curve based on InAs, $In_{0.53}Ga_{0.47}As$, and GaAs.

Figure 51 *Effective electron mobility versus effective electric field for $In_{0.53}Ga_{0.47}As$ (p-doping=1e17 cm^{-3} in channel) MOSFET with ALD Al_2O_3; dotted lines are obtained using 1D mobility calculator of Sentaurus Band Structure with alloy, surface roughness, phonon, and Coulomb scattering models, as well as interface trap model with charge density of 0, 2.0e12, and 4.3e12 cm^{-2}; solid lines are obtained using the NegInterfaceCharge and PosInterfaceCharge models with calibrated parameters suggested in the InGaAs.par file*



Strained Low-Field Mobility

This section discusses strained low-field mobility.

Multivalley Electron Mobility Model

Mechanical stress in devices can change the band structure and carrier mobility, and is considered an approach to enhance mobility. The multivalley electron mobility (`eSubband`) model with default parameters has been tested on several $In_{1-x}Ga_xAs$-based FETs on which either uniaxial or biaxial stress is applied [63][64][65][66]. The simulations reproduce the trend of electron mobility enhancement for tensile stress. However, the enhancement generally tends to be lower than for experimental data.

Since the investigated device structures and conditions are very diverse and incomplete, a more unified and better set of model parameters has not been established (for example, to

consider the dependencies on mole fraction and geometry for $In_{1-x}Ga_xAs$ materials) and default parameters have been kept.

Piezoresistance Mobility Model

Due to a lack of published experimental results for $In_{1-x}Ga_xAs$ alloys, the related piezoresistive coefficients are difficult to determine. Nainani *et al*. [67] demonstrated the conductivity extraction for n-$In_{0.2}Ga_{0.8}As$ material using the transfer length method and suggested a similar value for the longitudinal piezoresistive coefficient for n-$In_{0.2}Ga_{0.8}As$ as for n-Si.

Basic Properties of Silicon Germanium Devices

The `SiliconGermaniumc100.par` and `SiliconGermaniumc110.par` files contain the parameters for silicon germanium (SiGe) MOS devices. See Content of Parameter Files on page 11.

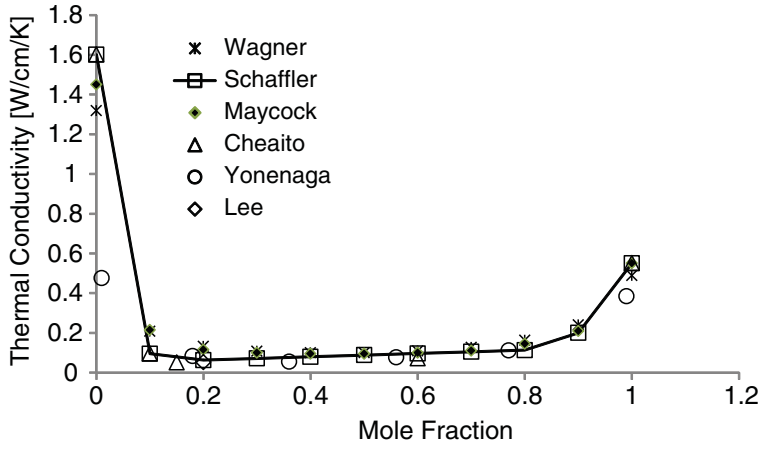
Permittivity

The low-frequency relative dielectric constant values are interpolated linearly between the corner values of 11.7 (silicon) and 16.2 (germanium) for each Ge mole fraction (see [68]).

Thermal Conductivity

The mole fraction dependency of the thermal conductivity is extracted from data from Schaffler [68] (see Figure 52). Data is implemented piecewise linearly in the parameter file.

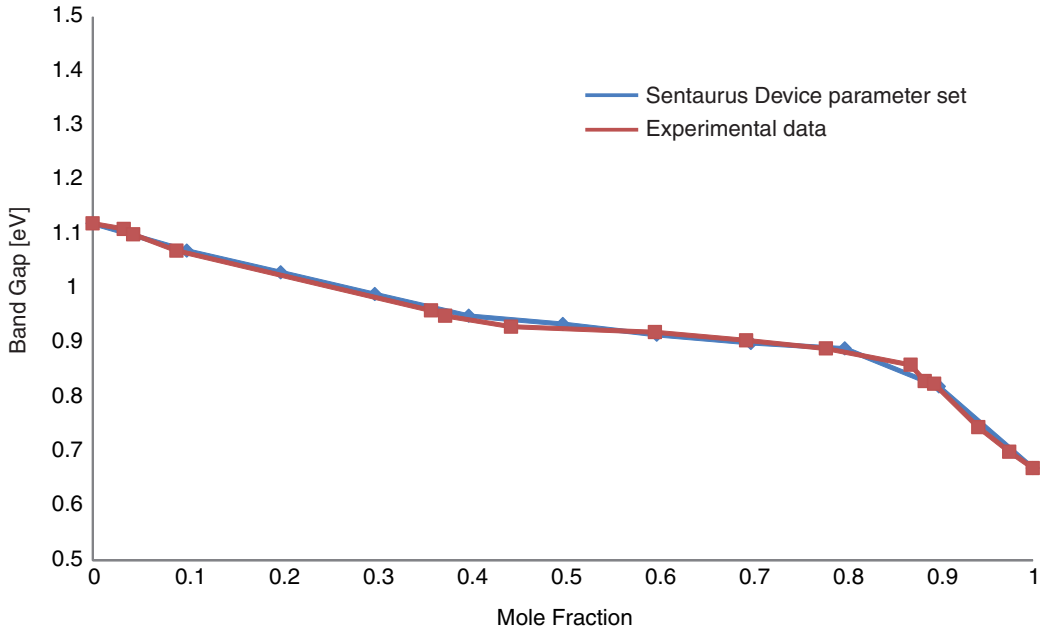
Figure 52 Thermal conductivity versus mole fraction [68][69][70][71][72][73]; solid line is the implementation in the parameter file



Band Structure

Band structure data (band gap and affinity) is extracted from [74] (see Figure 53 and Figure 54). The bandgap narrowing is extracted from Jain and Roulston [16] (see Figure 55) where the `OldSlotboom` model is used for fitting (see the *Sentaurus™ Device User Guide*).

Figure 53 Band gap as a function of mole fraction



Linear interpolation between silicon and germanium values is performed for the prefactor of the `OldSlotboom` model to describe the mole fraction dependency. All data is implemented piecewise linearly in the parameter file.

Figure 54 Conduction and valence bands as a function of mole fraction

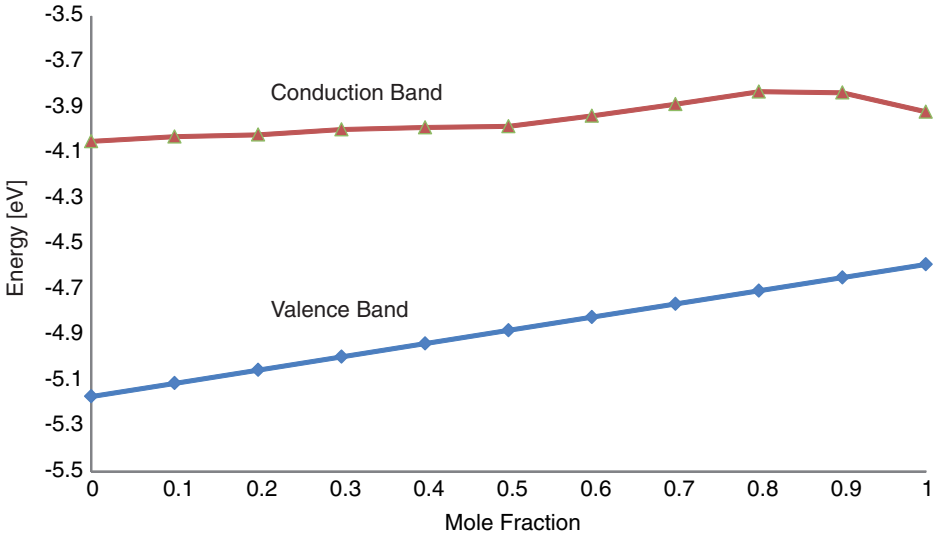
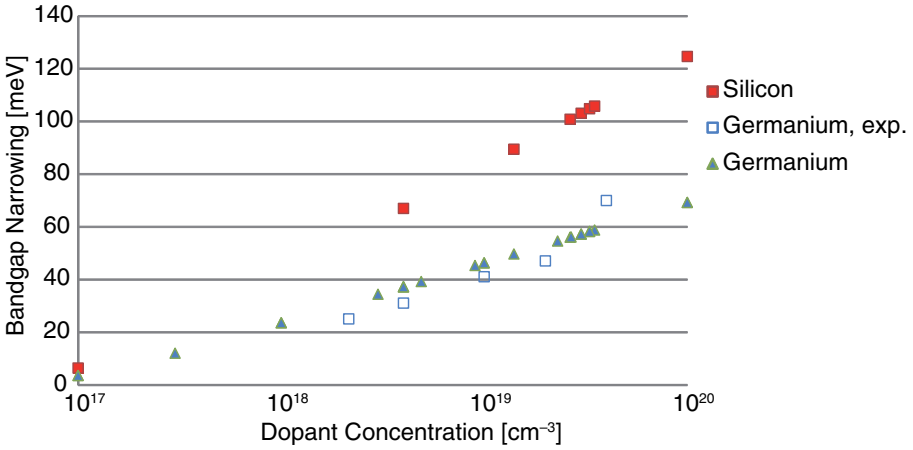


Figure 55 Bandgap narrowing for silicon and germanium; a prefactor of 5 meV for the OldSlotboom model is used

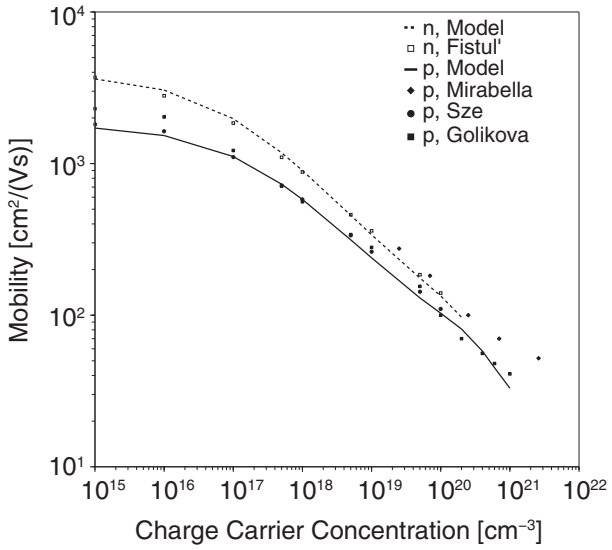


Bulk Low-Field Mobility

The bulk low-field mobility `PhuMob` model parameters for silicon, SiGe, and germanium are extracted from different measurements with the focus on the data from Golikova *et al*. [75]

and Fistul' *et al*. [76]. Figure 56 shows the results of this calibration for germanium. The model parameters are implemented piecewise linearly.

Figure 56 *Measurements from different sources [75][76][77][78], with [79] used as the reference measurement, for the doping dependency of electron and hole mobility in Ge and the implementation in the Sentaurus Device parameter files (lines)*



Ballistic Mobility

Quasiballistic transport is important and should be considered when scaling devices towards short channels. For this, it is required to extend the drift-diffusion (DD) model by combining an additional ballistic mobility with the standard drift-diffusion mobility. In Sentaurus Device, the ballistic mobility can be described by either the channel length-dependent model or the kinetic velocity model (KVM) [80]. The latter has the advantage of avoiding explicit channel length dependence and is used to extract the corresponding ballistic mobility parameters.

The ballistic mobility model parameters are calibrated with the solution of the subband-based Boltzmann transport equation (subband-BTE) as reference. The fitting parameter (`k`) in the KVM is mostly determined by fitting the on-state current at low field over a range of channel lengths between these two approaches. Moreover, the parameters `beta0` and the saturation velocity `vsat0` in the high-field saturation model are adjusted as well by fitting the on-state current at high field. In this way, ballistic mobility can be separated from high-field drift-diffusion mobility. Note that, the ballistic mobility model parameters and the high-field saturation model parameters can depend on the carrier polarity, the channel geometry, the layer thickness, the material composition, and the strain.

The devices used for parameter extraction are 3D square SiGe n-type and p-type MOSFET nanowires with a cross-sectional area of 5 nm × 5 nm. The gate length (L_{gate}) varies from 200 nm to 7 nm, and the channel direction is along the <110> crystal orientation.

To activate the KVM, specify the following command in the `Physics` section of the Sentaurus Device command file:

```
Mobility (
   BalMob ( KVM(Full) Fermi Frensley )
)
```

Here, the thermionic emission and free carrier acceleration terms are both considered in the KVM. As the first step, the long-channel mobility is calibrated to the subband-BTE simulation at on-state using the inversion and accumulation layer mobility (`IALMob`) model in drift-diffusion. Second, the channel access resistance (R_0) is extracted from the gate length dependence of the total resistance using the equation $R = R_0 + R'L_{\text{gate}}$. The difference of R_0 between the subband-BTE and drift-diffusion simulations can be added in the Sentaurus Device simulation either as an additional lumped resistance or by modifying the mobility model parameters in source/drain regions. After that, the parameters of the KVM and the high-field saturation model are optimized to match the on-state current at low-drain and high-drain bias voltages. It is likely that you must iterate the calibration procedures previously mentioned to optimize the parameters. For a detailed description, see Chapter 4 on page 46.

Figure 57 shows the on-state drain current obtained using the drift-diffusion simulation with the KVM at low-drain and high-drain bias voltages. The parameters of the KVM and the high-field saturation model have been optimized with respect to the simulation results of the Subband-BTE solver over a wide range of channel lengths and for germanium mole fractions from 0 to 1, as shown in Figure 58 on page 102.

Note that the slope of resistance as a function of gate length is found to vary nonlinearly at short channels when the KVM is implemented, and it then leads to a slightly lowered fitting accuracy for channels shorter than 10 nm. This behavior is under investigation and might result in a parameter update.

Figure 57 *Drain current versus channel length for 5 nm × 5 nm Si (left) n-type MOSFET nanowire and (right) p-type MOSFET nanowire, at low-drain and high-drain bias voltages; squares obtained using Subband-BTE solver in combination with surface roughness, phonon, and Coulomb scattering models; solid lines obtained using KVM with calibrated parameters for electrons (k=6.3, Tfactor=1.52, and Bfactor=0.3 in KVM; beta0=0.74 and vsat0=4e7 in high-field saturation model) and for holes (k=5.4, Tfactor=1.52, and Bfactor=0.3 in KVM; beta0=1.0 and vsat0=2e7 in high-field saturation model)*

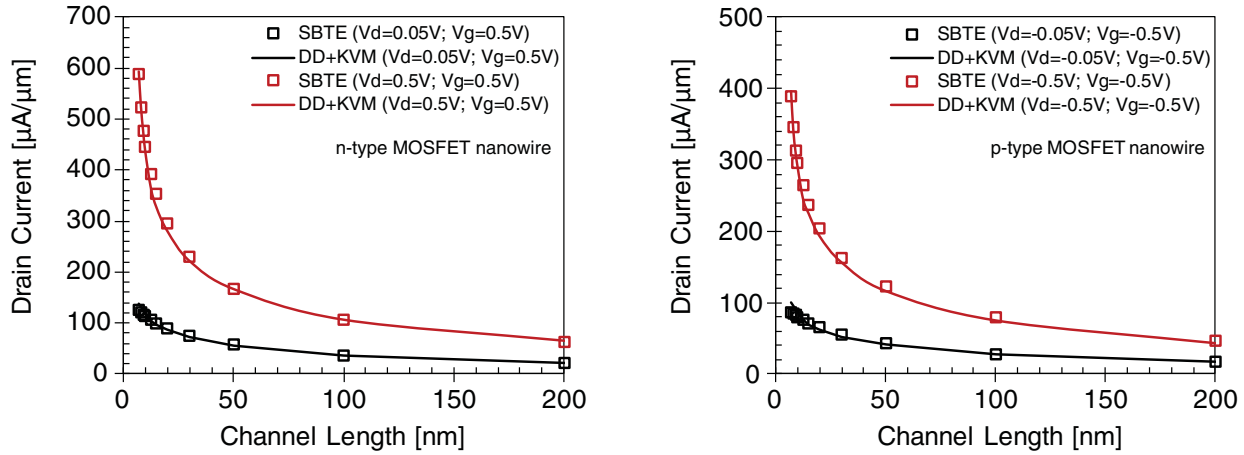
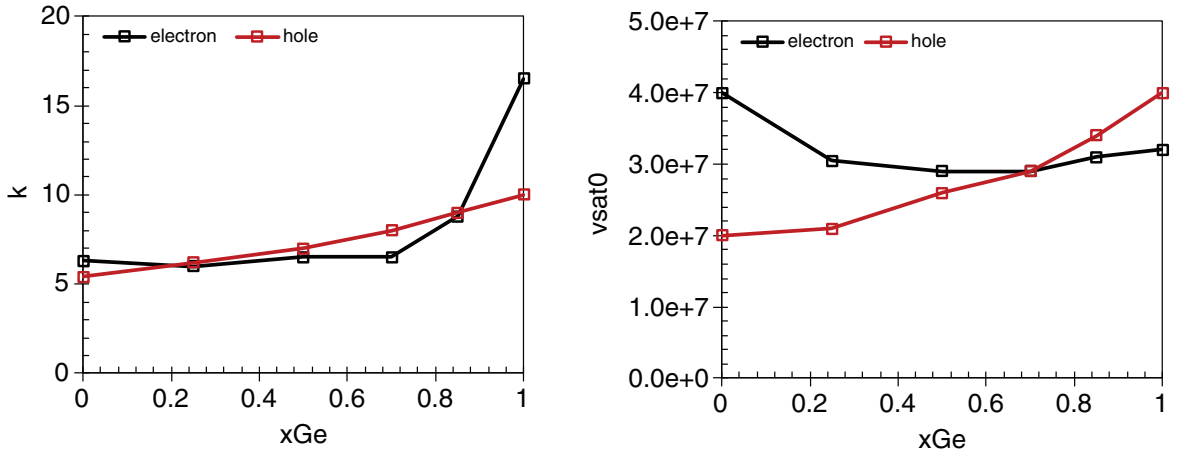


Figure 58 *Calibrated parameters (left) k in KVM and (right) vsat0 in high-field saturation model versus Ge mole fraction for 5 nm × 5 nm SiGe n-type and p-type MOSFET nanowires*



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A

Physical Model Interface Models

This appendix describes the physical model interface (PMI) models of Advanced Calibration Device.

Features of the MCmob and SBmob Models

The PMI models `MCmob` and `SBmob` of Sentaurus Device are based on nonlinear relations between the mobility enhancement and the stress components where the coefficients of the equations are extracted from reference tools such as Sentaurus Band Structure (`SBmob`) or Sentaurus Device Monte Carlo (`MCmob`). The corresponding command file sections are explained in Chapter 2 on page 23.

The features of the `MCmob` and `SBmob` models are as follows:

- Calculates the electron and hole mobility enhancement by stress (in the range [-6, 6] GPa) for SiGe at different mole fractions for different configurations. These configurations are defined by the channel orientation (in the crystal coordinates `X`, `Y`, `Z`) and the wafer crystallographic surface orientation.
- Determines the crystallographic surface or interface orientation automatically.

These models are designed for semiconductor–isolator interfaces as in a metal–insulator–semiconductor (MIS) structure where the bulk symmetry typical for the standard piezoresistance model is broken. This makes it necessary to determine the direction of the interface normal with respect to the device coordinate system (`x`, `y`, `z`).

Note:

This detection is not performed fully and automatically, and you must specify parameters in the Sentaurus Device parameter file (see Parameters).

Note:

If the absolute value of the orthogonal normal stress component is larger than 6 GPa, then the value is truncated at 6 GPa.

Parameters

You use the same parameters and parameter syntax for both the `MCmob` and `SBmob` models.

Table 16 Parameters of the MCmob and SBmob models

Parameter	Description
`autoorientation`	Switches on the detection of the crystallographic surface orientation. Valid options are: • `0`: No detection of crystallographic surface orientation, but crystallographic surface orientation is defined by `waferori`. • `1`: Detects crystallographic surface orientation. • `2`: Detects crystallographic surface orientation and adaptation of local coordinate system (coordinate system used to match the stress components to the MIS plane orientation and channel direction at each mesh node) for FinFET-like structures with <110> channel orientation. • `3`: Detects crystallographic surface orientation; local coordinate systems are defined using parameters in the Sentaurus Device parameter file. Default: `0`
`channelori`	Sets the channel direction. Valid options are: • `100`: <100> channel direction. • `110`: <110> channel direction. Default: `110`
`ConstantMoleFraction`	Switches between reading the mole-fraction distribution from a TDR file and setting the constant mole fraction in the parameter file. Valid options are: • `0`: Read from TDR file. • `1`: User sets mole fraction in Sentaurus Device parameter file.
`ConstantStress`	Switches between reading the distribution of stress components from a TDR file and setting the constant stress components in the parameter file. Valid options are: • `0`: Read from TDR file. • `1`: User sets constant stress components in Sentaurus Device parameter file.
`coordinatesystem`	Specifies the global coordinate system in Sentaurus Device. Default: `0`

Table 16 Parameters of the MCmob and SBmob models (Continued)

Parameter	Description
`corrections`	Controls interpolation in the stress space. Valid options are: • `0`: Results in a linear superposition of the mobility response to the uniaxial stress components. • `1`: Switches on cross-corrections (terms with mixed stress components). You must define the cross-corrections in the parameter file (default: all cross-corrections are set to zero). Default: `0` **Note:** This parameter is available only for the `MCmob` model.
`DeltaMuPrefactorx`	Scales the mobility enhancement caused by stress in the current flow direction. **Note:** This parameter is available only for the `MCmob` model.
`DeltaMuPrefactory`	Scales the mobility enhancement caused by stress in the normal to the MIS interface direction. **Note:** This parameter is available only for the `MCmob` model.
`DeltaMuPrefactorz`	Scales the mobility enhancement caused by stress in the direction normal to the current flow and in-plane with the MIS interface. **Note:** This parameter is available only for the `MCmob` model.
`DeltaMuPrefactors100c100x`	Scales the mobility enhancement caused by stress in the current flow direction for channel direction <100> and surface orientation (100). **Note:** This parameter is available only for the SBmob model.
`DeltaMuPrefactors100c100y`	Scales the mobility enhancement caused by stress normal to the MIS interface for channel direction <100> and surface orientation (100). **Note:** This parameter is available only for the SBmob model.

Table 16 Parameters of the MCmob and SBmob models (Continued)

Parameter	Description
`DeltaMuPrefactors100c100z`	Scales the mobility enhancement caused by stress in-plane with the MIS interface and normal to the current flow direction for channel direction <100> and surface orientation (100). **Note:** This parameter is available only for the SBmob model.
`DeltaMuPrefactors100c110x`	Scales the mobility enhancement caused by stress in the current flow direction for channel direction <110> and surface orientation (100). **Note:** This parameter is available only for the SBmob model.
`DeltaMuPrefactors100c110y`	Scales the mobility enhancement caused by stress normal to the MIS interface for channel direction <110> and surface orientation (100). **Note:** This parameter is available only for the SBmob model.
`DeltaMuPrefactors100c110z`	Scales the mobility enhancement caused by stress in-plane with the MIS interface and normal to the current flow direction for channel direction <110> and surface orientation (100). **Note:** This parameter is available only for the SBmob model.
`DeltaMuPrefactors110c110x`	Scales the mobility enhancement caused by stress in the current flow direction for channel direction <110> and surface orientation (110). **Note:** This parameter is available only for the SBmob model.
`DeltaMuPrefactors110c110y`	Scales the mobility enhancement caused by stress normal to the MIS interface for channel direction <110> and surface orientation (110). **Note:** This parameter is available only for the SBmob model.
`DeltaMuPrefactors110c110z`	Scales the mobility enhancement caused by stress in-plane with the MIS interface and normal to the current flow direction for channel direction <110> and surface orientation (110). **Note:** This parameter is available only for the SBmob model.
`MoleFraction`	SiGe mole fraction when `ConstantMoleFraction=1`.

Table 16 Parameters of the MCmob and SBmob models (Continued)

Parameter	Description
`referenceplanecoordinatex1`	Coordinate of the first reference plane in the x-direction of the device coordinate system (yz plane, default: 1e6).
`referenceplanecoordinatex2`	Coordinate of the second reference plane in the x-direction of the device coordinate system (yz plane, default: 1e6).
`referenceplanecoordinatey1`	Coordinate of the first reference plane in the y-direction of the device coordinate system (xz plane, default: 1e6).
`referenceplanecoordinatey2`	Coordinate of the second reference plane in the y-direction of the device coordinate system (xz plane, default: 1e6).
`referenceplanecoordinatez1`	Coordinate of the first reference plane in the z-direction of the device coordinate system (xy plane, default: 1e6).
`referenceplanecoordinatez2`	Coordinate of the second reference plane in the z-direction of the device coordinate system (xy plane, default: 1e6).
`referenceplanesystemx1`	Defines the local coordinate system for the reference plane parallel to the yz plane at the position x=`referenceplanecoordinatex1` (default: –1).
`referenceplanesystemx2`	Defines the local coordinate system for the reference plane parallel to the yz plane at the position x=`referenceplanecoordinatex2` (default: –1).
`referenceplanesystemy1`	Defines the local coordinate system for the reference plane parallel to the xz plane at the position y=`referenceplanecoordinatey1` (default: –1).
`referenceplanesystemy2`	Defines the local coordinate system for the reference plane parallel to the xz plane at the position y=`referenceplanecoordinatey2` (default: –1).
`referenceplanesystemz1`	Defines the local coordinate system for the reference plane parallel to the xy plane at the position y=`referenceplanecoordinatez1` (default: –1).
`referenceplanesystemz2`	Defines the local coordinate system for the reference plane parallel to the xy plane at the position y=`referenceplanecoordinatez2` (default: –1).
`Stress_xx`	S_{xx} stress component in the device coordinate system when `ConstantStress=1`.

Table 16 Parameters of the MCmob and SBmob models (Continued)

Parameter	Description
`Stress_yy`	S_{yy} stress component in the device coordinate system when `ConstantStress=1`.
`Stress_zz`	S_{zz} stress component in the device coordinate system when `ConstantStress=1`.
`UpperLimits110c110`	Sets the maximum polynomial value $f = \gamma - 1$ for the s100c110 configuration. Default: 10. See Parameter Interface and Equations of SBmob Model on page 122. **Note:** This parameter is available only for the SBmob model.
`UpperLimits110c110`	Sets the maximum polynomial value $f = \gamma - 1$ for the s110c110 configuration. Default: 10. See Parameter Interface and Equations of SBmob Model on page 122. **Note:** This parameter is available only for the SBmob model.
`UpperLimits100c100`	Sets the maximum polynomial value $f = \gamma - 1$ for the s100c100 configuration. Default: 10. See Parameter Interface and Equations of SBmob Model on page 122. **Note:** This parameter is available only for the SBmob model.
`waferori`	Sets the wafer orientation. Valid options are: • `100`: (001) wafer orientation. • `110`: (110) wafer orientation. Default: `100`

Note:

The set of `DeltaMuPrefactor*` parameters for different surface and channel orientations (for example, `DeltaMuPrefactors100c110y`) is available only for the `SBmob` model. For the `MCmob` model, only a reduced set is available (for example, `DeltaMuPrefactory`).

Specifying the Crystal Orientation Matrix

The conversion between the device Cartesian basis {$\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, $\hat{\mathbf{z}}$} and crystal Cartesian basis {$\hat{\mathbf{X}}$, $\hat{\mathbf{Y}}$, $\hat{\mathbf{Z}}$} is performed using the orthonormal matrix **Q**. The latter is named the crystal orientation matrix.

$$\mathbf{Q} = \begin{bmatrix} \hat{\mathbf{X}} \cdot \hat{\mathbf{x}} & \hat{\mathbf{X}} \cdot \hat{\mathbf{y}} & \hat{\mathbf{X}} \cdot \hat{\mathbf{z}} \\ \hat{\mathbf{Y}} \cdot \hat{\mathbf{x}} & \hat{\mathbf{Y}} \cdot \hat{\mathbf{y}} & \hat{\mathbf{Y}} \cdot \hat{\mathbf{z}} \\ \hat{\mathbf{Z}} \cdot \hat{\mathbf{x}} & \hat{\mathbf{Z}} \cdot \hat{\mathbf{y}} & \hat{\mathbf{Z}} \cdot \hat{\mathbf{z}} \end{bmatrix}$$

where ($\hat{\mathbf{X}}$, $\hat{\mathbf{Y}}$, $\hat{\mathbf{Z}}$) are the directions (unit vectors) of the crystal basis, that is *channel*, *normal* to MIS, and *width*. The PMI model allows choosing between six different **Q** matrices (crystal coordinate systems):

- `coordinate system=0` (default):

$$\mathbf{Q} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- `coordinatesystem=1`:

$$\mathbf{Q} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

- `coordinatesystem=2`:

$$\mathbf{Q} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- `coordinatesystem=3`:

$$\mathbf{Q} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

- `coordinatesystem=4`:

$$\mathbf{Q} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

- `coordinatesystem=5`

$$\mathbf{Q} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

Adapting the Crystal Orientation Matrix (autoorientation=2)

If `autoorientation=2`, the model checks whether the detected crystallographic surface orientation is different than the provided direction (`waferori`) and adjusts the crystal orientation accordingly, swapping the last two rows of the crystal orientation matrix. For example, if `coordinatesystem=0`, the adaptation changes the value to `coordinatesystem=1`.

Specifying Crystal Orientation Matrix Using Reference Planes

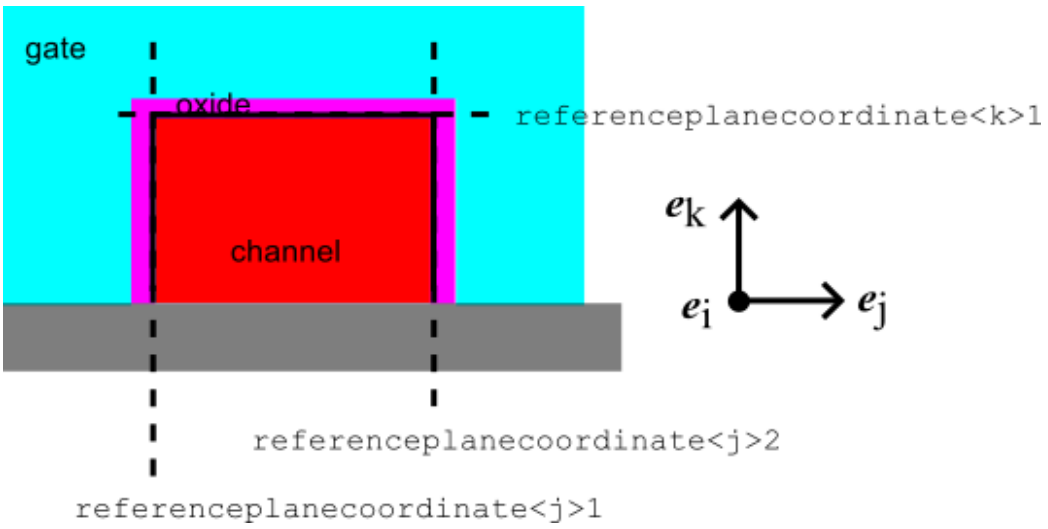
For devices having more than one MIS interface, for example, a FinFET or stacked nanosheet FET, it is necessary to assign the closest MIS interface to each vertex of the device for which mobility enhancement is prescribed. By setting `autoorientation=3`, the crystal orientation matrix is defined by reference planes and reference coordinates.

For each interface, you must define a `referenceplanecoordinate` (coordinate of the axis where the plane intersects the axis) and a `referenceplanesystem` (the axis specifying the plane).

Example of a FinFET

In a FinFET, the channel region is in contact with the gate over three sides. Figure 59 shows the cross-section of a simple FinFET device.

Figure 59 FINFET cross-section (solid line represents three interface orientations)



Using Figure 59, the following cases can be defined:

- The indices *i* = *x*, *j* = *z*, *k* = *y*, `referenceplanesystem<k>1=0`:

$$\mathbf{Q} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- The indices *i* = *x*, *j* = *z*, *k* = *y*, `referenceplanesystem<k>1=1`:
- The indices *i* = *x*, *j* = *y*, *k* = *z*, `referenceplanesystem<k>1=0`:

$$\mathbf{Q} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

- The indices *i* = *x*, *j* = *y*, *k* = *z*, `referenceplanesystem<k>1=1`:

$$\mathbf{Q} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

- The indices *i* = *y*, *j* = *z*, *k* = *x*, `referenceplanesystem<k>1=0`:

$$\mathbf{Q} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- The indices *i* = *y*, *j* = *z*, *k* = *x*, `referenceplanesystem<k>1=1`:

$$\mathbf{Q} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Specifying the Normal Stress Components in the Crystal Basis

The stress tensor in the crystal basis $\bar{\sigma}^*$ is determined from the stress tensor in the device basis $\bar{\sigma}$ and the crystal orientation matrix **Q**, that is:

$$\bar{\sigma}^* = \mathbf{Q}\bar{\sigma}\mathbf{Q}^T$$

Considering a diagonal stress tensor (no shear stress components) $\bar{\sigma}$:

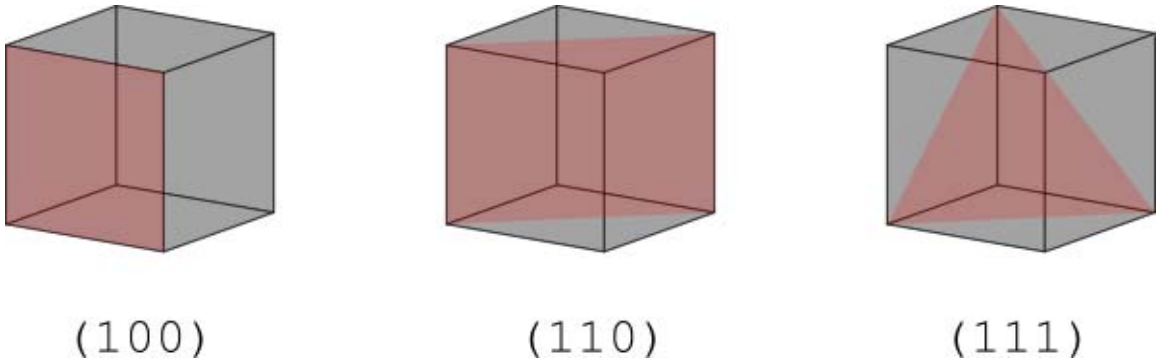
$$\bar{\sigma} = \begin{bmatrix} \sigma_x & 0 & 0 \\ 0 & \sigma_y & 0 \\ 0 & 0 & \sigma_z \end{bmatrix}$$

where the normal stress components (σ_x , σ_y , σ_z) are obtained from the TDR file or are directly specified by you – see `ConstantStress` in Table 16 on page 110.

Wafer Crystallographic Surface Orientation

The wafer crystallographic orientation represents the plane of the MIS interface (crystal basis). The plane is represented by a three-digit integer obtained by the inverse of the intersection of the plane with the three crystal axes *X*, *Y*, and *Z*.

Figure 60 Supported wafer crystallographic surface orientation (cubic crystal)



Assuming cubic crystal symmetry, you can either:

- Specify the wafer crystallographic surface orientation manually (`autoorientation=0`).
- Use auto-detection (`autoorientation>0`).

If `autoorientation>0`, Sentaurus Device returns the framework orientation that is closest to the actual orientation at the nearest interface vertex. The latter should be converted to the crystal basis using the crystal orientation matrix such that:

$$\hat{\mathbf{e}}_n^* = \mathbf{Q}\hat{\mathbf{e}}_n$$

where $\hat{e}_n$ is the normal to the MIS interface in the device basis.

Specifying the Crystal Configuration

Assuming cubic crystal symmetry, the PMI models compute the mobility enhancement factor γ – see Parameter Interface and Equations of MCmob Model on page 126 and Parameter Interface and Equations of MCmob Model on page 126, for four different configurations *k*:

- <100>/(100), channel orientation is 100, crystallographic surface orientation is 100.
- <110>/(100), channel orientation is 110, crystallographic surface orientation is 100.
- <100>/(110), channel orientation is 100, crystallographic surface orientation is 110.
- <110>/(110), channel orientation is 110, crystallographic surface orientation is 110.

where the orientations are in the crystal basis. The configuration is obtained using the variables `channelori` (channel direction) and `waferori` (crystallographic surface orientation) if `autoorientation=0`.

Using Parameters

The following examples illustrate how to use the parameters for different orientations and coordinate systems.

Example 1

Planar MISFET with (100) wafer orientation and device coordinate system where x=channel direction, y=wafer normal direction, and z=device width direction:

Use default parameter values.

Example 2

Planar MISFET with (100) wafer orientation and device coordinate system where x=channel direction, z=wafer normal direction, and y=device width direction, specify:

```
coordinatesystem=1
```

Example 3

Planar MISFET with (110) wafer orientation and device coordinate system where x=channel direction, z=wafer normal direction, and y=device width direction, specify either:

```
coordinatesystem=1
autoorientation=1
```

or:

```
coordinatesystem=1
waferori=110
```

Example 4

FinFET with (100) wafer orientation, <110> channel orientation, and device coordinate system where x=channel direction, y=wafer normal direction, and z=fin width direction, specify:

```
autoorientation=2
```

Example 5

FinFET with (100) wafer orientation, <110> channel orientation, and device coordinate system where x=channel direction, z=wafer normal direction, and y=fin width direction, specify:

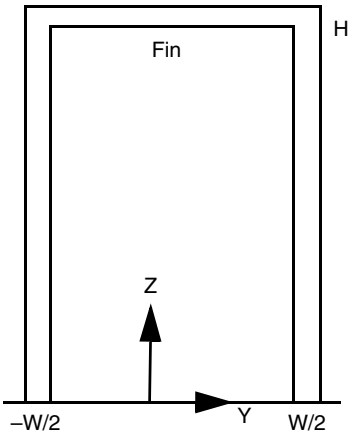
```
coordinatesystem=1
autoorientation=2
```

Example 6

FinFET with (100) wafer orientation, <100> channel orientation, and device coordinate system where x=channel direction, z=wafer normal direction, and y=fin width direction, specify:

```
coordinatesystem=1
autoorientation=3
referenceplanesystemy1=0
referenceplanesystemy2=0
referenceplanesystemz1=0
referenceplanecoordinatey1=-@W/2.0@
referenceplanecoordinatey2=@W/2.0@
referenceplanecoordinatez1=@H@
```

Figure 61 Cross section of a FinFET structure normal to the channel direction with height (H) and width (W); the origin of the coordinate system is at y=0 and z=0




```
referenceplanecoordinatey1=-W/2
referenceplanecoordinatey2=W/2
referenceplanecoordinatez1=H
referenceplanesystemy1=0
referenceplanesystemy2=0
referenceplanesystemz1=0
```

Parameter Interface

The `MCmob` and `SBmob` models have a parameter interface that allows you to change all the coefficients of the fit and interpolation functions. In this way, you can define the mobility response to mechanical stress using the Sentaurus Device parameter file. For historical reasons, the syntax of the interface differs for `MCmob` and `SBmob`.

Parameter Interface and Equations of SBmob Model explains the parameter interface of `SBmob` and its related implementations and equations.

Some settings must be specified in the Sentaurus Device command file. The `Type` parameter defines the device type: NMOS (`Type=0`) or PMOS (`Type=1`). The `Kanda` option switches on the Kanda model (see the *Sentaurus™ Device User Guide*).

Additional parameter files and explanations are available on request. Contact TCAD Support (see Contacting Your Local TCAD Support Team Directly) or Engineering & Consulting (tcad-services@synopsys.com).

Parameter Interface and Equations of SBmob Model

The `SBmob` model computes an isotropic enhancement factor γ by which the mobility or a parameter that determines the mobility changes due to stress in the device.

$$\gamma = 1 + f \tag{4}$$

where f is determined by nonlinear piezoresistive coefficients that are extracted from band structure simulations. The model has no accessible calibration parameters; it is precalibrated for SiGe.

Assuming cubic crystal symmetry, the factor f is defined for three configurations *k*, that is:

- <110>/(100), that is, `s100c110`.
- <110>/(110), that is, `s110c110`.
- <100>/(100), that is, `s100c100`.

where the numerator is the channel orientation (in the crystal coordinates) and the denominator is the wafer crystallographic surface orientation.

The piezo-factor for carrier type *q* (electrons or holes) and configuration *k* is defined by f_{kq}:

$$f_{kq} = \begin{cases} \sum_{j=X,Y,Z} \Delta\mu_j(p_{jkql}(\sigma_j) + C^{-}_{jkql}(\sigma_j)) & \text{if} \quad \sigma_j < \sigma_{-}, \\ \sum_{j=X,Y,Z} \Delta\mu_j(p_{jkql}(\sigma_j) + C^{+}_{jkql}(\sigma_j)) & \text{if} \quad \sigma_j < \sigma_{+}, \\ \sum_{J=X,Y,Z} \Delta\mu_j(p_{jkql}(\sigma_j)) & \text{otherwise.} \end{cases} \tag{5}$$

Where *X*, *Y*, *Z* are the coordinates in the crystal basis and where σ_{-} and σ_{+} are the minimum and maximum orthogonal stress component ($a_7, jkql$ and $a_9, jkql$) at which the polynomial model is extrapolated. The polynomial functions are:

$$p_{jkq} = \mu_j\left(\sum_{i=1}^{6} m_{i,jkql}\sigma_j\right) \tag{6}$$

where the polynomial coefficients $m_{i,kjql}$ are:

$$m_{i,jkql} = a_{i,jkql}\frac{a_{i,jkq(l+1)} - a_{i,jkql}}{\Delta s}(X_{Ge} - s_l)\sigma_j^{6-i} \tag{7}$$

where s_l is the lower edge of the mole fraction interval $\Delta s = 0.1$ at mole index l for which $s_{l+1} \leq X_{Ge} < s_l$. The coefficients $C^-_{jkql}(\sigma_j)$ and $C^+_{jkql}(\sigma_j)$ are:

$$C^-_{jkql}(\sigma_j) = a_{8,jkql}\frac{a_{8,jkq(l+1)} - a_{8,jkql}}{\Delta s}(X_{Ge} - s_l)(\sigma_j - a_{7,jkql}) \tag{8}$$

$$C^+_{jkql}(\sigma_j) = a_{10,jkql}\frac{a_{10,jkq(l+1)} - a_{10,jkql}}{\Delta s}(X_{Ge} - s_l)(\sigma_j - a_{9,jkql}) \tag{9}$$

Note:

The model truncates the orthogonal stress component values to be in the range -6 GPa to 6 GPa. In addition, the upper limit for the factor f is set to 10 by default. To adjust the upper limit, use the parameter `UpperLimits<k>`, where k is one of the three available configurations, that is, `s100c110`, `s100c100`, or `s110c110`.

It is possible to override the default coefficient $a_{i,jkq}$ using the following command pattern:

Command	Description
`auni<j= c, n or w>s<waferori>c<channelori><i=1-10><q= Type>`	$a_{i,jkq}$

Note:

The directions c, n, and w repectively refer to the *X*, *Y*, and *Z* directions.

The following polynomial describes the mobility response to the uniaxial stress components s_{xx}, s_{yy}, and s_{zz}:

$$\frac{\Delta\mu_i}{\mu_i} = a_{6,i}s_{ii}^1 + a_{5,i}s_{ii}^2 + a_{4,i}s_{ii}^3 + a_{3,i}s_{ii}^4 + a_{2,i}s_{ii}^5 + a_{1,i}s_{ii}^6 \tag{10}$$

where i is the index for the x-, y-, or z-component of the mobility.

The parameter interface allows you to define the coefficients of the polynomials for the mobility response to uniaxial stress. Table 17 lists most of these coefficients. Those not listed can be derived from information in the table.

Table 17 Coefficients of polynomials for mobility response to uniaxial stress

Coefficient	Description
`aunics100c1106h`	Coefficient a_6 of the hole mobility response to stress in the channel direction for (100) surface orientation and <110> channel direction. The other five coefficients are named as follows: • `aunics100c1105h` • `aunics100c1104h` • `aunics100c1103h` • `aunics100c1102h` • `aunics100c1101h`
`aunins100c1101h`	Coefficient a_1 of the hole mobility response to the stress normal to the MIS interface for (100) surface orientation and <110> channel direction.
`auniws100c1001h`	Coefficient a_1 of the hole mobility response to stress in the width direction of the device for (100) surface orientation and <100> channel direction.
`auniws100c1101h`	Coefficient a_1 of the hole mobility response to stress in the width direction of the device for (100) surface orientation and <110> channel direction.
`auniws110c1101h`	Coefficient a_1 of the hole mobility response to stress in the width direction of the device for (110) surface orientation and <110> channel direction.

All a_i coefficients are mole fraction dependent. The coefficients must be defined for each of the 11 points of the mole fraction discretization, that is, for germanium or silicon content of 0, 0.1, 0.2, ..., 1.0. For example, to set the coefficient a_6 for holes with stress applied in the channel direction with channel orientation 110 and wafer orientation 100, use:

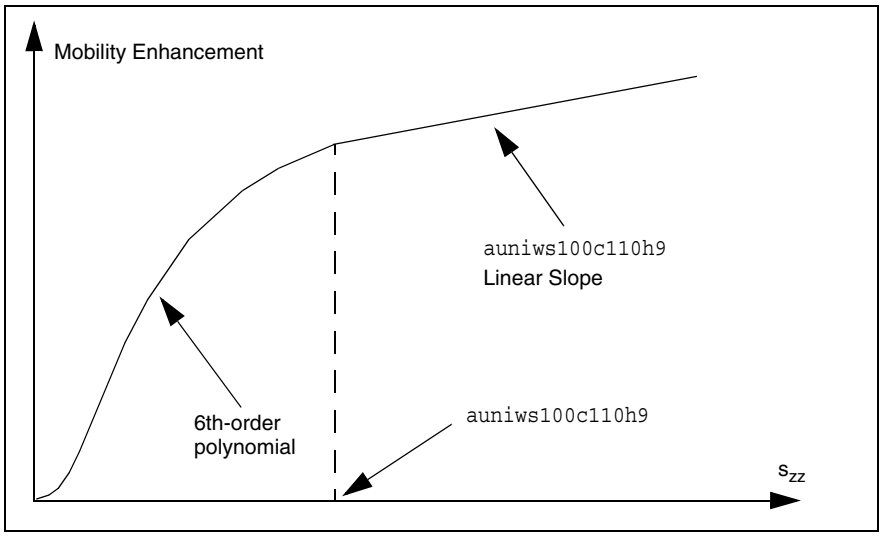
```
aunics100c1101h=(-0.20156, -0.21273, -0.19696, -0.18257, -0.18225,
-0.17145, -0.16738, -0.16675, -0.16368, -0.15079, -0.15600)
```

To improve interpolation for high stress and for cases where the mobility response becomes independent of the stress or behaves linearly, additional coefficients for the mobility response to uniaxial stress are introduced. In the parameter file, these coefficients have the same name and syntax as the parameters for the a_i coefficients but have higher numbers (see Table 18).

Table 18 Additional coefficients of polynomials for mobility response to uniaxial stress

Coefficient	Description
`auniws100c1107h`	Sets the negative stress value in GPa for the switch between the 6th-order polynomial and linear slope for the hole mobility response to stress in the width direction of the device for (100) surface orientation and <110> channel direction (see Figure 62 on page 126).
`auniws100c1108h`	Sets the slope in 1/Pa for the linear slope for stress values less than `auniws100c1107h` for the hole mobility response to stress in the width direction of the device for (100) surface orientation and <110> channel direction (see Figure 62).
`auniws100c1109h`	Sets the positive stress value in GPa for the switch between the 6th-order polynomial and linear slope for the hole mobility response to stress in the width direction of the device for (100) surface orientation and <110> channel direction (see Figure 62).
`auniws100c11010h`	Sets the slope in 1/Pa for the linear slope for stress values greater than `auniws100c1109h` for the hole mobility response to stress in the width direction of the device for (100) surface orientation and <110> channel direction (see Figure 62).

Figure 62 Use of the ai *parameters in fitting uniaxial stress curves*



Parameter Interface and Equations of MCmob Model

The MCmob model computes the factor f (see Equation 4) using nonlinear piezoresistive coefficients and higher-order cross-term corrections that are extracted from Monte Carlo simulations. The model has no accessible calibration parameters; it is precalibrated for SiGe.

Assuming cubic crystal symmetry, the factor f is defined for four configurations *k*, that is:

<100>/(100), that is, `s100c100`

<110>/(100), that is, `s100c110`

<100>/(110), that is, that is, `s110c100`

<110>/(110), that is, `s110c110`

where the numerator is the channel orientation (in the crystal coordinates) and the denominator is the wafer crystallographic surface orientation.

The piezo-factor for carrier type *q* (electrons or holes) and configuration *k* is defined by f_{kq} .

$$f_{kq} = \begin{cases} \sum_{j = X, Y, Z} \Delta\mu_j(p_{jkq}(\sigma_j) + C_j) & \text{if corrections = 0,} \\ \sum_{j = X, Y, Z} \Delta\mu_j(p_{jkq}(\sigma_j) + C_j) + \sum_{i = 1}^{19} f_{i, kq} & \text{otherwise.} \end{cases} \quad (11)$$

where the polynomial functions are

$$p_{jkq} = \mu_j\left(\sum_{i = 1}^{6} m_{i, jkq}\sigma_j\right) \quad (12)$$

with the polynomial coefficients $m_{i, jkq}$ depends on the local germanium mole fraction X_{Ge}.

$$m_{1, jkq} = (a_{1, jkq} + b_{1, jkq}X_{Ge} + c_{1, jkq}X_{Ge}^2)\sigma_j^5 \quad (13)$$

$$m_{2, jkq} = (a_{2, jkq} + b_{2, jkq}X_{Ge} + c_{2, jkq}X_{Ge}^2)\sigma_j^4 \quad (14)$$

$$m_{3, jkq} = (a_{3, jkq} + b_{3, jkq}X_{Ge} + c_{3, jkq}X_{Ge}^2)\sigma_j^3 \quad (15)$$

$$m_{4, jkq} = (a_{4, jkq} + b_{4, jkq}X_{Ge} + c_{4, jkq}X_{Ge}^2)\sigma_j^2 \quad (16)$$

$$m_{5, jkq} = (a_{5, jkq} + b_{5, jkq}X_{Ge} + c_{5, jkq}X_{Ge}^2)\sigma_j^1 \quad (17)$$

$$m_{6, jkq} = (a_{6, jkq} + b_{6, jkq}X_{Ge} + c_{6, jkq}X_{Ge}^2) \quad (18)$$

and where the cross-correction terms $f_{i, kq}$ are:

$$f_{1, kq} = d_{1, kq}\sigma_X\sigma_Y \quad (19)$$

$$f_{2, kq} = d_{2, kq}\sigma_X\sigma_Y \quad (20)$$

$$f_{3, kq} = d_{3, kq}\sigma_X\sigma_Y \quad (21)$$

$$f_{4, kq} = d_{4, kq}\sigma_X\sigma_Z^2 \quad (22)$$

$$f_{5, kq} = d_{5, kq}\sigma_X\sigma_Y^2 \quad (23)$$

$$f_{6, kq} = d_{6, kq}\sigma_Y\sigma_X^2 \quad (24)$$

$$f_{7,kq} = d_{7,kq}\sigma_Y\sigma_Z^2 \quad (25)$$

$$f_{8,kq} = d_{8,kq}\sigma_Z\sigma_X^2 \quad (26)$$

$$f_{9,kq} = d_{9,kq}\sigma_Z\sigma_Y^2 \quad (27)$$

$$f_{10,kq} = d_{10,kq}\sigma_X\sigma_Y\sigma_Z \quad (28)$$

$$f_{11,kq} = d_{11,kq}\sigma_X\sigma_Z^3 \quad (29)$$

$$f_{12,kq} = d_{12,kq}\sigma_X\sigma_Y^3 \quad (30)$$

$$f_{13,kq} = d_{13,kq}\sigma_Y\sigma_X^3 \quad (31)$$

$$f_{14,kq} = d_{14,kq}\sigma_Y\sigma_Z^3 \quad (32)$$

$$f_{15,kq} = d_{15,kq}\sigma_Z\sigma_X^3 \quad (33)$$

$$f_{16,kq} = d_{16,kq}\sigma_Z\sigma_Y^3 \quad (34)$$

$$f_{17,kq} = d_{17,kq}\sigma_Y^2\sigma_Z^2 \quad (35)$$

$$f_{18,kq} = d_{18,kq}\sigma_X^2\sigma_Z^2 \quad (36)$$

$$f_{19,kq} = d_{19,kq}\sigma_X^2\sigma_Y^2 \quad (37)$$

Note:

The model truncates the orthogonal stress component values to be in the range -6 GPa to 6 Pa.

It is possible to override the default coefficient using the following patterns

Command	Description
`a<j= c, n, or w>si0ges<waferori>c<channelori><i=1-10><q= Type>`	$a_{i,jkq}$
`a<j= c, n, or w>si1ges<waferori>c<channelori><i=1-10><q= Type>`	$b_{i,jkq}$
`a<j= c, n, or w>si2ges<waferori>c<channelori><i=1-10><q= Type>`	$c_{i,jkq}$
`as<waferori>c <channelori><i=1-20><q= Type>`	$d_{i,jkq}$

Note:

The directions `c`, `n`, and `w` respectively refer to the *X*, *Y*, and *Z* directions.

Example: Overriding the Default Coefficient

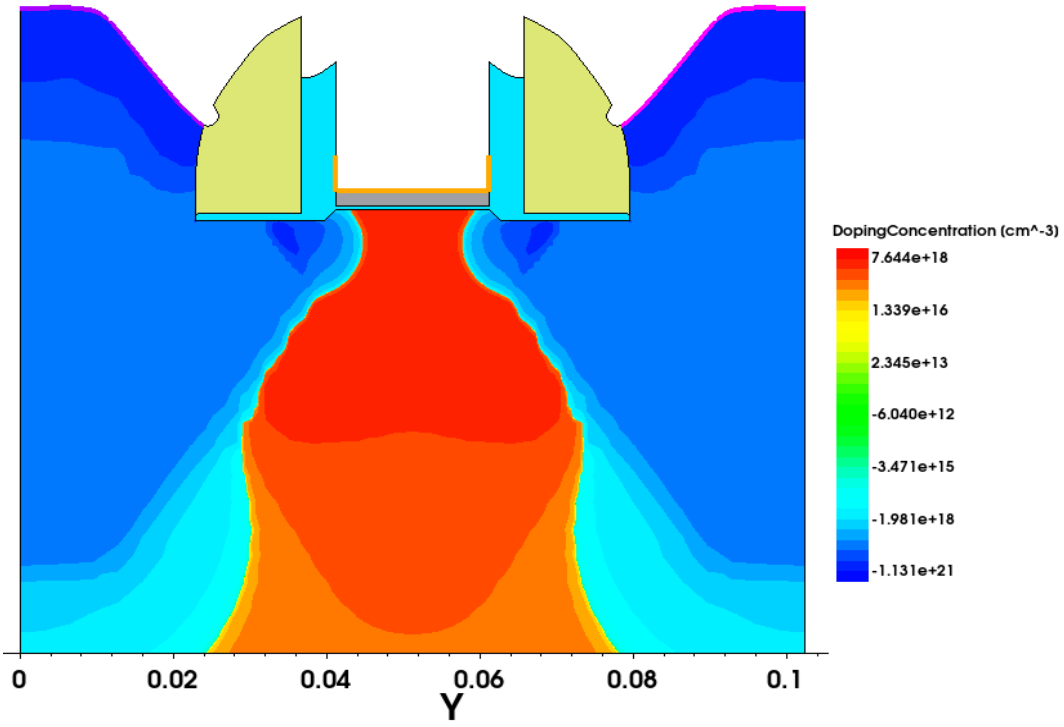
Overriding the coefficient `a1` for holes in the channel direction for the <110>/(100) configuration:

```
acsi0ges100c1101h = 3.3204e-02
```

Parameter File Example

Figure 63 shows a planar FET device with wafer crystallographic surface orientation (100), and channel orientation <110> along the y-direction (device basis).

Figure 63 Planar FET and doping concentration



The following snippet shows the content of the `.par` file.

```
SBmob:
{
   ConstantStress = 1 # specify stress in .par file
   Stress_yy = -3.26e+07 # stress [Pa]
   channelori = 100 # channel orientation in crystal basis
   coordinatesystem = 2 # definition of the crystal orientation matrix
   autoorientation = 1 # auto-detect the wafer crystallographic
orientation
   # Set a1 for holes in the channel direction for the <110>/(100)
   # configuration. Each number of the list represents the value of a1 at
   # the different mole fractions 0, 0.1, 0.2, .., 1.0
   aunics100c1101h = (-0.20156, -0.21273, -0.19696, -0.18257, -0.18225,
     -0.17145, -0.16738, -0.16675, -0.16368, -0.15079, -0.15600)
}
```

PMI for Calculating the Sheet Resistance

For the calculation of sheet resistance, you can use the Sentaurus Device PMI `sheetresistance`. In this approach, Sentaurus Device solves the Poisson equation or the complete set of drift-diffusion equations using an arbitrary doping profile as input. Then, the PMI calculates the sheet resistance using Equation 38:

$$R_s = \frac{1}{e \cdot \int_0^t \mu_n \cdot n \cdot dx} \qquad \text{in } \Omega/\text{square} \tag{38}$$

All mobility models that are available in Sentaurus Device can be used together with this PMI. R_s is written as a Sentaurus Workbench design-of-experiments parameter into the log file of Sentaurus Device and can be used by Sentaurus Workbench for further analysis. The input syntax for the Sentaurus Device command file is:

```
CurrentPlot {
   PMIModel (
      Name = "sheetresistance"
      rs_name = "sheetres"
      # BoxMin = (-1.0) BoxMax = (0.05)   # With this command, you can
                                 # change the integration interval,
                                 # for example, in cases where an
                                 # adjustment of the integration depth
                                 # is necessary.
   )
}
```

where `sheetres` is the name of the design-of-experiments parameter (default: `Rs_sim`).